SQL Retail Optical Business Analytics Project

A comprehensive RDBMS-driven analysis of Customer journeys,
Staff performance, Product & Brand outcomes, Referrers impacts on business

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Tools used:

- SQL (Data Analysis)
- Advance Excel (Data Visualization)
- Git Hub (Project Management)

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Problem Statement:

An independent optical retailer wants to understand which customers generate the highest lifetime value, which referral channels drive sustainable growth, how staff performance varies, and which operational improvements can increase revenue and retention.

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2. Project Introduction & Overview

2.1 Background & Motivation

The idea for this project was inspired by a personal connection, a family member who owns and runs a retail optical store (family business). Observing the day-to-day operations of that business raised a natural question: *beyond selling glasses, what does the data actually tell us about how the business is really performing?*

This project was built with three goals in mind:

- To learn SQL through a real-world business scenario
- To build a strong end-to-end portfolio project
- To apply Data analysis to a domain that felt familiar and meaningful.

An optical retail store turned out to be an ideal domain for data analysis. It involves multiple layers clinical records, customer walk-ins, staff interactions, product sales, and post-sale delivery all of which generate rich, interconnected data.

2.2 Project Overview

This is a mid-scale SQL analytics project built around a simulated retail optical store. The project covers the full cycle from database design to data generation to business analysis and is organized into the following components:

- **Database Design** A relational database named retail optical business consisting of 9 interconnected tables, designed in Third Normal Form (3NF)
- **Synthetic Dataset** Over 35,000 SQL statements generating approximately 8,000 customers, 8,000 prescriptions, and 5,000 invoices, built using a Python-based data seeding engine with the Faker library
- **SQL Analysis** Approximately 2,500 lines of structured SQL queries across five analysis modules
- **Business Insights** Actionable findings extracted from the data across customer behaviour, staff performance, referrer impact, and product outcomes

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2.3 Business Objectives

This project performs a 360° analysis of the optical store business, addressing four core business questions:

1. **Customer Behaviour** Who are our customers? How many walk-in visitors actually make a purchase? Who comes back repeatedly?
2. **Staff Performance** Which staff members are driving the most sales and revenue?
3. **Referrer Impact** Are word-of-mouth referrals bringing in real buyers, or just window shoppers?
4. **Product & Brand Performance** Which product types and brands are selling the most?

2.4 Disclaimer

This project is intended solely for educational and portfolio purposes. All names, contact details, and transactional records are synthetically generated using the Python Faker library and do not correspond to any real individuals or businesses. No real-world Personally Identifiable Information (PII) is included in this dataset.

3. Database Schema Design (EER Diagram)

3.1 Database Schema Overview

- The database for this project is named “retail optical business”.
- It is built using MySQL and designed in Third Normal Form (3NF), which eliminates data redundancy and ensures data integrity across all tables.
- The schema consists of 9 interconnected tables that together capture the complete journey of a customer from their first walk-in and eye checkup, all the way through to product purchase and post-sale delivery.

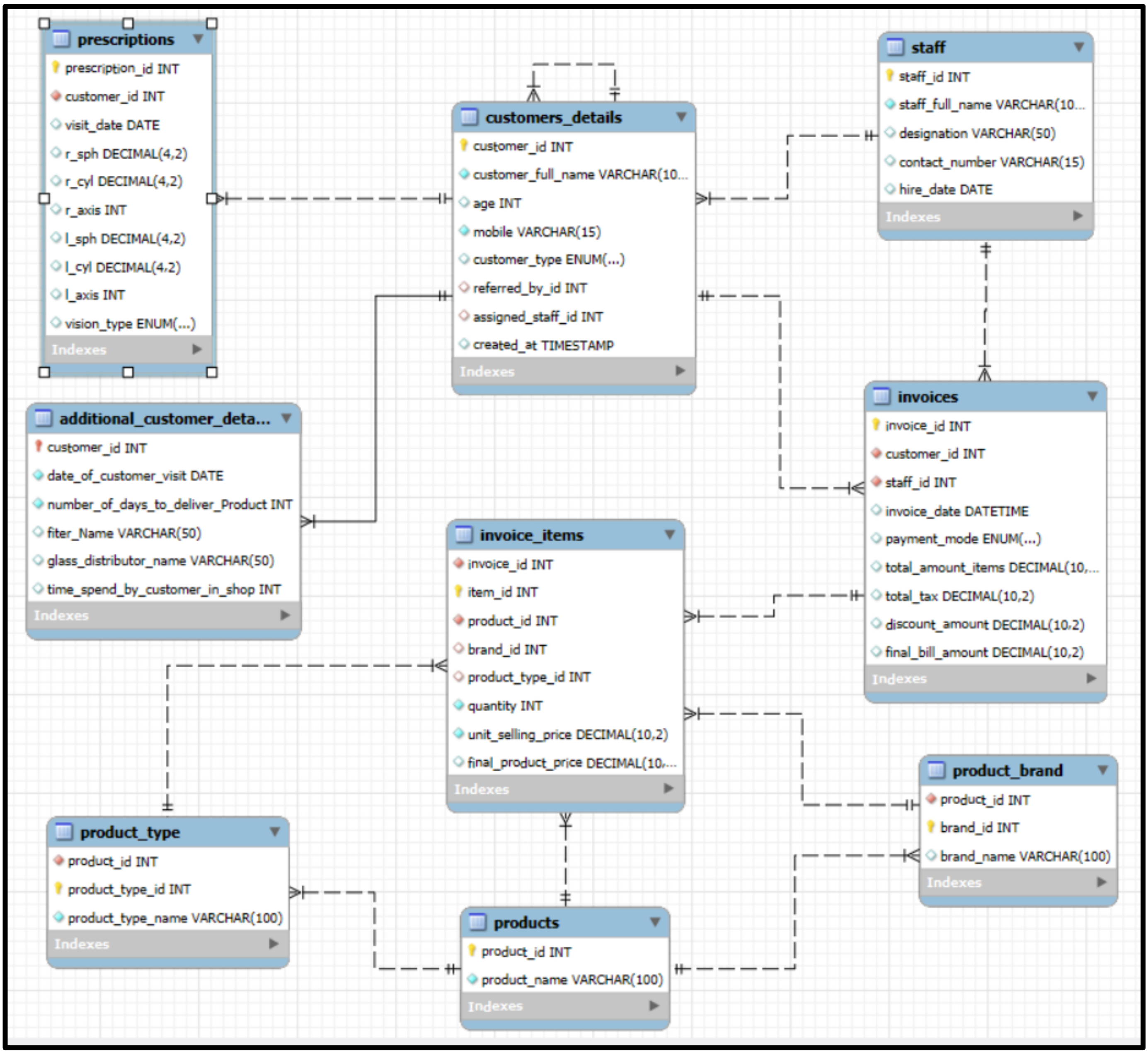
3.2 Database Layers

The 9 tables are organized across 5 logical layers, each serving a distinct business purpose:

Layer	**Tables Included**
Core Layer	Customers details, Staff details
Medical Layer	Prescriptions
Operations Layer	Additional customers details
Sales Layer	Invoices, Invoice Items
Product Layer	Products, Product Type, Product Brand

3. EER Diagram

The Enhanced Entity-Relationship (EER) Diagram below provides a visual map of the entire database. Each table is connected through Primary Keys (PK) and Foreign Keys (FK), ensuring every record is properly linked and no data is stored in isolation.



2.2) Database Schema

```sql
-- Table 1 : Staff Table (Stores the Satff data)

CREATE TABLE staff (
    staff_id INT PRIMARY KEY AUTO_INCREMENT,
    staff_full_name VARCHAR(100) NOT NULL,
    designation VARCHAR(50),
    contact_number VARCHAR(15) UNIQUE,
    hire_date DATE DEFAULT (CURRENT_DATE)
);
```

```sql
-- Table 2 : Customer_details (Stores the Customer_data)

CREATE TABLE customers_details (
    customer_id INT PRIMARY KEY AUTO_INCREMENT,
    customer_full_name VARCHAR(100) NOT NULL,
    age INT CHECK (age > 0 AND age < 120),
    mobile VARCHAR(15) UNIQUE NOT NULL,
    customer_type ENUM('New', 'Repeated', 'Referral'),
    referred_by_id INT, -- Links to another customer_id
    assigned_staff_id INT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (referred_by_id) REFERENCES customers_details(customer_id),
    FOREIGN KEY (assigned_staff_id) REFERENCES staff(staff_id)
);
```

```sql
-- Table 3 : Prescription Table
-- Note: Prescriptions change over time. We store them as a history.
CREATE TABLE prescriptions (
    prescription_id INT PRIMARY KEY AUTO_INCREMENT,
    customer_id INT NOT NULL,
    visit_date DATE DEFAULT (CURRENT_DATE),
    r_sph DECIMAL(4,2), r_cyl DECIMAL(4,2), r_axis INT,
    l_sph DECIMAL(4,2), l_cyl DECIMAL(4,2), l_axis INT,
    vision_type ENUM('Near', 'Distance', 'Bifocal', 'Progressive'),
    FOREIGN KEY (customer_id) REFERENCES customers_details (customer_id)
    ON DELETE CASCADE
);
```

```sql
-- Table 4 : Products Table

CREATE TABLE products (
    product_id INT PRIMARY KEY AUTO_INCREMENT,
    product_name VARCHAR(100) NOT NULL
);

INSERT INTO products
( product_id, product_name)
VALUES
( 1, "Frame"),
( 2, "Glass"),
( 3, "Sunglass"),
( 4, "Contact_Lens"),
( 5, "Accessories"),
( 6, "Lens Solution");
```

```sql
-- Table 5: Product_Brand

CREATE TABLE product_brand (
        product_id INT NOT NULL,
        brand_id INT PRIMARY KEY AUTO_INCREMENT,
        brand_name VARCHAR(100),
        FOREIGN KEY(product_id) REFERENCES products(product_id)
);

INSERT INTO product_brand
( product_id, brand_id, brand_name)
VALUES
( 1, 1, "Local Brand"),
( 1, 2, "K & D"),
( 1, 3, "NOVA"),
( 1, 4, "Ray Ban"),
( 1, 5, "Prada"),
( 1, 6, "OAKLEY"),
```

```sql
-- Table 6: Product Type Table
CREATE TABLE product_type (
    product_id INT NOT NULL,
    product_type_id INT PRIMARY KEY,
    product_type_name VARCHAR(100) NOT NULL,
    FOREIGN KEY(product_id) REFERENCES products(product_id)
);

INSERT INTO product_type
( product_id, product_type_id, product_type_name)
VALUES
( 1,1, "Metal+Square"),
( 1,2, "Metal+Rectangle"),
( 1,3, "Metal+Round"),
( 1,4, "Metal+Hexagon"),
( 1,5, "Metal+Pilot"),
( 1,6, "Metal+Oval"),
( 1,7, "Plastic+Square")
```

```sql
-- Table 7 : Invoice Table

CREATE TABLE invoices (
    invoice_id INT PRIMARY KEY AUTO_INCREMENT,
    customer_id INT NOT NULL,
    staff_id INT NOT NULL,
    invoice_date DATETIME DEFAULT CURRENT_TIMESTAMP,
    payment_mode ENUM('Cash', 'UPI', 'Card'),
    total_amount_items DECIMAL(10,2),   -- SUM of all item prices
    total_tax DECIMAL(10,2) DEFAULT 0.00,
    discount_amount DECIMAL(10,2) DEFAULT 0.00,
    final_bill_amount DECIMAL(10,2) GENERATED ALWAYS AS
    (COALESCE(total_amount_items,0) + COALESCE(total_tax,0) - COALESCE(discount_amount,0) ) STORED,
    FOREIGN KEY (customer_id) REFERENCES customers_details(customer_id),
    FOREIGN KEY (staff_id) REFERENCES staff(staff_id)
);
```

```sql
-- Table 8: Item_in_Invoice

CREATE TABLE invoice_items (
    invoice_id INT NOT NULL,
    item_id INT PRIMARY KEY AUTO_INCREMENT,
    product_id INT NOT NULL,
    brand_id INT,
    product_type_id INT,
    quantity INT NOT NULL DEFAULT 1,
    unit_selling_price DECIMAL(10,2) NOT NULL,
    final_product_price DECIMAL(10,2) GENERATED ALWAYS AS (unit_selling_price * quantity) STORED,
    FOREIGN KEY (invoice_id) REFERENCES invoices(invoice_id)
    ON DELETE CASCADE,
    FOREIGN KEY (product_id) REFERENCES products(product_id)
    ON DELETE CASCADE,
    FOREIGN KEY (brand_id) REFERENCES product_brand(brand_id)
    ON DELETE CASCADE,
    FOREIGN KEY (product_type_id) REFERENCES product_type(product_type_id)
    ON DELETE CASCADE
);
```

```sql
-- Table 9: Additional Details about the Customer

CREATE TABLE additional_customer_details(
     customer_id INT PRIMARY KEY,
     date_of_customer_visit DATE NOT NULL,
     number_of_days_to_deliver_Product INT NOT NULL,
     fiter_Name VARCHAR(50),
     glass_distributor_name VARCHAR(50),
     time_spend_by_customer_in_shop INT,
     FOREIGN KEY(customer_id) REFERENCES customers_details(customer_id)
     ON DELETE CASCADE
);
```

2.3.2 Database Tables Description (9 Interconnected Tables)

Table 1: Customers Details

Stores basic information about every customer, such as name, age, mobile number, customer type, assigned staff, and referral details. It acts as the central table of the database and connects to most other tables. Used for customer tracking and behaviour analysis.

Table 2: Staff

Stores details of employees, including their name, designation, contact number, and hire date. It helps track which staff member served customers and generated sales. Used for staff performance analysis.

Table 3: Prescriptions

Stores eye examination records, including visit date, eye power measurements, and vision type. Each record is linked to a customer. Used to analyse eye checkups and customer conversion into sales.

Table 4: Additional Customer Details

Stores operational information such as visit date, delivery time, filter used, distributor name, and time spent in the shop. Extends customer information beyond sales. Used for customer experience and operational analysis.

Table 5: Invoices

Stores summary information for each sale, including invoice date, payment mode, taxes, discounts, and final bill amount. Each invoice is linked to a customer and staff member. Used for revenue and sales analysis.

Table 6: Invoice Items

Stores detailed information about products sold in each invoice, including product, brand, type, quantity, and selling price. Provides item-level transaction data. Used for detailed sales and product analysis.

Table 7: Products

Stores the master list of products available in the optical store. Each product is assigned a unique ID and name. Serves as the foundation for product-related tables.

Table 8: Product Type

Stores product categories such as Types of frames, sunglasses, single-vision lenses, and bifocal lenses, etc (total 36 types of products). Each type is linked to a product. Used to analyse sales and performance by product category.

Table 9: Product Brand

Stores brand information for products, such as Ray-Ban, Titan, etc (total 25 brands) . Each brand record is linked to a product. Used for brand-wise sales and customer preference analysis.

2.3.3Foreign Key Relationships

1) Customer-Centric Relationships

FK 1: prescriptions.customer_id → customers_details.customer_id

Relationship: One-to-Many
Purpose: Links each prescription to a customer.
Business Reasoning: Helps track customer eye checkup history and analyze conversion from checkups to purchases.

FK 2: additional_customer_details.customer_id → customers_details.customer_id

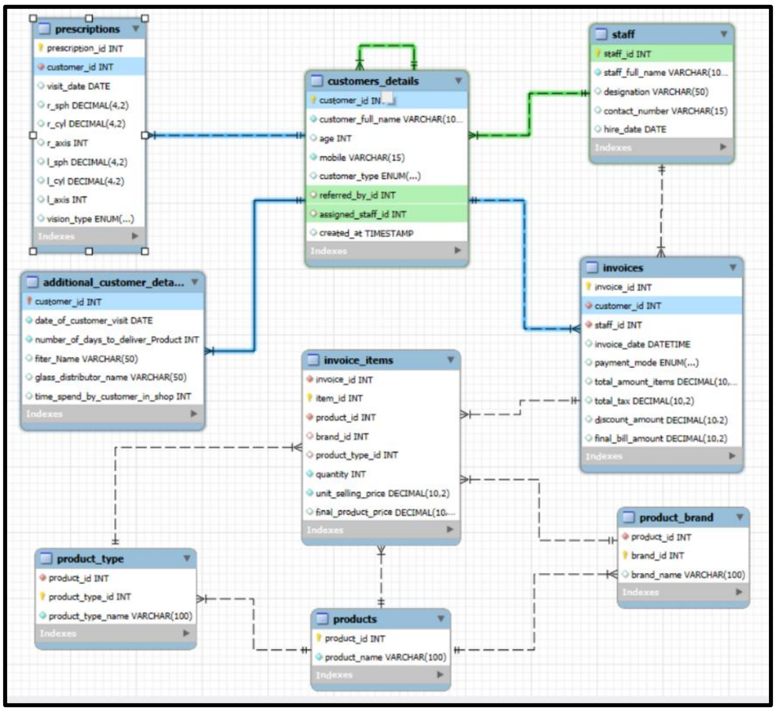
Relationship: One-to-One
Purpose: Links operational and CRM details to a customer.
Business Reasoning: Supports delivery, distributor, and customer experience analysis.

FK 3: customers_details.referred_by_id → customers_details.customer_id

Relationship: Self-Referencing
Purpose: Connects customers with the customers who referred them.
Business Reasoning: Enables referral tracking and analysis of referral-based customer acquisition.

FK 4: customers_details.assigned_staff_id → staff.staff_id

Relationship: Many-to-One
Purpose: Links customers to the staff member who attended them.
Business Reasoning: Helps measure staff workload and customer handling performance.



2) Invoice & Sales Relationships

FK 5: invoices.customer_id → customers_details.customer_id

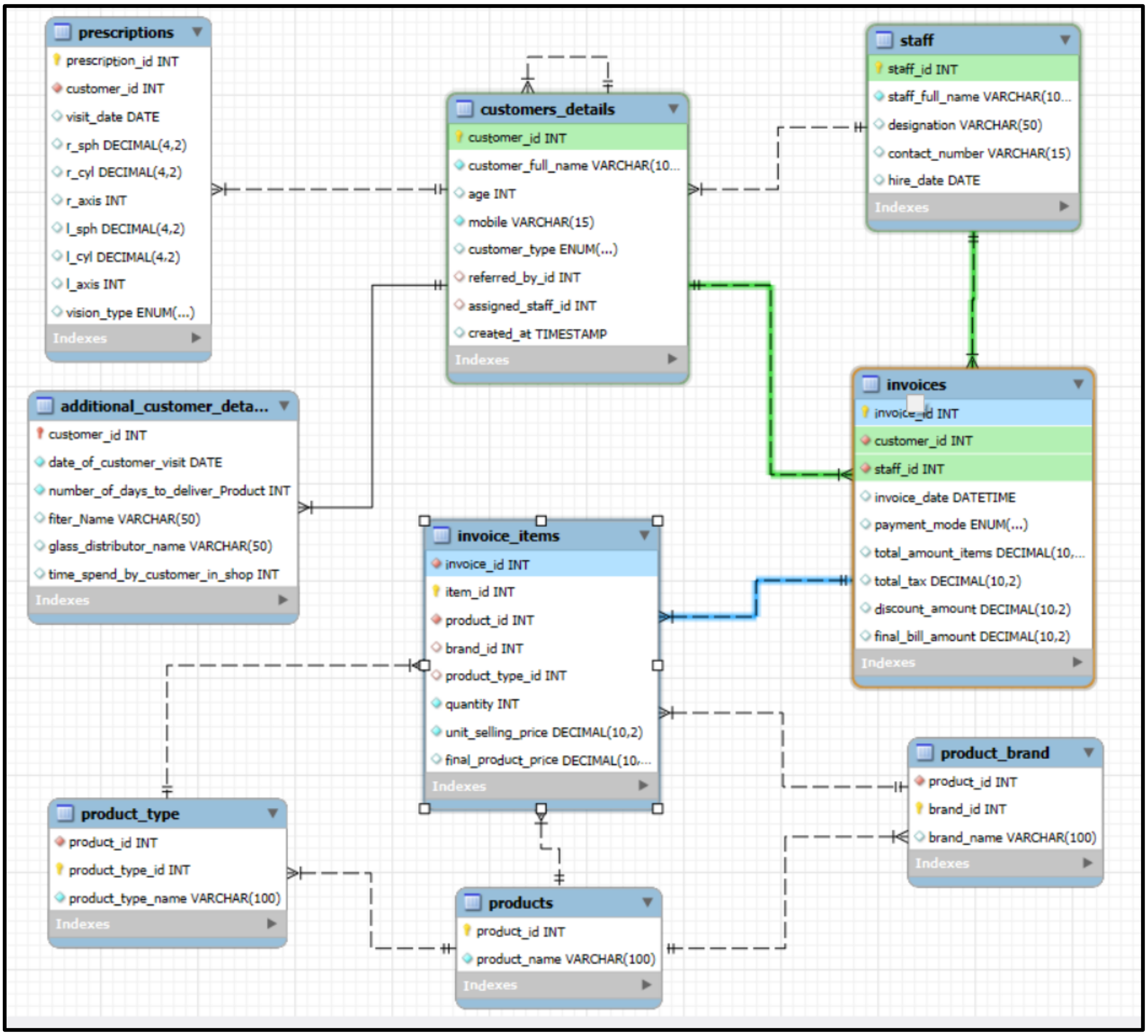
Relationship: One-to-Many
Purpose: Links invoices to customers.
Business Reasoning: Enables customer spending, repeat purchase, and lifetime value analysis.

FK 6: invoices.staff_id → staff.staff_id

Relationship: Many-to-One
Purpose: Links invoices to staff members.
Business Reasoning: Supports staff sales and revenue performance analysis.

FK 7: invoice_items.invoice_id → invoices.invoice_id

Relationship: One-to-Many
Purpose: Links invoice items to their parent invoice.
Business Reasoning: Enables detailed product-level sales analysis within each transaction.



3) Invoice Items - Centric Relationships

FK 8: invoice_items.product_id → products.product_id

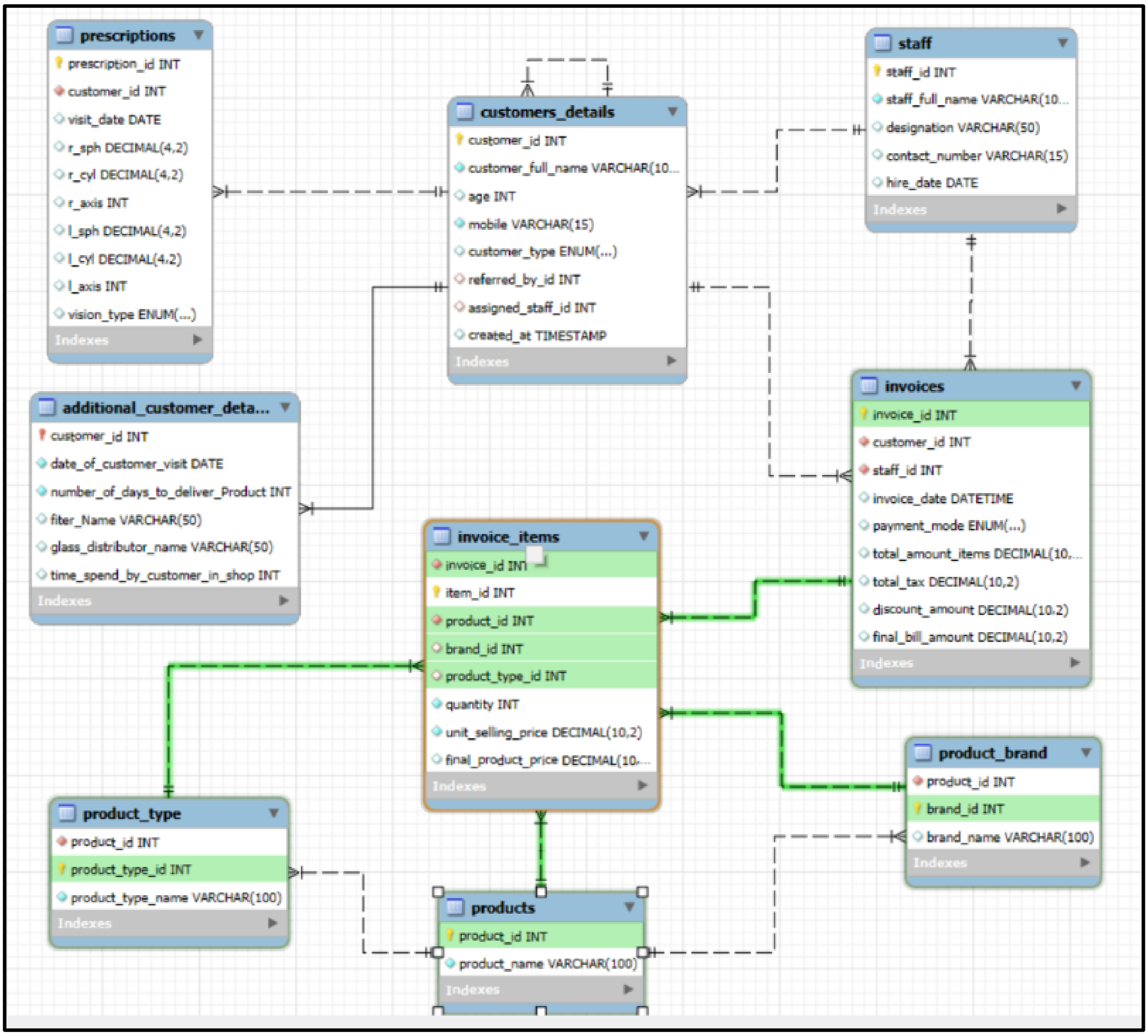
Relationship: Many-to-One
Purpose: Links sold items to products.
Business Reasoning: Helps analyze product sales, demand, and revenue.

FK 9: invoice_items.brand_id → product_brand.brand_id

Relationship: Many-to-One
Purpose: Links sold items to brands.
Business Reasoning: Enables brand-wise sales and customer preference analysis.

FK 10: invoice_items.product_type_id → product_type.product_type_id

Relationship: Many-to-One
Purpose: Links sold items to product categories.
Business Reasoning: Supports category-level sales and inventory analysis.



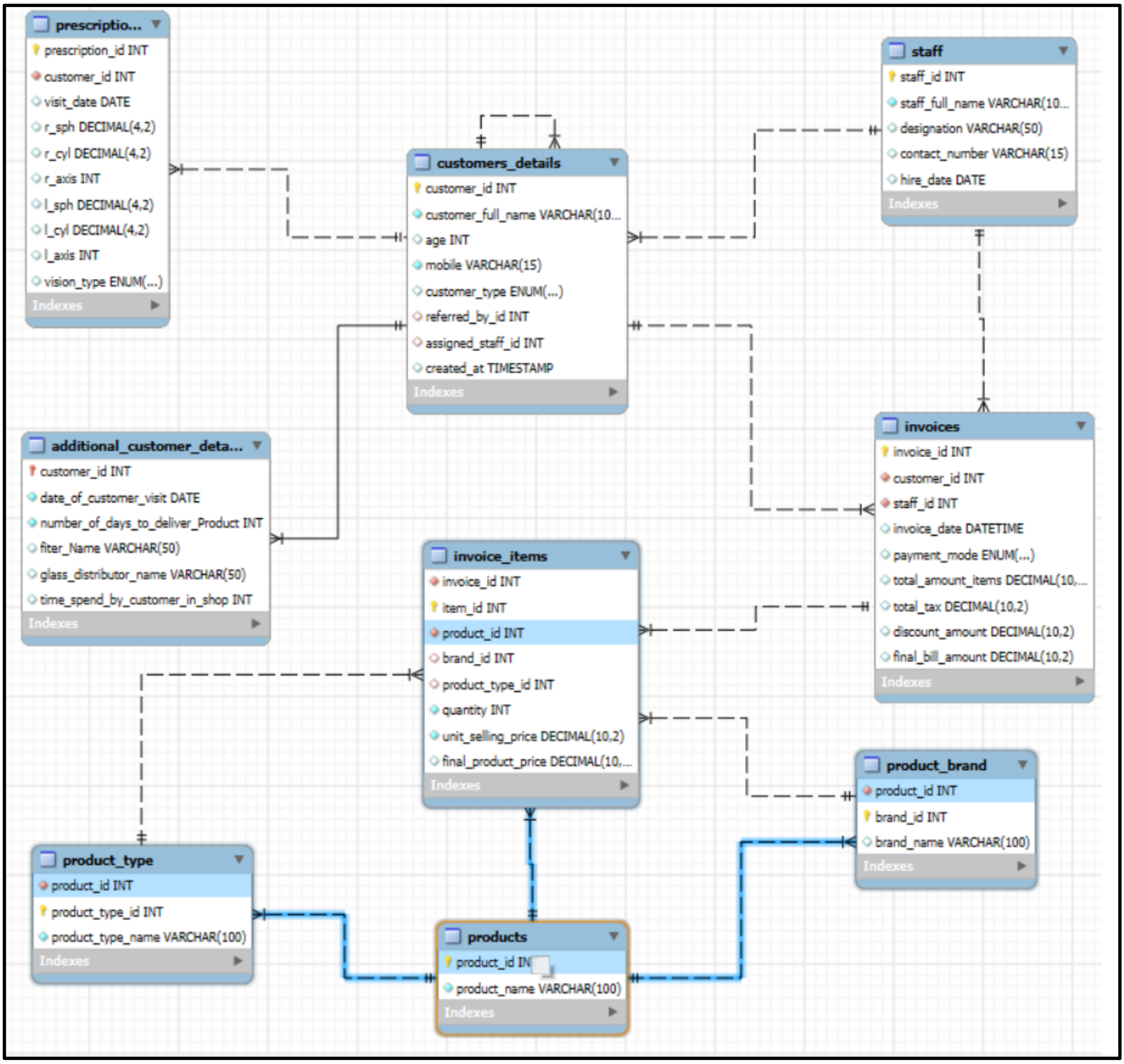
4) Products – Centric Relationships

FK 11: product_brand.product_id → products.product_id

Relationship: Many-to-One
Purpose: Links brands to products.
Business Reasoning: Maintains the product-brand hierarchy and data consistency.

FK 12: product_type.product_id → products.product_id

Relationship: Many-to-One
Purpose: Links product types to products.
Business Reasoning: Maintains the product-category hierarchy and supports product classification.



3. SQL Techniques Used

1. Basic SQL Clauses

- SELECT : Retrieve data from one or more tables
- WHERE : Filter rows based on specific conditions
- GROUP BY : Group rows for aggregation
- GROUP BY ... WITH ROLLUP : Generate subtotals and grand totals alongside grouped results
- HAVING : Filter grouped results after aggregation
- ORDER BY : Sort results in ascending or descending order
- LIMIT : Restrict output to top N rows
- DISTINCT : Remove duplicate values and count unique records
- AS (Alias) : Rename columns and tables for better readability

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2. Aggregate Functions

- COUNT() : Count total rows or records
- COUNT(DISTINCT ...) : Count only unique values
- SUM() : Calculate totals (revenue, quantity, tax, discount)
- AVG() : Calculate averages (bill amount, time spent in shop)
- MAX() : Find the highest value (latest date, maximum age)
- MIN() : Find the lowest value (earliest date, minimum age)
- ROUND() : Round decimal values to a specified number of places

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3. Conditional Logic & Pivot Queries

- CASE WHEN ... THEN ... ELSE ... END : Apply conditional logic inside queries
- CASE inside GROUP BY : Categorize customers into age groups, customer segments
- SUM(CASE WHEN ...) : Build SQL Pivot tables by converting rows into columns
- COUNT(CASE WHEN ...) : Count conditional occurrences across categories
- Customer Revenue Segmentation : Classifying customers as Platinum, Gold, Silver using CASE

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4. JOINs

- INNER JOIN: Fetch only matching records between two tables
- LEFT JOIN: Fetch all records from the left table including non-matching rows
- LEFT Exclusive JOIN: LEFT JOIN ... WHERE right_table.id IS NULL to extract non-matching records only (used for window shoppers and non-buyer referrers)
- Self JOIN: Joining a table to itself (used on customers_details via referred_by_id for referral network analysis)
- Multiple JOINS : Chaining 3, 4, or 5 tables in a single query (e.g. customers → invoices → invoice_items → products → product_brand)

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5. Subqueries

- Inline Subquery inside SELECT : Used to calculate window shopper rate against total customer base
- Scalar Subquery : A subquery that returns a single value used directly inside a calculation

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6. CTEs Common Table Expressions

- Single CTE Used to simplify intermediate steps and improve readability
- Multiple Chained CTEs Building complex logic step by step where each CTE builds on the previous one (up to 4–5 CTEs chained in a single query)
- CTE used as a reusable filter e.g. non_buyer_referrers, repeated_customers, unique_referrer_buyers defined once and reused across the query
- CTE + Multiple JOINs + GROUP BY + HAVING — Combined in single complex queries for referrer analysis, staff leaderboard, and product-age analysis

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7. MySQL VIEWS

- CREATE VIEW Save a reusable query result as a virtual table
- repeated_customers VIEW Stores all customers with more than 1 invoice, reused across customer, staff, and product analysis modules
- top_50_customers VIEW Stores the top 50 customers by total revenue, used for VIP and at-risk customer analysis

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8. Date Functions

- YEAR() : Extract year from a date for yearly trend analysis
- MONTH() : Extract month from a date for monthly trend analysis
- DATEDIFF() : Calculate the number of days between two dates (used to find at-risk VIP customers who have not purchased in over 180 days)

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9. Set Operations

- UNION ALL : Combine row counts from all 9 tables into a single output in the EDA section

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10. Database Design Constraints

- PRIMARY KEY : Uniquely identifies each record in a table
- AUTO_INCREMENT : Automatically generates sequential ID values
- FOREIGN KEY : Enforces referential integrity between tables
- ON DELETE CASCADE : Automatically deletes child records when the parent record is deleted
- UNIQUE : Ensures no duplicate values in a column (mobile number, contact number)
- NOT NULL : Prevents critical fields from being left empty
- CHECK : Validates data at entry (e.g. age must be between 0 and 120)
- DEFAULT : Sets a default value when none is provided (hire date, invoice date, tax, discount)
- ENUM : Restricts a column to a fixed set of allowed values (customr_type, payment_mode, vision_type)
- GENERATED ALWAYS AS ... STORED : Auto-calculates derived columns (final_bill_amount in invoices, final_product_price in invoice_items)

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11. Database Normalization

- 3NF (Third Normal Form) : Every non-key column depends only on the primary key, eliminating transitive dependencies
- BCNF (Boyce-Codd Normal Form) : Every determinant is a candidate key, applied in product, brand, and type tables
- 4NF (Fourth Normal Form) : No multi-valued dependencies product type and product brand are stored in separate tables

4) Analysis Structure & Queries

4.1) Exploratory Data Analysis (EDA)

1.1) Counting Number of rows in each table

```
SELECT 'customers_details' AS TABLE_NAME, COUNT(*) AS Number_of_Rows FROM customers_details
UNION ALL
SELECT 'additional_customer_details', COUNT(*) AS Number_of_Rows FROM additional_customer_details
UNION ALL
SELECT 'invoices', COUNT(*) AS Number_of_Rows FROM invoices
UNION ALL
SELECT 'invoice_items', COUNT(*) AS Number_of_Rows FROM invoice_items
UNION ALL
SELECT 'prescriptions', COUNT(*) AS Number_of_Rows FROM prescriptions
UNION ALL
SELECT 'product_brand', COUNT(*) AS Number_of_Rows FROM product_brand
UNION ALL
SELECT 'product_brand', COUNT(*) AS Number_of_Rows FROM product_brand
UNION ALL
SELECT 'products', COUNT(*) AS Number_of_Rows FROM products
UNION ALL
SELECT 'staff', COUNT(*) AS Number_of_Rows FROM staff
ORDER BY Number_of_Rows DESC;
```

OUTPUT:

Result Grid | Filter Rows:

TABLE_NAME	Number_of_Rows
invoice_items	10020
customers_details	8000
prescriptions	8000
invoices	5000
additional_customer_details	3721
product_brand	25
product_brand	25
staff	20
products	6

1.2) Identifying the null values in each column of the Table

```
-- customer_details
SELECT
    SUM(CASE WHEN customer_full_name IS NULL THEN 1 ELSE 0 END) AS missing_names,
    SUM(CASE WHEN mobile IS NULL THEN 1 ELSE 0 END) AS missing_mobile,
    SUM(CASE WHEN customer_type IS NULL THEN 1 ELSE 0 END) AS missing_type,
    SUM(CASE WHEN assigned_staff_id IS NULL THEN 1 ELSE 0 END) AS missing_staff_assignment
FROM customers_details;
```

OUTPUT :

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

missing_names	missing_mobile	missing_type	missing_staff_assignment
0	0	0	0

1.3) Identifying the “NULL” Values in the tables

```
-- 5) Checking the duplicates in the table

SELECT COUNT(customer_id) AS number_of_duplicates FROM customers_details
GROUP BY customer_id
HAVING COUNT(customer_id) > 1;

SELECT COUNT(invoice_id) AS number_of_duplicates FROM invoices
GROUP BY invoice_id
HAVING COUNT(invoice_id) >1;

SELECT COUNT(prescription_id) AS number_of_duplicates FROM prescriptions
GROUP BY prescription_id
HAVING COUNT(prescription_id) > 1;

SELECT COUNT(staff_id) AS number_of_duplicates FROM staff
GROUP BY staff_id
HAVING COUNT(staff_id)>1;
```

1.4) EDA on “customers details” Table

```
SELECT COUNT( DISTINCT customer_id) AS Unique_customers FROM customers_details;

-- Age ranges from 18 to 85
SELECT DISTINCT Age FROM customers_details
ORDER BY Age DESC;

SELECT MAX(Age), MIN(age) FROM customers_details;

SELECT DISTINCT customer_type FROM customers_details;

SELECT COUNT(DISTINCT referred_by_id) AS COUNT_reffered FROM customers_details;

SELECT COUNT(DISTINCT assigned_staff_id) AS COUNT_staff FROM customers_details;
```

1.5) EDA on “Prescriptions” Table

```sql
-- 6.2) EDA for prescriptions
SELECT COUNT( DISTINCT prescription_id) AS Unique_prescription FROM prescriptions;

SELECT COUNT(DISTINCT customer_id) AS COUNT_unique_customers FROM prescriptions;

SELECT MAX(visit_date), MIN(visit_date) FROM prescriptions;

SELECT DISTINCT vision_type FROM prescriptions;
```

1.6) EDA on “Invoices” Table

```sql
SELECT COUNT(DISTINCT staff_id) FROM invoices;

SELECT MAX(invoice_date), MIN(invoice_date) FROM invoices;

SELECT DISTINCT payment_mode FROM invoices;

SELECT SUM(total_tax) FROM invoices;

SELECT SUM(discount_amount) FROM invoices;

SELECT SUM(final_bill_amount) FROM invoices;

SELECT DISTINCT YEAR(invoice_date) AS Year, SUM(final_bill_amount) FROM invoices
GROUP BY Year;

SELECT DISTINCT YEAR(invoice_date) AS Year, MONTH(invoice_date) AS Month, SUM(final_bill_amount) FROM invoices
GROUP BY Year, month
ORDER BY Year, month;

SELECT DISTINCT payment_mode, SUM(final_bill_amount) FROM invoices
GROUP BY payment_mode
ORDER BY SUM(final_bill_amount) DESC;
```

1.7) EDA on “Invoices Item” Table

```sql
-- 6.4) EDA for invoice_items
SELECT COUNT(DISTINCT item_id) AS Unqiue_item FROM invoice_items;

SELECT SUM(quantity) AS Total_Qunatity_sold FROM invoice_items;

SELECT SUM(final_product_price) FROM invoice_items;
```

4.2) Customer Analysis

The Customer Analysis is the largest and most detailed module of this project. It is divided into four sub-sections:

- Customer Demographics
- Window Shoppers
- Actual Buyer Customers
- Repeated Customers
- Top 50 Customers
- Churn Customers

each addressing a specific business question about who our customers are, how they behave, and how much value they bring to the business.

- 4.2.1) Customer Demographics

Business Questions Answer about the customers demographics are as follows:

1) What is the age distribution of our customer base, and which age segment represents the largest share of walk-in visitors?

2) Which age group demonstrates the highest conversion rate from walk-in visitor to actual buyer, and what does this reveal about our sales effectiveness across generations?

3) Which age group contributes the most revenue to the business, and is there a revenue concentration risk in any particular segment?

4) How are New and Referral customers distributed across different age groups, and which age group produces the most referral-driven customers?

5) Which customer type New or Referral generates higher revenue, and what does this indicate about the value of our referral network?

6) "Which payment method Cash, UPI, or Card is preferred by each age group, and does the mode of payment reveal any generational shift in spending and payment behaviour among our customers?"

7) What is the distribution of vision types: Near, Distance, Bifocal, and Progressive; across our entire customer base?

8) How does vision type vary between New and Referral customers, and does the type of vision problem influence how a customer arrives at our store?

Why Customer Demographics Analysis Matters?

Understanding customer demographics is the foundation of every customer strategy. Before optimizing sales, marketing, retention, or referral programs, the business must first understand who its customers are, how they behave, and which customer segments contribute the greatest long-term value. This demographic analysis examines customer characteristics including age distribution, purchasing behaviour, referral patterns, payment preferences, and vision requirements to identify the primary customer segments driving business performance.

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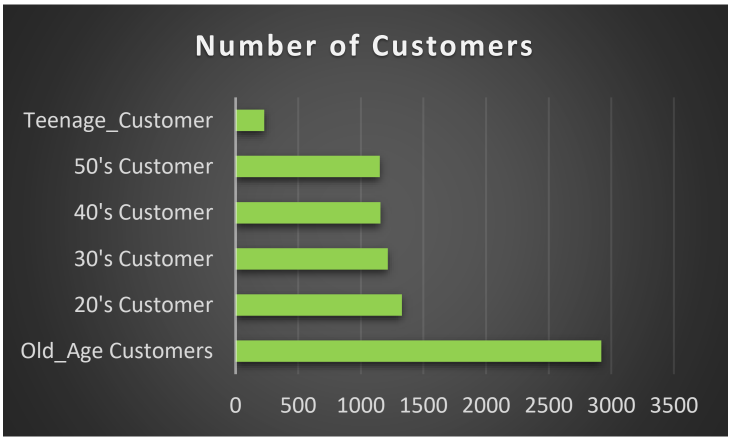
1) Count of Visited Customers across different Age groups

1) *What is the age distribution of our customer base, and which age segment represents the largest share of walk-in visitors?*

```sql
SELECT
CASE
    WHEN age < 20 THEN "Teenage_Customer"
    WHEN age BETWEEN 20 AND 30 THEN "20's Customer"
    WHEN age BETWEEN 31 AND 40 THEN "30's Customer"
    WHEN age BETWEEN 41 AND 50 THEN "40's Customer"
    WHEN age BETWEEN 51 AND 60 THEN "50's Customer"
    WHEN age > 60 THEN "Old_Age Customers"
  END AS Age_group,
COUNT(*) AS Count_of_Customers,
ROUND( COUNT(*) * 100 / 8000, 2) AS Percentage
FROM customers_details
GROUP BY Age_group WITH ROLLUP
ORDER BY Count_of_Customers DESC;
```

Result Grid | Filter Rows: | Export:

Age_group	Count_of_Customers	Percentage
Old_Age Customers	2920	36.50
20's Customer	1328	16.60
30's Customer	1214	15.18
40's Customer	1157	14.46
50's Customer	1152	14.40
Teenage_Customer	229	2.86



- Largest customer segment is Old Age Customers (above 60 years), accounting for 2,920 customers (36.5%)
- The second-largest segment is 20's Customers with 1,328 customers (16.6%), followed by 30's Customers (1,214; 15.2%), 40's Customers (1,157; 14.5%), and 50's Customers (1,152; 14.4%). These four age groups contribute relatively similar customer volumes.
- The smallest segment is Teenage Customers, with only 229 customers (2.9%)

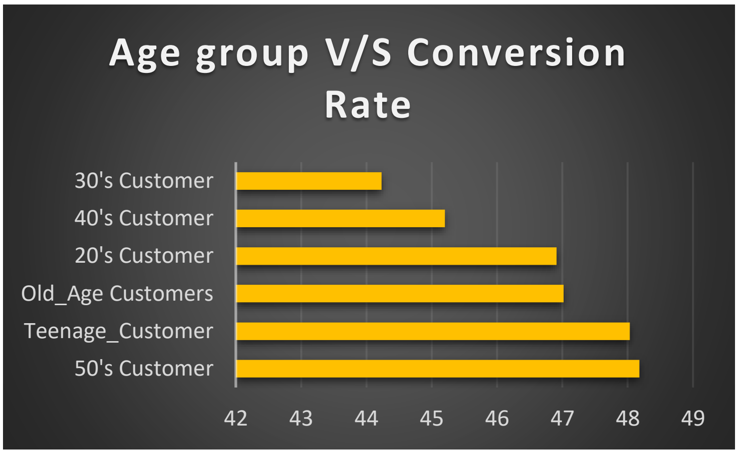
Business Insight:

- The data shows that the store's customer base is heavily concentrated in the 60+ age group, making senior citizens the most important target segment.
- Marketing campaigns, product inventory, and services such as progressive lenses, bifocals, and regular eye-checkup programs should continue to focus on this segment while parallelly exploring opportunities to attract more younger customers.

2) Customers Conversion Rate across different Age Groups

2) Which age group demonstrates the highest conversion rate from walk-in visitor to actual buyer, and what does this reveal about our sales effectiveness across generations?

```
SELECT
CASE
    WHEN age < 20 THEN "Teenage_Customer"
    WHEN age BETWEEN 20 AND 30 THEN "20's Customer"
    WHEN age BETWEEN 31 AND 40 THEN "30's Customer"
    WHEN age BETWEEN 41 AND 50 THEN "40's Customer"
    WHEN age BETWEEN 51 AND 60 THEN "50's Customer"
    WHEN age > 60 THEN "Old_Age Customers"
  END AS Age_group,
ROUND( COUNT( DISTINCT i.customer_id)*100 /
COUNT( DISTINCT c.customer_id), 2) AS Conversion_Rate
FROM customers_details as c
LEFT JOIN invoices as i
ON c.customer_id=i.customer_id
GROUP BY Age_group
ORDER BY Conversion_Rate DESC;
```



The conversion rate across all age groups is remarkably consistent, ranging between 44% and 48% meaning roughly 1 in every 2 customers who walks into the store ends up making a purchase.

Result Grid | Filter Rows:

Age_group	Conversion_Rate
50's Customer	48.18
Teenage_Customer	48.03
Old_Age Customers	47.02
20's Customer	46.91
40's Customer	45.20
30's Customer	44.23

Business Insights:

- The store is already performing well with older and mature age segments that is above 50 age.
- Although Teenage Customers represent the smallest visitor group (only 229 customers), they demonstrate one of the highest conversion rates (48.03%), indicating strong purchase intent once they enter the store.
- Also it's a strong signal that the sales staff is doing an excellent job at converting young walk-in visitors into actual buyers. (must take feedback from the staff, regarding the Genz customers behaviour and requirements in the store)
- There is a clear opportunity to increase footfall among younger and Gen Z customers. The high teenage conversion rate proves that once young customers walk in, the staff can close the sale.
- Hence, If our Future Goal is to enter into the Gen-z Retail Category here are the following solution and Risk mitigation:

Strategic Solution & Recommendation:

- To expand this segment, the business may need to invest in targeted marketing initiatives such as social media advertising, influencer collaborations, and digital engagement campaigns that resonate with younger consumers.
- The store may also need to maintain a more fashion-oriented inventory, including trendy frames, sunglasses, and eyewear collections that align with rapidly changing consumer preference

Risk Mitigation:

- However, entering this segment requires careful inventory management. Fashion trends among younger consumers evolve quickly, increasing the risk of slow-moving stock and inventory obsolescence.
- Therefore, the business should focus on faster inventory turnover, regular collection updates, and demand-driven purchasing decisions to minimize inventory risk while capturing growth opportunities in this market.

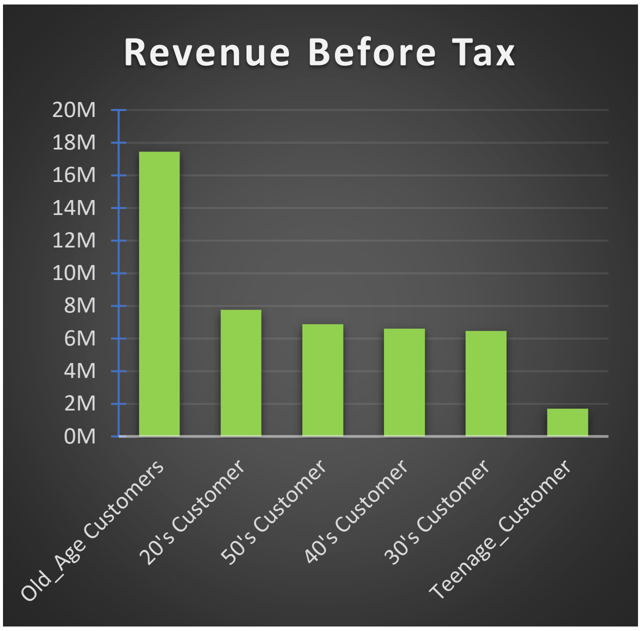
3) Contribution of different Age Groups in Revenue before Tax

3) Which age group contributes the most revenue to the business, and is there a revenue concentration risk in any particular segment?

```sql
SELECT
CASE
    WHEN age < 20 THEN "Teenage_Customer"
    WHEN age BETWEEN 20 AND 30 THEN "20's Customer"
    WHEN age BETWEEN 31 AND 40 THEN "30's Customer"
    WHEN age BETWEEN 41 AND 50 THEN "40's Customer"
    WHEN age BETWEEN 51 AND 60 THEN "50's Customer"
    WHEN age > 60 THEN "Old_Age Customers"
  END AS Age_group,
SUM(i.final_bill_amount) AS Revenue,
ROUND( SUM(i.final_bill_amount) * 100 / 46811413.00, 2)
AS Contribution_in_Revenue
FROM customers_details as c
INNER JOIN invoices as i
ON c.customer_id=i.customer_id
GROUP BY Age_group WITH ROLLUP
ORDER BY Revenue DESC;
```

Result Grid | Filter Rows: | Export:

Age_group	Revenue	Contribution_in_Revenue
NULL	46811413.00	100.00
Old_Age Customers	17440454.00	37.26
20's Customer	7751143.00	16.56
50's Customer	6870009.00	14.68
40's Customer	6598649.00	14.10
30's Customer	6458010.00	13.80
Teenage_Customer	1693148.00	3.62



Old Age customers (age > 60) are the top revenue generators at approximately ₹1.8 Crore, which is a significant lead over every other age group.

The remaining five age groups 20s, 30s, 40s, 50s, and Teenage all fall within a relatively narrow band of ₹60 Lakhs to ₹80 Lakhs each, with 20s customers leading that mid-tier group at approximately ₹80 Lakhs.

Business Insights:

Positives: It validates that the business has built a strong and loyal base of older customers who trust the store, return regularly, and spend more per visit.

Negative: Over-dependence on a single age segment is a long-term business risk. As this customer base ages further, their purchasing frequency may decline

Meanwhile, the 20s, 30s, 40s, and 50s segments which together represent the majority of the customer count are each generating only ₹60–80 Lakhs individually.

The gap between Old Age revenue and everyone else is too wide to be comfortable from a business sustainability standpoint.

Strategic Solution & Recommendation:

The business needs to actively work on increasing the average transaction value and purchase frequency among the 20s, 30s, and 40s segments.

These are working-age customers with disposable income who are likely buying basic single vision lenses and standard frames. Introducing premium lens options, blue-light blocking glasses, and branded frame collections targeted at this demographic could meaningfully close the revenue gap between the Old Age segment and the rest.

Additionally, the Teen age segment, despite being the smallest in count, should not be overlooked.

As discussed in the conversion rate analysis, teenage customers convert at a very high rate they simply do not walk in enough. Increasing their footfall alone could add meaningful revenue given how well the sales team handles this segment.

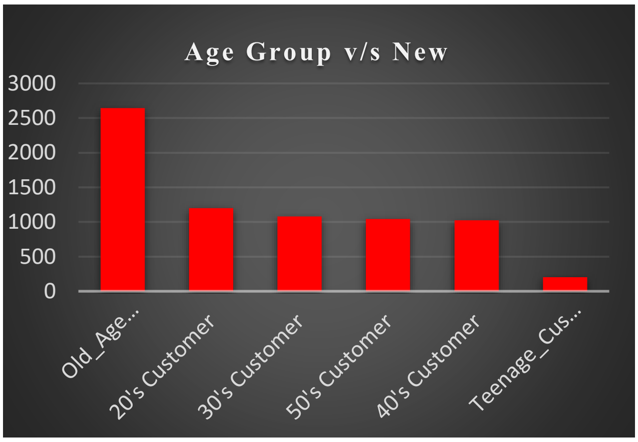
4) Customer Type across different Age Groups

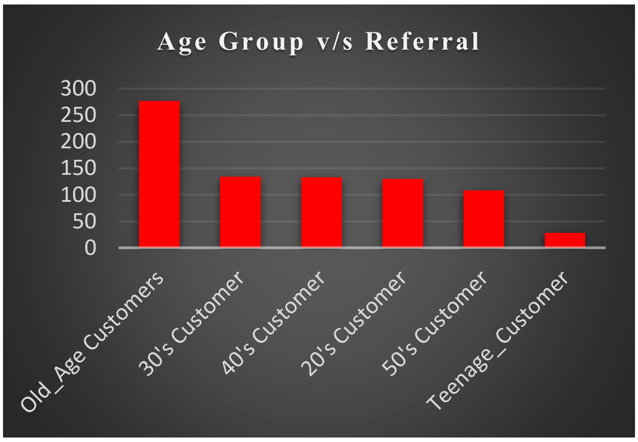
4) How are New and Referral customers distributed across different age groups, and which age group produces the most referral-driven customers?

```sql
SELECT
CASE
    WHEN age < 20 THEN "Teenage_Customer"
    WHEN age BETWEEN 20 AND 30 THEN "20's Customer"
    WHEN age BETWEEN 31 AND 40 THEN "30's Customer"
    WHEN age BETWEEN 41 AND 50 THEN "40's Customer"
    WHEN age BETWEEN 51 AND 60 THEN "50's Customer"
    WHEN age > 60 THEN "Old_Age Customers"
  END AS Age_group,
  SUM(CASE WHEN customer_type="New" THEN 1 ELSE 0 END)
  AS Count_New,
  SUM(CASE WHEN customer_type="Referral" THEN 1 ELSE 0 END)
  AS Count_Refferal
  FROM customers_details
GROUP BY Age_group;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Conte

Age_group	Count_New	Count_Refferal	Percentage_of_Referrals
20's Customer	1198	130	10.85
30's Customer	1080	134	12.41
40's Customer	1024	133	12.99
50's Customer	1044	108	10.34
Old_Age Customers	2643	277	10.48
Teenage_Customer	201	28	13.93
NULL	7190	810	11.27





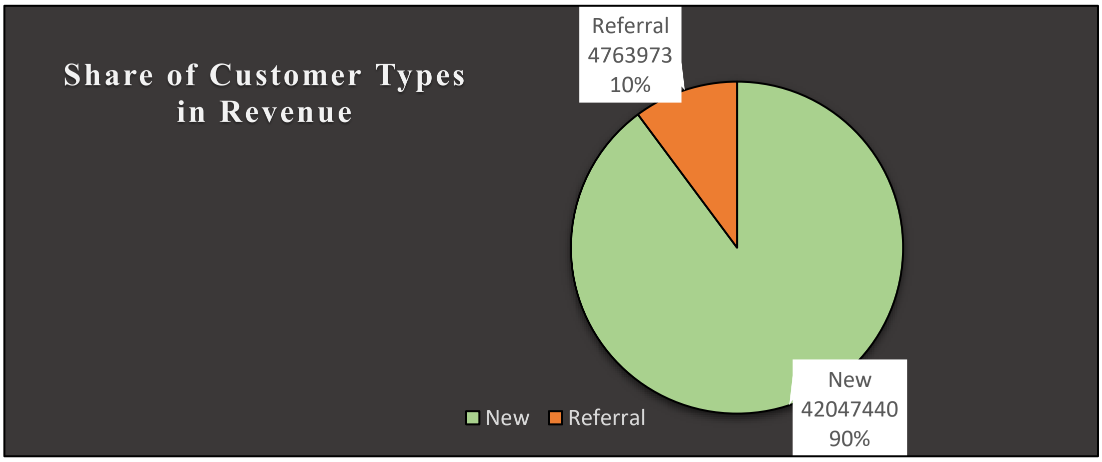
5) Revenue before Tax by different Customer Type

5) Which customer type New or Referral generates higher revenue, and what does this indicate about the value of our referral network?

```sql
SELECT c.customer_type,
    SUM(i.final_bill_amount) AS Revenue,
    ROUND(SUM(i.final_bill_amount)*100/
    (SELECT SUM(final_bill_amount) FROM invoices), 2)
    AS Revenue_Contribution
FROM customers_details as c
INNER JOIN invoices as i
ON c.customer_id=i.customer_id
GROUP BY customer_type
ORDER BY Revenue DESC;
```

Result Grid | Filter Rows: | Export

customer_type	Revenue	Revenue_Contribution
New	42047440.00	89.82
Referral	4763973.00	10.18



Business Insight:

1) The analysis shows that the business primarily acquires customers through New Customers, who account for approximately 90% of the total customer base, while Referral Customers contribute the remaining 10%.
2) From a volume perspective, the Old Age Customer (>60 years) segment records the highest number of referral customers, followed by customers in their 30s, 40s, 20s, and 50s, while teenage customers contribute the fewest referrals. However, this largely reflects the overall customer distribution, as the older age segment is also the store's largest customer group.
3) When referral activity is compared as a percentage within each age group, the differences are minimal. Most age segments have referral shares between 10% and 12%, indicating that referral behaviour is fairly consistent across generations rather than being concentrated within a specific demographic.
4) Overall, the findings suggest that referrals are generated uniformly across the customer base, with no single age group displaying significantly stronger referral behaviour. Although referral customers represent only a small share of total visitors, they remain an important customer acquisition channel, as demonstrated in the dedicated Referral Analysis.

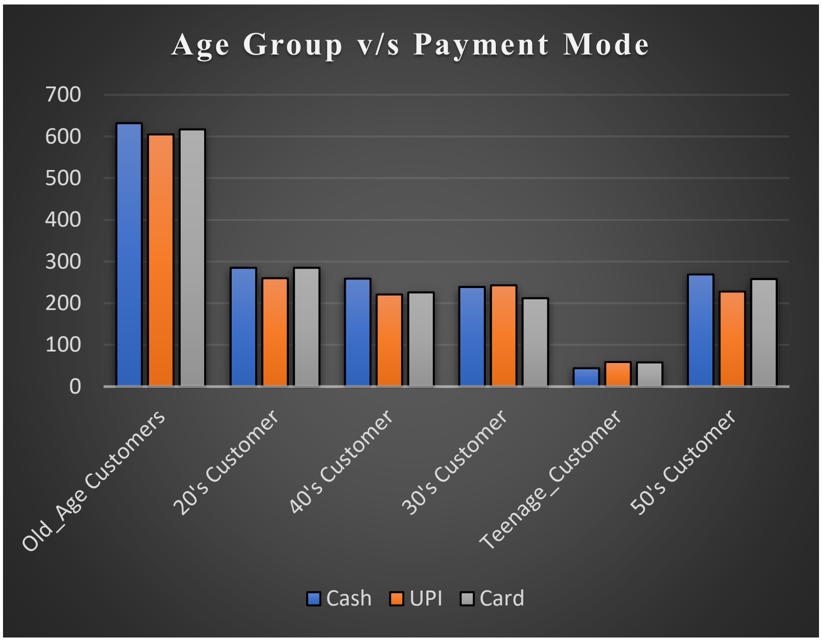
Business Recommendation:

- Although Referral Customers represent only around 10% of the customer base, they should not be evaluated solely by their volume. As demonstrated in the dedicated Referral Analysis, this relatively small segment contributes approximately ₹47.64 lakh, representing 10.18% of the store's total revenue. This confirms that the referral network is a meaningful source of business growth achieved with virtually no direct marketing expenditure.
- Therefore, the business should continue investing in customer experience, trust-building, and referral initiatives to encourage satisfied customers to recommend the store. Even modest improvements in referral generation could produce a meaningful increase in customer acquisition while maintaining low acquisition costs.

6) Count of different Payments Mode usage by each Age group

6) Which payment method Cash, UPI, or Card is preferred by each age group, and does the mode of payment reveal any generational shift in spending and payment behaviour among our customers?

```
SELECT
     CASE
     WHEN c.age < 20 THEN "Teenage_Customer"
   WHEN c.age BETWEEN 20 AND 30 THEN "20's Customer"
   WHEN c.age BETWEEN 31 AND 40 THEN "30's Customer"
   WHEN c.age BETWEEN 41 AND 50 THEN "40's Customer"
   WHEN c.age BETWEEN 51 AND 60 THEN "50's Customer"
   WHEN c.age > 60 THEN "Old_Age Customers"
  END AS Age_group,
COUNT( CASE WHEN i.payment_mode="Cash" THEN 1 END) AS Cash,
COUNT( CASE WHEN i.payment_mode="UPI" THEN 1 END) AS UPI,
COUNT( CASE WHEN i.payment_mode="Card" THEN 1 END) AS Card
FROM customers_details as c
INNER JOIN invoices as i
ON c.customer_id=i.customer_id
GROUP BY Age_group;
```



Result Grid | Filter Rows:

Age_group	Cash	UPI	Card	Total
Old_Age Customers	632	605	617	1854
20's Customer	285	260	285	830
40's Customer	259	221	226	706
30's Customer	239	243	212	694
Teenage_Customer	44	59	58	161
50's Customer	269	228	258	755

Result Grid | Filter Rows:

payment_mode	Count_Payment_Modes
Cash	1535
UPI	1465
Card	1502

Result Grid | Filter Rows:

payment_mode	Revenue_Collected
Cash	16196941.00
Card	15674327.00
UPI	14940145.00

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

payment_mode	Count_Payment_Modes	Revenue_Collected	Contribution_in_Revenue	Average_order_value
Cash	1535	16196941.00	34.60	9373.229745
UPI	1465	14940145.00	31.92	9245.139233
Card	1502	15674327.00	33.48	9465.173309

Business Insight:

Understanding customers' preferred payment methods helps the business ensure a smooth checkout experience, optimize payment infrastructure, and reduce operational dependency on any single payment channel.

The analysis shows that the business has a well-diversified payment ecosystem. Cash, UPI, and Card transactions each contribute almost equally to the total number of payments and revenue collected.

Similarly, the Average Order Value (AOV) remains highly consistent across all payment methods (approximately ₹9,200–₹9,500), indicating that customers spend nearly the same amount regardless of how they choose to pay. This suggests that payment preference does not influence purchase value.

The age-group analysis also reveals a similar pattern. Across every customer segment, payment usage is relatively balanced, with no age group displaying a strong dependence on Cash, UPI, or Card. Even among senior customers (60+), transactions are distributed almost evenly across all three payment methods.

Business Recommendation

The business is not operationally dependent on any single payment method, which reduces payment collection risk and provides flexibility for customers during checkout. Going forward, the store should continue supporting all major payment options while ensuring a seamless payment experience across each channel rather than prioritizing one method over another.

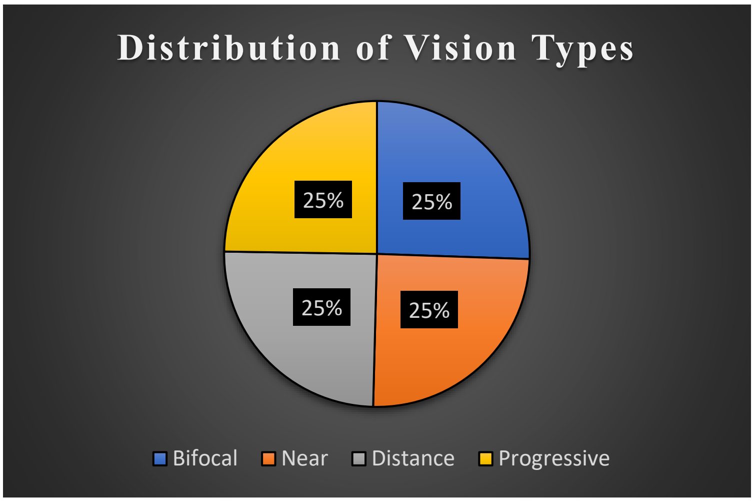
7) Distribution of Vision Types across all the Visited Customers

7) What is the distribution of vision types: Near, Distance, Bifocal, and Progressive; across our entire customer base?

```sql
-- Vision Type COUNT
SELECT p.vision_type,
COUNT(customer_id)
AS COUNT_Vision_Type,
ROUND( COUNT(customer_id)*100/
(SELECT COUNT(customer_id) FROM prescriptions), 2)
AS Percentage_of_Vision_Types
FROM prescriptions as p
GROUP BY p.vision_type
ORDER BY COUNT_Vision_Type DESC;
```

Result Grid | Filter Rows: | Export:

vision_type	COUNT_Vision_Type	Percentage_of_Vision_Types
Bifocal	2043	25.54
Near	1989	24.86
Distance	1988	24.85
Progressive	1980	24.75



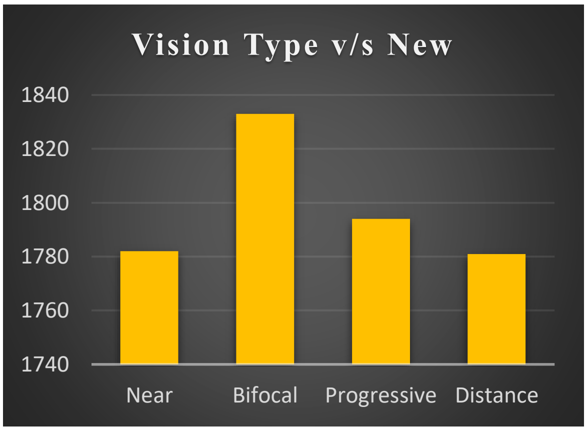
8) Count of Customer Type in different Vision type

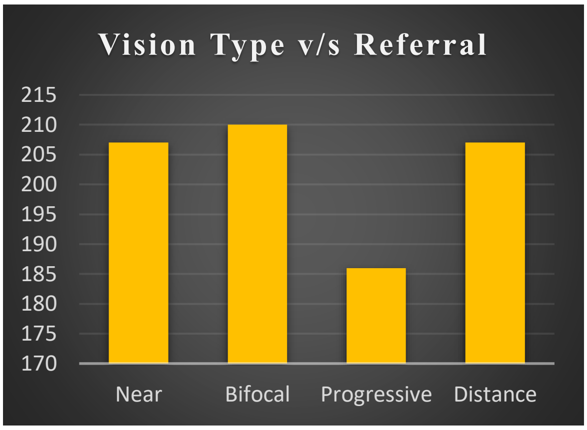
8) How does vision type vary between New and Referral customers, and does the type of vision problem influence how a customer arrives at our store?

```sql
SELECT p.vision_type,
    SUM(CASE WHEN c.customer_type="New" THEN 1 ELSE 0 END)
    AS COUNT_New,
    SUM(CASE WHEN c.customer_type="Referral" THEN 1 ELSE 0 END)
    AS COUNT_Referral
FROM prescriptions as p
LEFT JOIN customers_details as c
ON c.customer_id=p.customer_id
GROUP BY p.vision_type;
```

Result Grid | Filter Rows:

vision_type	COUNT_New	COUNT_Referral
Near	1782	207
Bifocal	1833	210
Progressive	1794	186
Distance	1781	207





Business Insight

Understanding the distribution of customers' vision types is critical for an optical business because it directly influences clinical infrastructure requirements, inventory planning, staff expertise, and customer experience. Delivering an accurate prescription is the foundation of customer satisfaction, as even a minor prescription error (such as ±0.25 di-opters) can result in blurred vision, eye strain, headaches, and ultimately loss of customer trust.

No single vision category dominates the customer base, indicating that the business must maintain expertise, inventory, and testing capabilities across all major vision correction requirements rather than specializing in only one category.

However, this finding should be interpreted alongside the previous demographic analyses. Earlier analysis identified customers above 60 years of age as the store's largest customer segment. This is particularly important because senior customers are more likely to require complex prescriptions such as progressive and bifocal lenses, and their eye prescriptions naturally change every two to three years due to age-related vision conditions such as presbyopia.

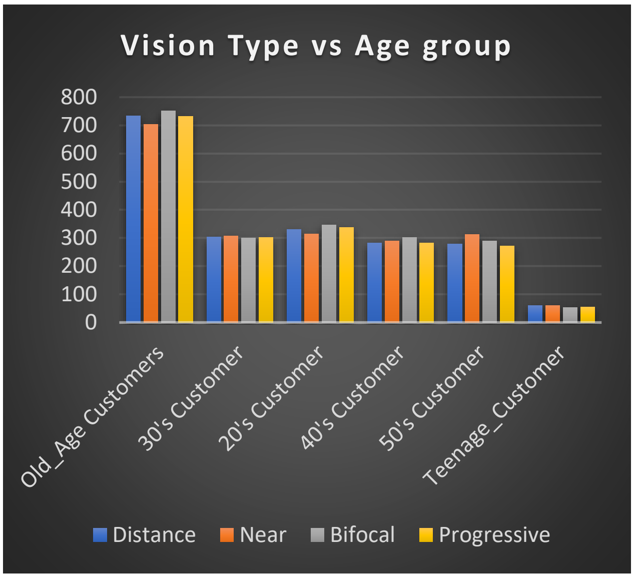
Although all vision categories contribute almost equally to the business, the large senior customer base makes prescription accuracy a strategic priority. Retaining these customers depends not only on product quality but also on consistently delivering precise vision correction during every visit.

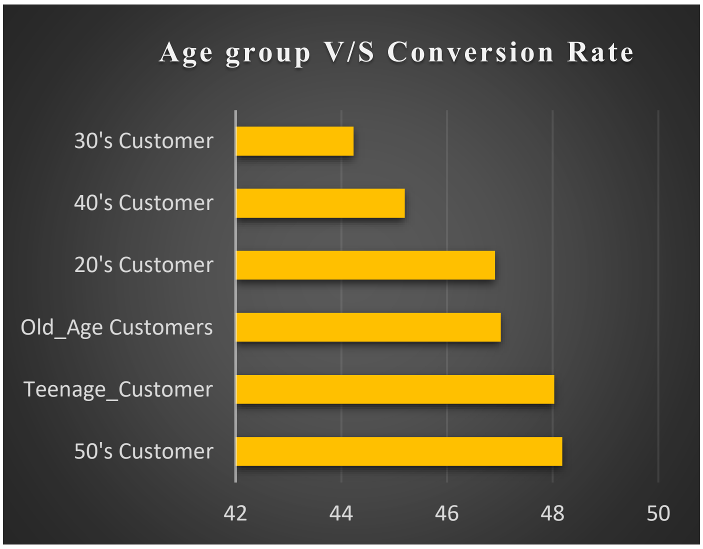
Business Recommendation

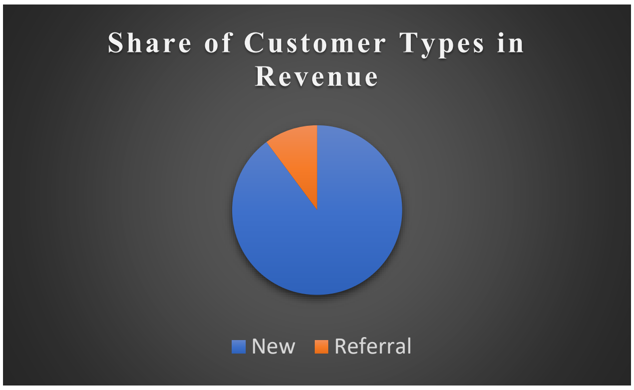
To support long-term customer retention and maintain a high-quality customer experience, the business should continue investing in its vision testing infrastructure. Key priorities include:

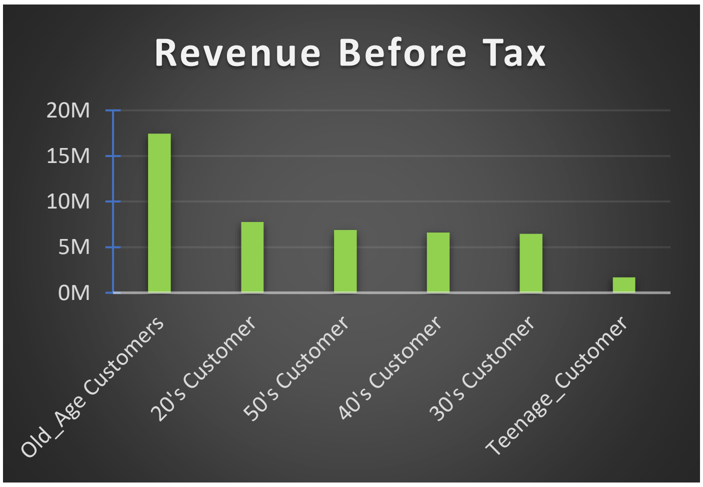
- Upgrading to high-precision digital eye examination equipment.
- Maintaining accurate PD (Pupillary Distance) measurement tools, particularly for progressive lenses.
- Hiring experienced optometrists capable of handling complex prescriptions.
- Ensuring skilled spectacle fitting to maximize comfort and visual accuracy.
- Conducting periodic calibration of testing equipment to minimize prescription errors.

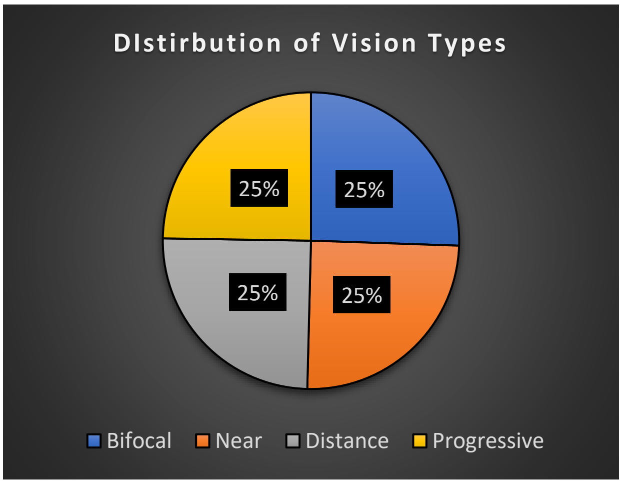
1. Dashboard of Customer Demographics

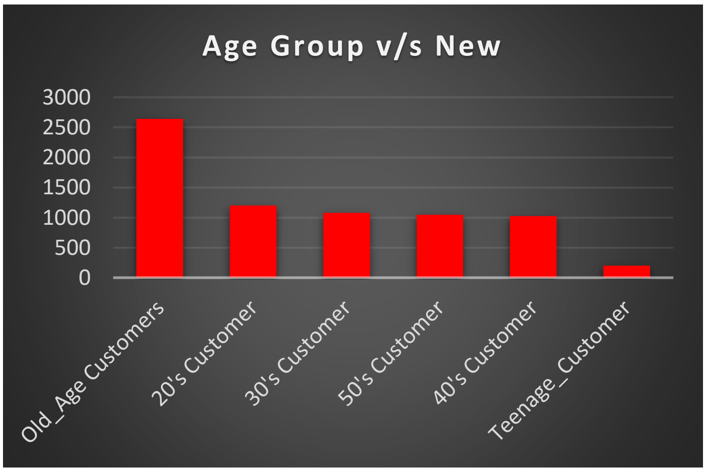


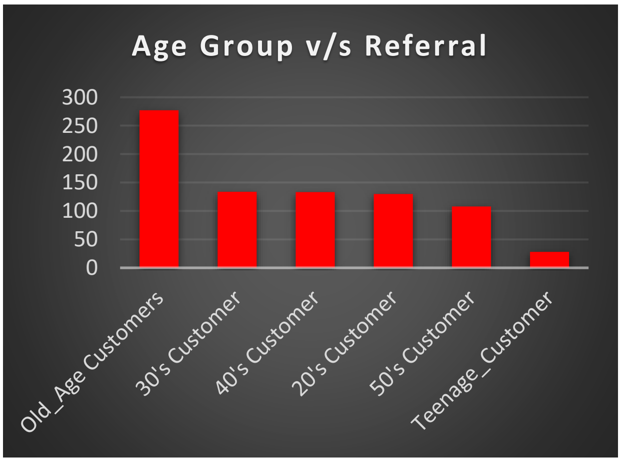












Customer Demographics Conclusion:

The Customer Demographics Analysis provides a comprehensive understanding of who the business serves before evaluating how customers behave throughout their purchasing journey. By analysing customer age, vision requirements, payment preferences, customer acquisition channels, and revenue contribution, we established the demographic profile of the store's customer base and identified the key segments driving business performance.

The analysis reveals that the business has developed a particularly strong presence among customers above 60 years of age, who consistently lead across customer volume, revenue contribution, and referral activity. Customers between 20 and 50 years also represent a stable and well-balanced customer base, while teenage customers, although smaller in volume, demonstrate encouraging conversion and retention characteristics that present future growth opportunities.

From an operational perspective, the business benefits from a well-diversified customer portfolio. Revenue is generated across multiple vision correction categories, payment methods are evenly distributed between Cash, UPI, and Card, and referral customers contribute a meaningful share of overall revenue despite representing only a small proportion of the customer base. These findings indicate that the business is not overly dependent on a single customer characteristic or payment channel, reducing operational risk while supporting sustainable growth.

One of the most significant observations throughout this section is the strategic importance of customer experience. Given the large proportion of senior customers and the nature of vision correction services, long-term success depends on delivering highly accurate prescriptions, maintaining customer trust, and providing exceptional service quality. As vision prescriptions naturally evolve over time particularly

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```
1. Customer Demographics
       ↓
Who are our customers?

2. Customer Analysis
       ↓
How do they purchase?

3. Churn Analysis
       ↓
Who are we losing?

4. Staff Analysis
       ↓
Who is driving business performance?

5. Referrer Analysis
       ↓
How are we acquiring customers organically?

6. Product Analysis
       ↓
What are customers buying?

7. Final Business Conclusion
       ↓
What should the business do next?
```

- Overall, the Customer Demographics Analysis establishes the foundation for the remainder of this report. While it explains who the customers are…
- The following sections will examine how they behave including purchasing patterns, customer conversion, repeat purchases, churn behaviour, referral networks, and customer lifetime value to identify actionable opportunities for improving business performance.

Customer Lifecycle Analysis

Problem Statement:

Although the store receives a significant number of walk-in visitors, not every visitor becomes a buyer, not every buyer returns for future purchases, and some loyal customers eventually stop visiting altogether. Without understanding where customers drop off during their journey, management cannot accurately identify weaknesses in customer acquisition, sales conversion, retention, or staff performance. Therefore, the business requires a comprehensive customer lifecycle analysis to identify bottlenecks, improve customer retention, reduce churn, and maximize customer lifetime value.

Why Customer Lifecycle Analysis Matters?

The success of a retail business is not determined solely by the number of customers visiting the store, but by how effectively those customers progress through each stage of the customer lifecycle. Every walk-in visitor represents a potential opportunity, yet not every visitor becomes a buyer, not every buyer returns for future purchases, and not every loyal customer remains with the business over time; without understanding where customers convert, where they disengage, and why they eventually churn, management cannot accurately identify operational weaknesses or opportunities for sustainable growth.

This analysis aims to evaluate the complete customer lifecycle in five distinct stages are as follows:

Stage 1 → Walk-in Visitors — Who came to the store?

↓

Stage 2 → Window Shoppers — Who left without buying?

↓

Stage 3 → Actual Buyers — Who made a purchase?

↓

Stage 4 → Repeated Customers — Who came back?

↓

Stage 5 → Churned Customers — Who stopped coming?

Each stage is analysed across four dimensions Age Group, Vision Type, Customer Type (New vs Referral), and Staff to identify not just what is happening at each stage of the funnel, but who it is happening to, and what the business can do about it.

Executive Summary on Customer Lifecycle Analysis

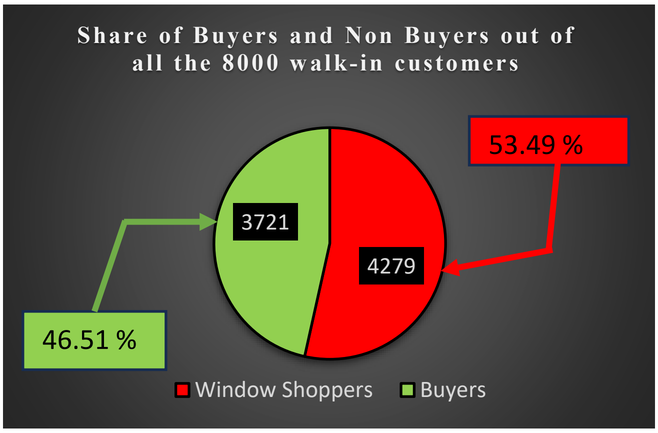
8000
Visited Customers

→

4279
Window Shoppers

→

3721
Buyer Customers



46.51%
Customer Conversion Rate

46.8M Rs
Total Revenue

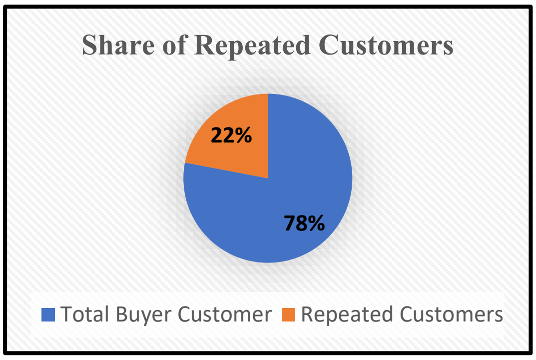
- The customer lifecycle analysis reveals that the store attracted 8,000 walk-in visitors, of which 3,721 customers completed a purchase, resulting in an overall customer conversion rate of 46.51%.
- While this indicates that nearly one out of every two visitors convert into a buyer, 4,279 visitors (53.49%) left the store without making a purchase, highlighting a significant opportunity to improve sales conversion and recover lost revenue.
- Collectively, these customers generated approximately ₹4.68 crore in revenue during the analysis period.

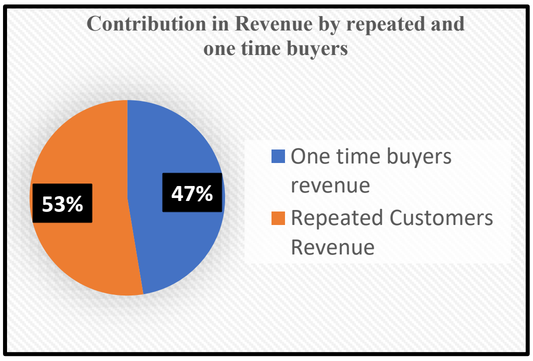
3721
Buyer Customers

→

1051
Repeated Customers

28.25%
Customers Retention Rate





- Customer retention presents the next major stage of the lifecycle. Among the 3,721 buyer customers, only 1,051 customers returned for repeat purchases, resulting in a customer retention rate of 28.25
- Although repeated customers represent only 28.25% of the total buyer base, they contribute approximately 53% of the store's total revenue, while regular (one-time) customers contribute the remaining 47%.
- This indicates that customer loyalty has disproportionately large impact on business performance. In other words, a relatively small segment of retained customers is responsible for generating the majority of the store's revenue, highlighting customer retention as one of the most critical drivers of long-term revenue growth and profitability.

3721 Buyer Customers → **235** Churn Customers → **6.32%** Churning Rate

- Out of the 3,721 customers who made a purchase, 235 customers have churned, defined as customers who have not returned to the store for more than 180 days since their last purchase. This represents a churn rate of 6.32%, which is relatively low and indicates that the business has been reasonably successful in retaining its customer base.
- Nevertheless, these churned customers represent recoverable revenue, making targeted re-engagement campaigns and follow-up initiatives an important opportunity for future business growth.

8000 Walk-in-visitors → **721** Referrers → **810** Referrals → **381** Buyer Referrals

47% Referrals Conversion Rate

10% Contribution in Revenue

47.63L Rs Revenue by Referrals

- The business also benefits from a strong word-of-mouth network. Out of 8,000 walk-in customers, 721 customers (9.01%) became referrers, effectively acting as unpaid brand advocates.
- These referrers generated 810 referral customers, of which 381 converted into buyers, resulting in a 47.04% referral conversion rate.
- Although referral customers represent only around 10% of the customer base, they contributed approximately 10% of the store's total revenue, demonstrating that the referral ecosystem is a valuable and cost-effective source of customer acquisition.
- The referral ecosystem plays a strategically important role in the business despite its relatively small size. A detailed evaluation of referral performance, customer behaviour, and business opportunities is presented in the Referral Analysis section.

Customer Lifecycle Dashboard – Age Group Analysis

8000
Walk-in-Visitors

4279
Window Shoppers

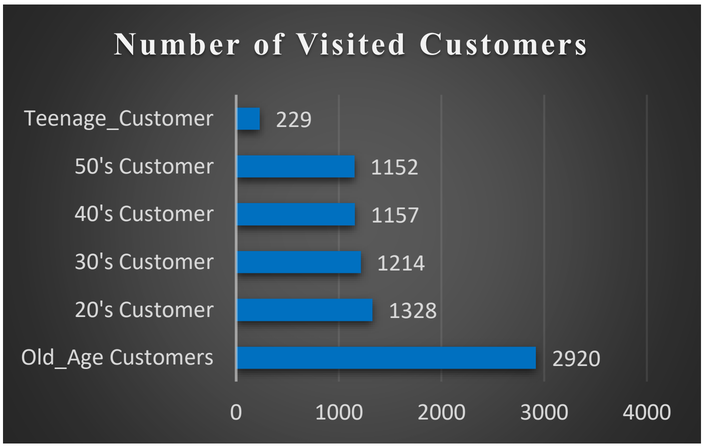
3721
Converted Buyers

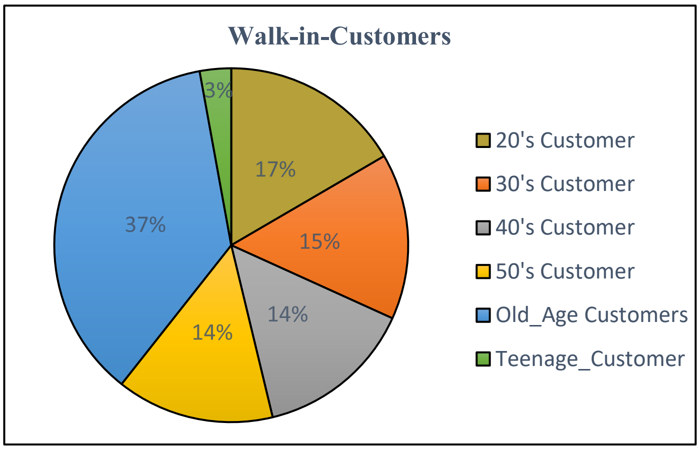
1051
Repeated Customers

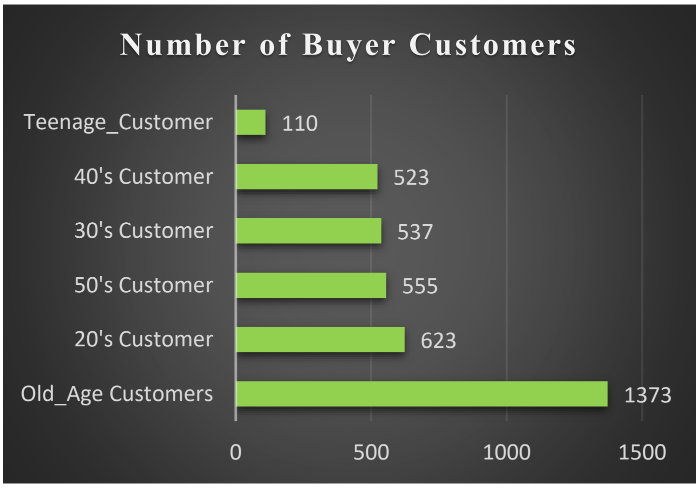
46.51%
Customer Conversion Rate

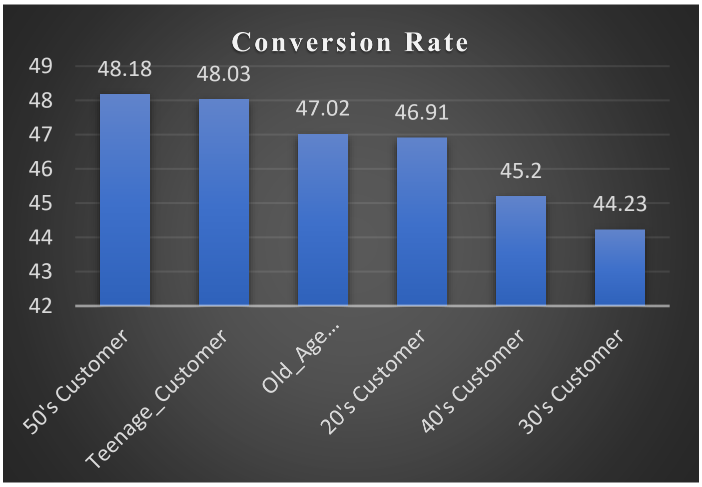
28.25%
Customers Retention Rate

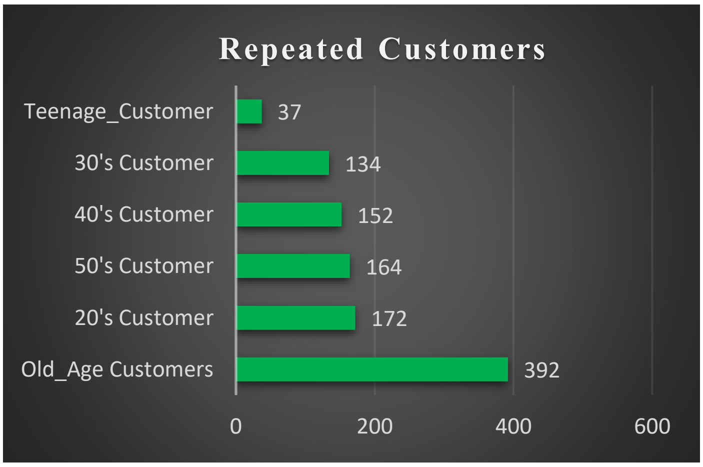
4.68 Cr
Revenue Generated

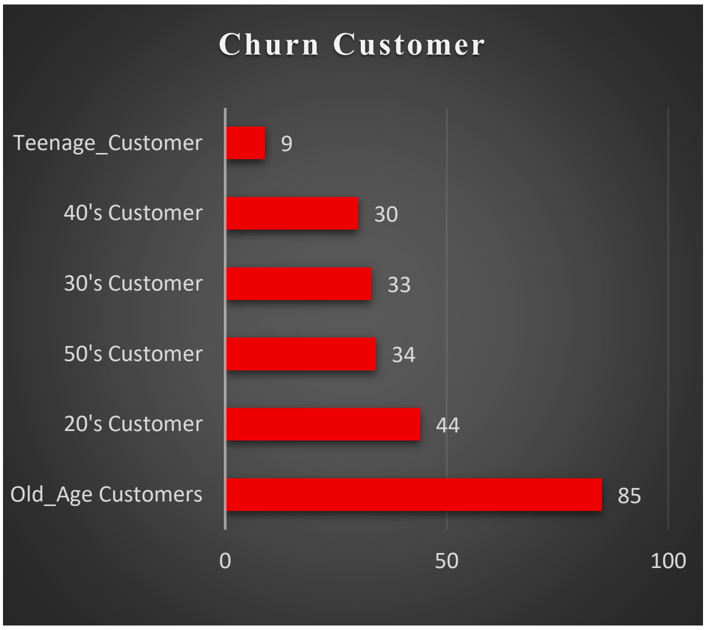


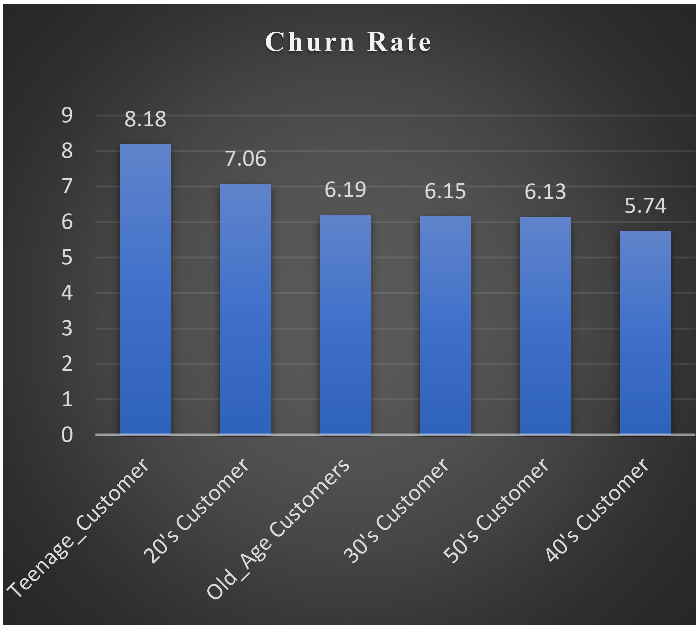


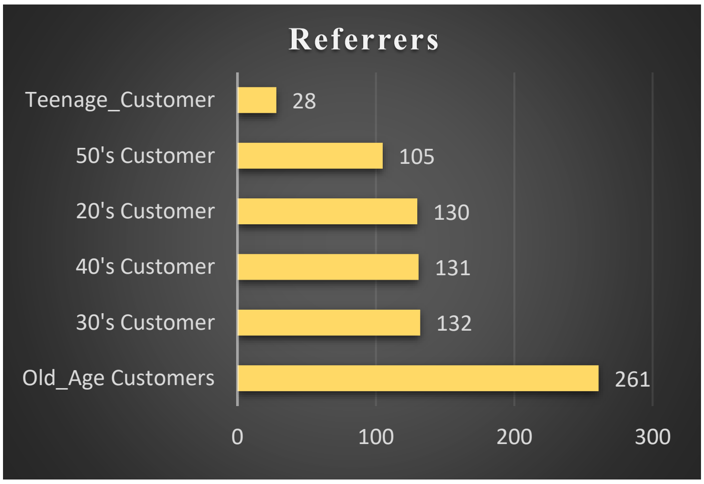


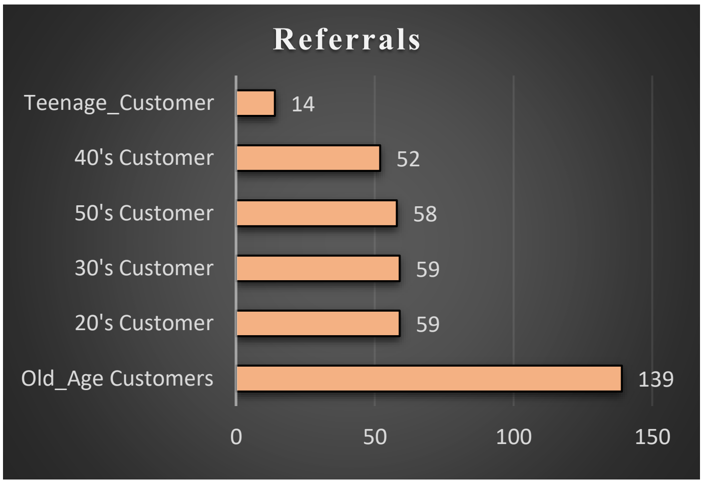


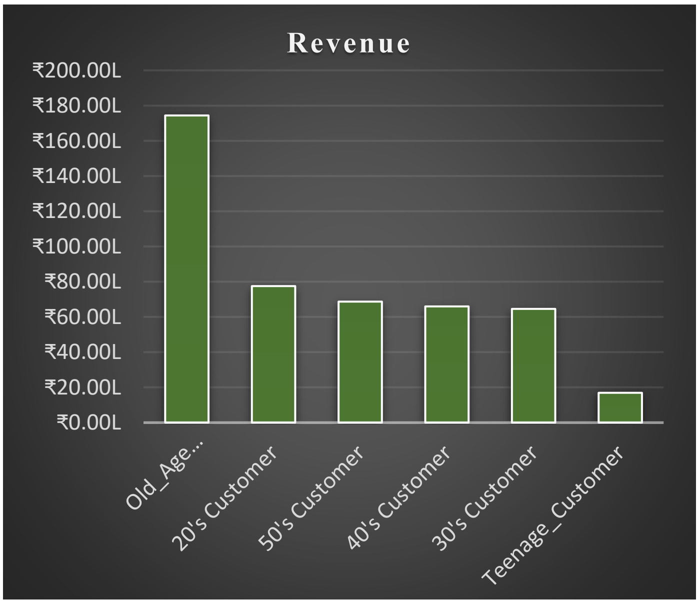


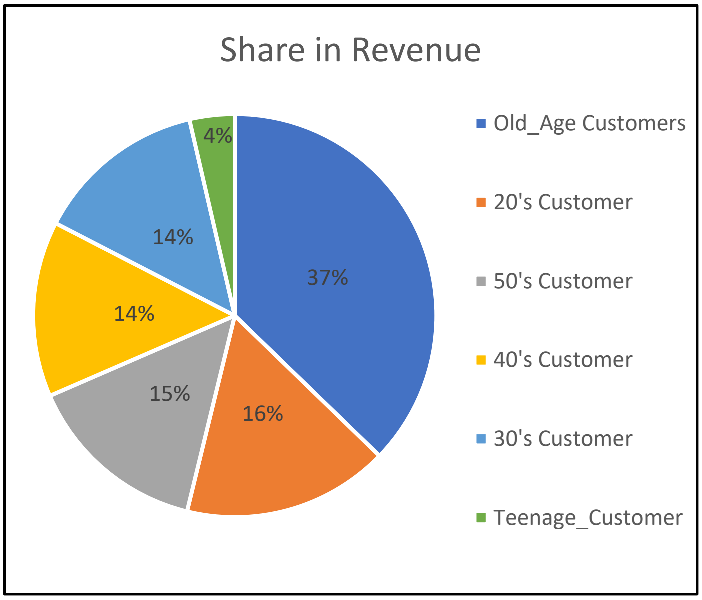












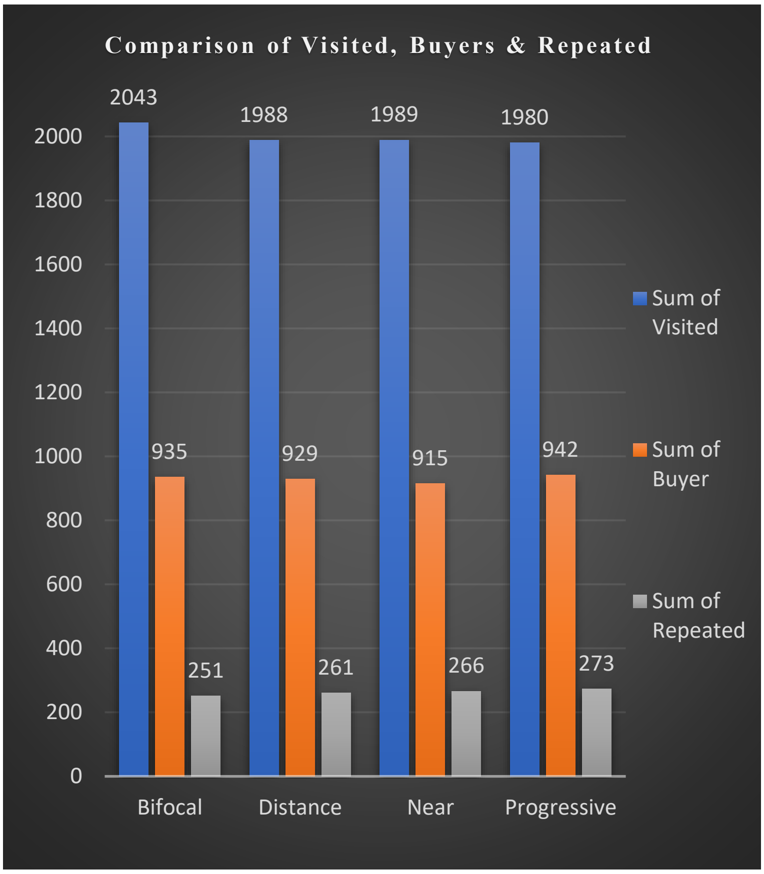
Query Output:

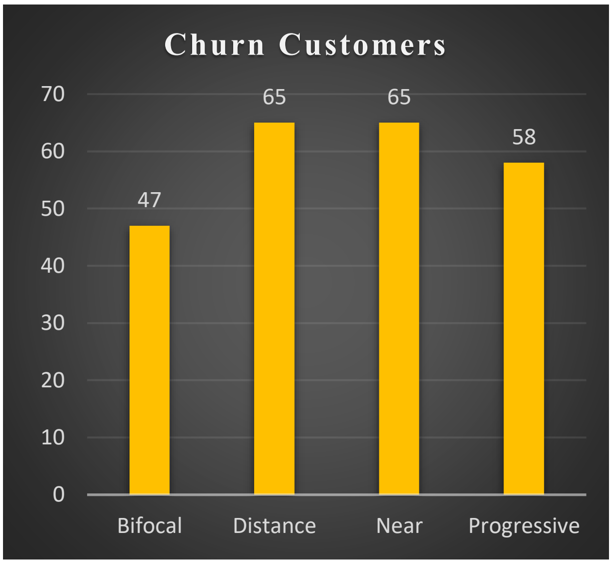
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

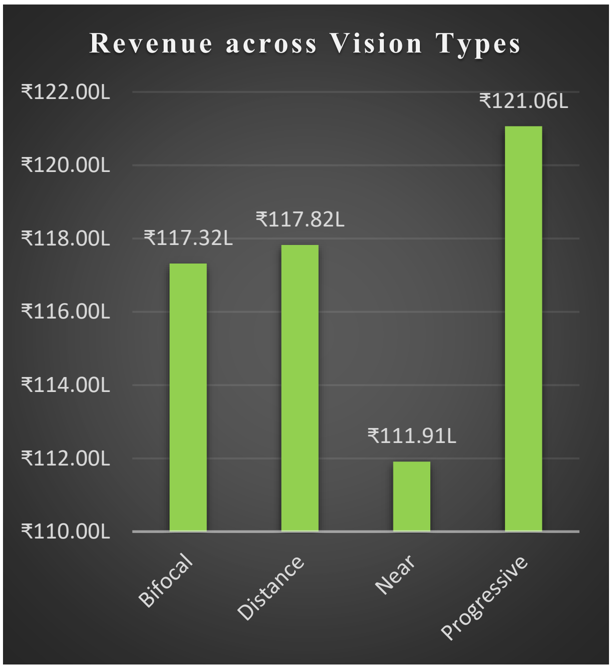
Age_group	Visited_Customers	Buyer_Customers	Conversion_Rate	Referral_Customers	Repeated_Customers	Retention_Rate	Churn_Customers	Churn_Rate	Referrers_Count	Referrers_Rate	Revenue	Contribution_in_revenue
20's Customer	1328	623	46.91	59	172	27.61	44	7.06	130	9.79	7751143.00	16.56
30's Customer	1214	537	44.23	59	134	24.95	33	6.15	132	10.87	6458010.00	13.80
40's Customer	1157	523	45.20	52	152	29.06	30	5.74	131	11.32	6598649.00	14.10
50's Customer	1152	555	48.18	58	164	29.55	34	6.13	105	9.11	6870009.00	14.68
Old_Age Customers	2920	1373	47.02	139	392	28.55	85	6.19	261	8.94	17440454.00	37.26
Teenage_Customer	229	110	48.03	14	37	33.64	9	8.18	28	12.23	1693148.00	3.62
NULL	8000	3721	46.51	381	1051	28.25	235	6.32	721	9.01	46811413.00	100.00

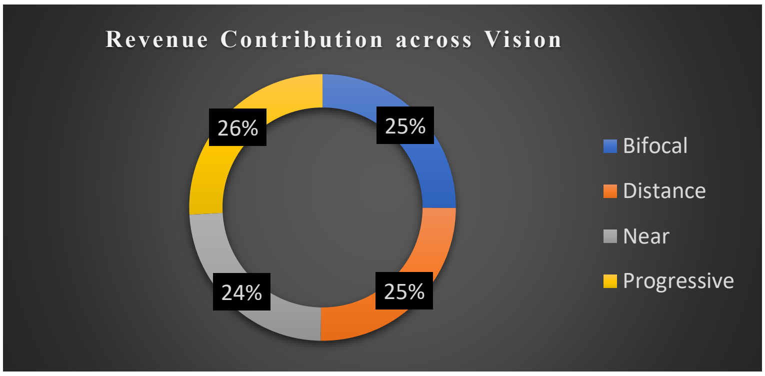
Customer Lifecycle Dashboard – Vision Type Analysis

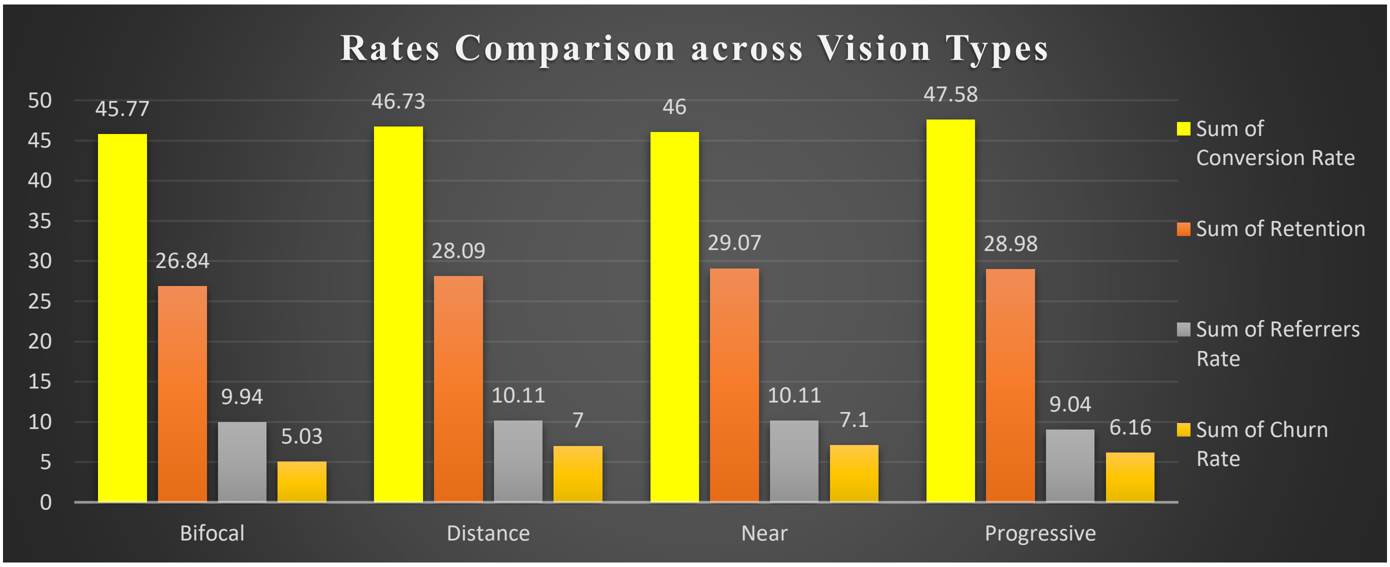
Distance | **Near** | **Bifocal** | **Progressive**











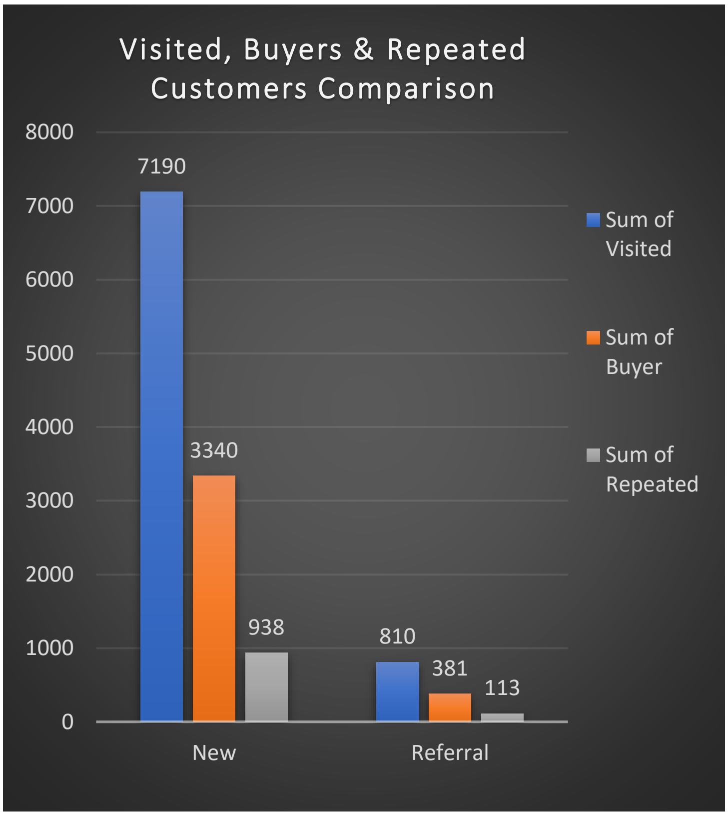
Customer Lifecycle Dashboard – New v/s Referral Customers

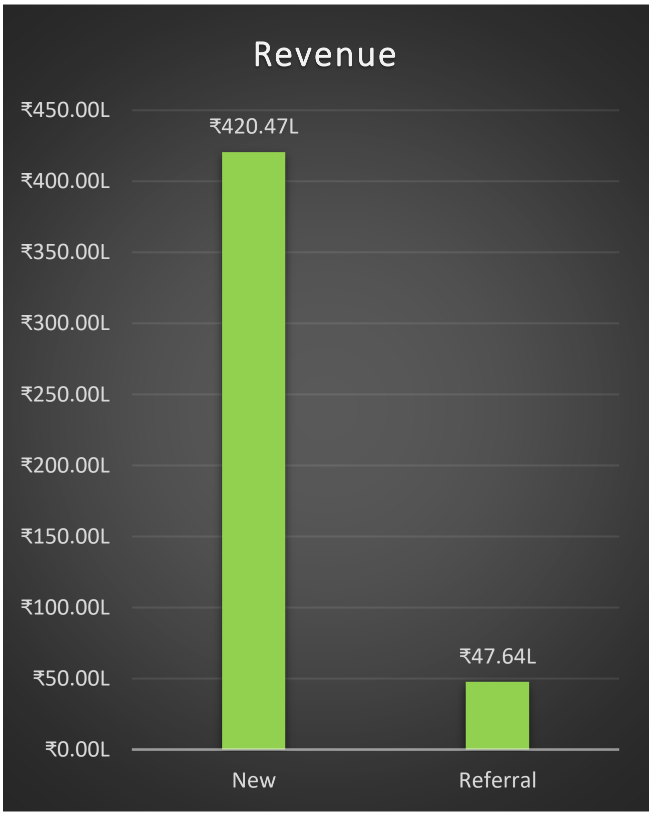
8000 Walk-in-visitors → **721** Referrers → **810** Referrals → **381** Buyer Referrals

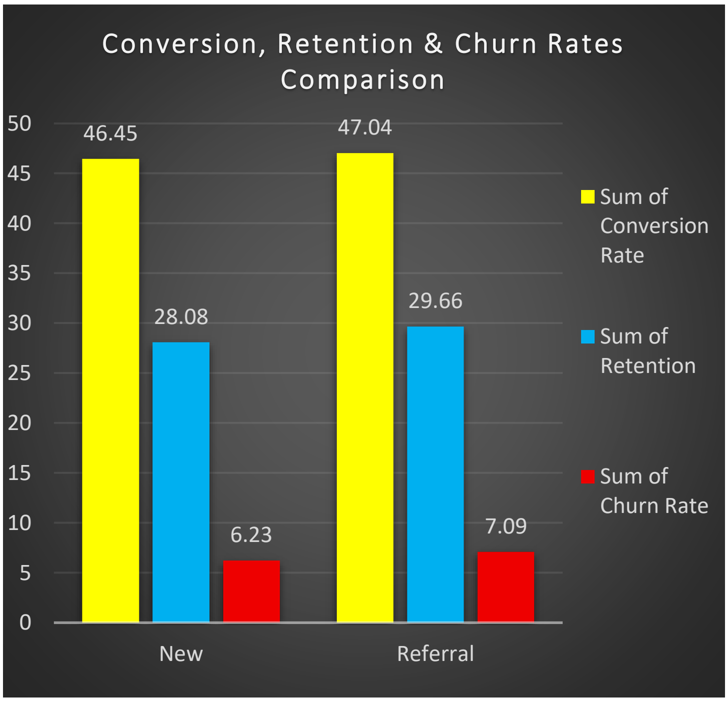
47% Referrals Conversion Rate

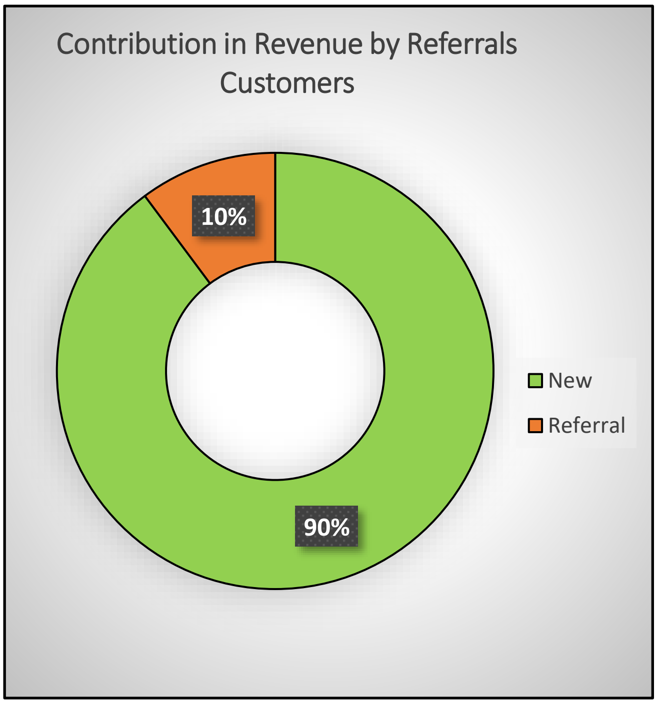
10% Contribution in Revenue

47.63L Rs Revenue by Referrals









Age Group Analysis

Why this Analysis Matters?

Understanding customer behaviour across different age groups enables the business to identify which customer segments contribute the most value throughout the customer lifecycle from first-time visits and purchases to repeat business, referrals, and long-term retention. This analysis highlights where the business is already performing well, where customer leakage occurs, and which customer segments should be prioritized to maximize sustainable revenue growth.

Key Insights

Insight 1 – Senior Customers (60+) are the Core Growth Driver of the Business

The analysis clearly identifies Senior Customers (above 60 years) as the store's most valuable customer segment.

Across almost every stage of the customer lifecycle, this age group consistently records the highest performance, including:

- Highest number of walk-in visitors
- Highest number of converted buyers
- Highest revenue contribution
- Highest number of repeat customers
- Highest number of referrers
- Highest number of referral customers

Rather than dominating only one metric, senior customers consistently lead across the entire customer journey, making them the primary driver of business performance.

Strategic Recommendation

Senior customers should be treated as the business's highest-priority customer segment.

Management should continue investing in services specifically designed for this demographic, including:

- Personalized customer assistance
- Premium consultation experience
- High-quality progressive and bifocal lens solutions
- Comfortable store experience
- Regular eye examination reminders

Protecting this customer base should become a long-term strategic objective rather than simply acquiring new customers.

Insight 2 – The 20–50 Age Groups Form a Stable and Diversified Customer Base

Although senior customers dominate by volume, customers in their 20s, 30s, 40s, and 50s display remarkably balanced performance.

Across visitors, buyers, repeat customers, referrals, and revenue, these age groups contribute relatively similar numbers without any major fluctuations.

This indicates that the business has successfully diversified its customer base instead of relying exclusively on one working-age demographic.

Strategic Recommendation

Rather than aggressively shifting focus toward one specific working-age segment, management should continue maintaining balanced marketing efforts across these age groups.

Since these customers represent a stable source of business, incremental improvements in customer experience and service quality are likely to improve overall conversion and retention without requiring major strategic changes.

Insight 3 – Teenage Customers Represent the Smallest Segment but the Largest Future Opportunity

Teenage customers contribute the lowest customer volume across every lifecycle stage.

However, despite their small size, they demonstrate:

- The highest conversion rate
- The highest customer retention rate

This suggests that once teenage customers enter the store, they are highly likely to purchase and return.

Therefore, the limitation is not customer quality it is customer acquisition.

Strategic Recommendation

The business should increase its marketing efforts specifically toward younger customers by focusing on:

- Social media marketing
- Student-focused promotions
- Fashion eyewear campaigns
- Digital engagement
- Local influencer collaborations

Increasing customer acquisition within this segment has the potential to generate long-term customer relationships at an early age

Insight 4 – Senior Customer Retention Represents the Largest Revenue Growth Opportunity

Although senior customers generate the highest business volume, they also account for the highest number of churned customers.

This finding deserves particular attention.

From an optical retail perspective, this behaviour is expected because customers above 50–60 years typically experience age-related vision changes such as presbyopia, resulting in prescription changes approximately every 2–3 years. Consequently, this customer segment naturally requires periodic eye examinations, updated prescriptions, and replacement eyewear. This recurring demand makes customer retention significantly more valuable than customer acquisition within this segment.

The analysis also aligns with the earlier churn analysis, which identified senior customers as the largest contributor to customer churn.

Strategic Recommendation

Along with focusing on primarily acquiring new senior customers, management should also prioritize retaining an existing ones by:

- Conducting regular post-purchase follow-ups
- Collecting customer feedback
- Sending prescription reminder campaigns every 2–3 years
- Scheduling preventive eye-checkup reminders
- Strengthening trust through personalized consultations
- Recommending products based on customer needs rather than short-term sales targets

Even a modest improvement in retention within this segment could produce substantial long-term revenue growth because of its already large customer base.

Insight 5 – Senior Customers are the Centre of the Referral Network

One of the strongest patterns identified throughout the referral analysis is that customers from nearly every age group predominantly refer Senior Customers (60+) to the business.

This indicates that senior customers are not only active referrers themselves but are also the primary recipients of referrals from family members and acquaintances.

This behaviour strongly suggests that the business has developed a trusted reputation for serving senior customers.

Families appear confident recommending the store for their parents and grandparents, making senior customers the central node of the referral ecosystem.

Strategic Recommendation

The business should capitalize on this competitive advantage by strengthening its positioning as a trusted optical care provider for senior citizens.

Recommended initiatives include:

- Senior citizen loyalty programs
- Family referral campaigns
- Annual eye health awareness initiatives
- Personalized follow-up services
- Dedicated consultation support for elderly customers

Since senior customers naturally revisit every few years due to prescription changes, retaining them today increases the likelihood that they become future repeat customers, referral generators, and long-term brand advocates.

Overall Business Conclusion

The age group analysis reveals a healthy and diversified customer portfolio. The business serves customers across every age segment, with the working-age population (20–50 years) providing consistent business volume, while senior customers form the primary engine of long-term growth.

Two strategic priorities emerge from the analysis:

1. Protect and retain Senior Customers, as they generate the highest lifetime value, referrals, repeat purchases, and revenue. Improving retention within this segment represents the largest opportunity for sustainable revenue growth.
2. Expand acquisition among Teenage Customers, who exhibit the strongest conversion and retention behaviour despite their small customer base. Targeted marketing initiatives could transform this segment into a significant future growth engine.

By simultaneously strengthening retention among high-value senior customers and increasing acquisition of younger customers, the business can improve both immediate profitability and long-term customer lifetime value while maintaining a balanced and sustainable customer portfolio.

Vision Type Analysis – Business Insights & Strategic Recommendations

Why this analysis matters

Understanding customer vision types enables the business to align its infrastructure, inventory, clinical capabilities, and customer service with actual market demand. Different vision categories require different levels of expertise, equipment, consultation time, and lens technology. By identifying which vision conditions generate the highest customer volume, revenue, and long-term loyalty, the store can prioritize investments that maximize customer satisfaction and long-term profitability.

Key Insight 6 – The Store Serves a Well-Diversified Vision Portfolio

The analysis shows that customer demand is remarkably balanced across all four major vision categories.

Vision Type	Share of Customers
Bifocal	~25.5%
Near Vision	~24.9%
Distance Vision	~24.9%
Progressive	~24.8%

A similar distribution is observed across:

- Walk-in visitors
- Buyer customers
- Repeated customers

This indicates that the business is not dependent on a single vision segment, reducing concentration risk while demonstrating the capability to serve a broad spectrum of optical needs.

Key Insight 7 – Progressive Lenses Generate the Highest Revenue Opportunity

Although customer volume is nearly identical across all vision categories, Progressive vision customers generate the highest revenue, contributing approximately ₹1.21 crore, followed closely by Distance and Bifocal customers (approximately ₹1.17 crore each), while Near Vision contributes the least.

This is primarily because Progressive lenses command:

- Higher selling prices
- Better profit margins
- More customization
- Higher-value lens technologies

From a commercial perspective, Progressive lenses represent the store's highest-value product category rather than simply another vision type.

Strategic Recommendation

Instead of merely increasing Progressive sales volume, the business should strengthen its capability to deliver premium Progressive lens experiences through investments such as:

- Advanced computerized eye testing equipment
- High-precision digital refraction systems
- Accurate PD (Pupillary Distance) measuring devices
- Experienced optometrists
- Skilled spectacle fitters
- Modern lens fitting technologies

Improving precision in Progressive dispensing will increase customer satisfaction, reduce remakes, and strengthen long-term trust.

Key Insight 8 – Progressive Vision Aligns Perfectly with the Largest Customer Segment

One of the strongest patterns across the entire project is the close relationship between:

- Senior Customers (60+)
- Progressive Vision
- Repeat Purchases
- Referrals

Previous demographic and referral analyses consistently showed that senior customers represent the largest customer segment.

This analysis explains why.

Age-related vision conditions such as Presbyopia naturally increase the need for Progressive lenses.

Unlike younger customers, senior customers typically require:

- Regular prescription updates
- New lenses every 2–3 years
- Continued optometrist consultation
- Ongoing eyewear replacement

This creates a naturally recurring customer lifecycle.

Key Insight 9 – Repeat Customers Drive Long-Term Profitability

The customer lifecycle analysis reveals an important business reality.

Although only 22% of buyer customers become repeated customers, they contribute approximately 47% of the store's total revenue.

This demonstrates that long-term profitability depends far more on retaining existing customers than continually acquiring new ones.

Every retained customer represents multiple future purchases over their lifetime.

Key Insight 10 – Referral Customers Deliver High-Value Revenue Without Marketing Cost

Referral customers account for only 10% of total buyers, yet contribute approximately 10% of overall business revenue, generating nearly ₹47.6 lakh.

More importantly, this revenue is acquired without any direct marketing or advertising expenditure.

This makes referrals one of the highest-return customer acquisition channels available to the business.

The analysis also shows that today's satisfied customers become tomorrow's referral generators, creating a self-sustaining growth cycle.

Key Insight 11- Customer Lifetime Value is Concentrated in a Small Segment

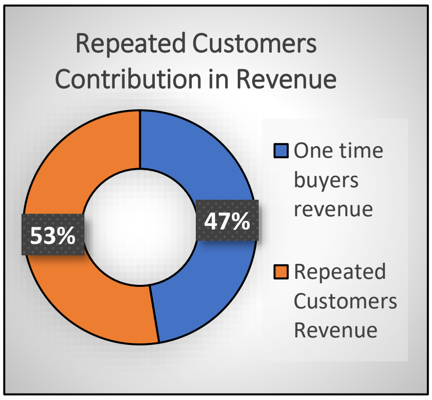
One of the most important strategic findings from the entire project is the disproportionate revenue contribution of loyal customers.

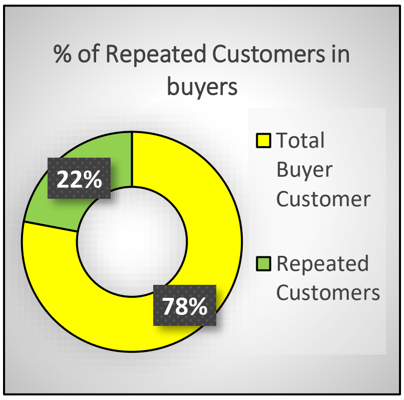
- 22% of buyers become repeated customers and generate 47% of total revenue.
- 10% of buyers arrive through referrals and generate an additional 10% of revenue.

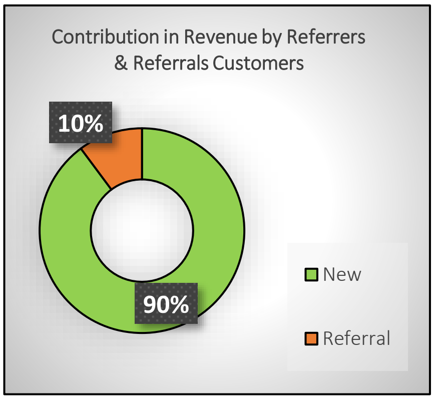
Together, these two segments represent only 32% of the buyer base, yet contribute approximately 57% of total business revenue (around ₹2.66 crore).

By comparison, the remaining 68% of customers primarily first-time buyers generate only 43% of revenue.

This highlights that the store's most valuable customers are not necessarily its largest customer segment, but its most loyal and engaged customers.







Strategic Business Recommendations

Based on the complete Customer Lifecycle, Demographic, Referral, and Vision analyses, the business should prioritize the following strategic initiatives.

1. Prioritize Senior Customer Retention

Senior customers represent the strongest long-term revenue opportunity because they:

- Generate the highest customer volume
- Generate the highest revenue
- Produce the highest number of repeat customers
- Generate the largest referral network
- Naturally require prescription updates every 2–3 years

Retention programs for this segment should include:

- Periodic eye examination reminders
- Prescription renewal campaigns
- Post-purchase follow-ups
- Personalized customer relationship management
- Trust-based consultation instead of sales-driven recommendations

---

2. Strengthen Progressive Lens Capability

Progressive lenses represent the highest-value product category.

The business should continue investing in:

- Advanced diagnostic technology
- Premium dispensing processes
- Staff training
- Progressive lens consultation expertise
- Precision fitting

These investments will improve customer outcomes while increasing margins.

---

3. Treat Repeat Customers as Strategic Assets

Repeat customers should receive dedicated retention initiatives such as:

- Loyalty programs
- Annual vision check reminders
- Personalized offers
- Follow-up communication
- After-sales service

Since they generate nearly half of the business revenue, even modest improvements in retention can significantly increase long-term profitability.

---

4. Expand the Referral Ecosystem

Referral customers have already demonstrated their ability to generate substantial revenue at virtually zero acquisition cost.

The business should encourage referrals by:

- Delivering exceptional customer experiences
- Rewarding referrals appropriately
- Maintaining strong relationships with existing customers
- Encouraging satisfied customers to recommend friends and family

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5. Do Not Ignore the Teenage Segment

Although teenagers represent the smallest customer segment by volume, they consistently demonstrate:

- The highest conversion rate
- The highest retention rate
- One of the lowest churn rates

This indicates strong customer quality despite lower customer numbers.

Targeted digital marketing, school and college outreach, and stronger social media engagement could significantly expand this high-potential segment.

Overall Strategic Conclusion

The analyses collectively show that the store's future growth should not rely solely on acquiring more walk-in customers. Instead, the greatest opportunities also lie in maximizing Customer Lifetime Value (CLV) by retaining high-value customers and strengthening referral-driven growth.

The highest-return strategic priorities are:

1. Senior Customers (60+) Largest customer base with recurring prescription needs and the strongest long-term value.
2. Progressive Lens Customers Highest revenue and margin opportunity requiring continued investment in clinical expertise and dispensing infrastructure.
3. Repeat Customers Only 22% of buyers, yet responsible for 47% of total revenue.
4. Referrers and Referral Customers Low-cost acquisition channel contributing approximately 10% of business revenue while reinforcing sustainable organic growth.
5. Teenage Customers A smaller but highly efficient segment with exceptional conversion and retention rates, representing a valuable opportunity for future expansion through targeted marketing.

Executive Summary of Business Recommendations

Priority 1 (Highest ROI)

Improve retention among senior customers.

Expected impact:

- Higher lifetime value
- Increased repeat purchases
- Stronger referral generation

Priority 2

Invest in Progressive lens infrastructure.

Expected impact:

- Higher margins
- Better customer satisfaction
- Reduced fitting errors

Priority 3

Build a referral reward program.

Expected impact:

- Increase low-cost customer acquisition
- Improve organic growth

Priority 4

Target teenagers through digital campaigns.

Expected impact:

- Expand a segment with the highest conversion and retention rates

---

Priority 5

Reduce window shoppers through CRM follow-ups and sales training.

Expected impact:

- Improve overall conversion rate

- **Top 50 Customers**

```sql
CREATE VIEW top_50_customers AS
SELECT
      i.customer_id,
      SUM(i.final_bill_amount)
      AS Total_Revenue_by_Customer
FROM invoices as i
GROUP BY i.customer_id
ORDER BY Total_Revenue_by_Customer DESC
LIMIT 50;
```

The Top 50 customers VIEW stores the top 50 customers ranked by total revenue generated. Instead of recalculating this ranking in every subsequent query, the VIEW is created once and reused across age group, customer type, vision type, product, and at-risk analysis ensuring consistency across all metrics and keeping individual queries clean and readable.

Click → Insights and Strategic Recommendations

Business Questions Answer about the Actual Buyer Customers are as follows:

1) Who are the top 50 highest revenue generating customers of the store, and how much total revenue do they collectively contribute to the business? Total Revenue per Customer

2) How much total revenue by the Top 50 customers collectively contribute to the business?

3) Which age group dominates among our top 50 customers and does our highest value customer segment align with the overall age distribution of the store?

4) Which products are most frequently purchased by our top 50 customers, and do high-value customers show a different product preference compared to the average buyer?

5) Which of our top 50 customers have not made a purchase in over 180 days and are now at risk of being lost and what is the urgency of re-engaging them?

6) At what point does the overlap between our top revenue generating customers and our repeated customers begin to break down and what does this reveal about the relationship between customer loyalty and revenue contribution?

1) Top 50 Highest Revenue generating Customers and their details

1) Who are the top 50 highest revenue generating customers of the store, Total Revenue per Customer?

```
WITH top_50_customers AS (
SELECT
      i.customer_id,
      SUM(i.final_bill_amount)
      AS Total_Revenue_by_Customer
FROM invoices as i
GROUP BY i.customer_id
ORDER BY Total_Revenue_by_Customer DESC
LIMIT 50
)
SELECT tc5.customer_id,
       c.customer_full_name,
       c.age, c.mobile, c.customer_type,
       tc5.Total_Revenue_by_Customer
FROM customers_details as c
INNER JOIN top_50_customers as tc5
ON tc5.customer_id=c.customer_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	customer_full_name	age	mobile	customer_type	Total_Revenue_by_Customer
5669	Samar Parmer	77	0348246068	New	56508.00
4728	Neelima Garde	27	5787144903	New	54250.00
2316	Yug Borde	75	6289378796	New	50246.00
7640	Isaac Murthy	18	3515909626	New	48900.00
7360	Neha Dara	56	7392907576	New	48654.00
4679	Matthew Bawa	50	6909930534	New	48648.00
6978	Ojasvi Dass	48	9143874605	New	48191.00
3293	Vedant Saxena	22	5480227836	New	47956.00
6348	Om Sanghvi	41	4031571008	New	47926.00
2489	Darika Sami	80	6520985027	New	47082.00
3652	Naksh Borra	66	4949030136	New	46748.00
1872	Yashvi Rout	51	8421374081	New	46577.00
5575	Aditya Goel	18	9978426471	New	45704.00
1967	Isaiah Sahni	62	6071770882	New	45647.00
7413	Gaurav Mahajan	59	6628280530	New	45163.00
5561	Mekhala Boase	71	3805756164	New	45083.00
5123	Reyansh Chahal	30	9368062251	New	44796.00
7720	Vrishti Karnik	27	2729898132	New	44287.00
1532	Akshay Shanker	72	6047570873	New	43305.00
5586	Ekta Narayanan	73	0141383975	New	43226.00
7328	Aishani Tailor	49	4224891606	New	43075.00
5205	Eesha Dhawan	37	8420036932	New	42818.00
6091	Ayushman Mohanty	37	6788475484	New	42630.00
2141	Tarak Garg	23	1670533401	New	42550.00
7111	Vritti Acharya	35	8450100202	New	42134.00
3568	Ishwar Verma	30	3756137095	New	42095.00

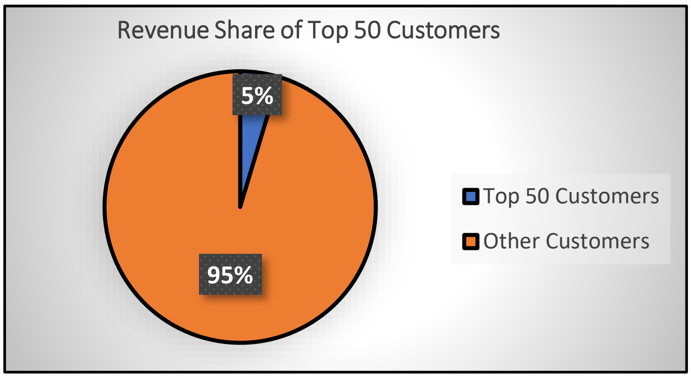
2) Top 50 Customer revenue contribution in total revenue collected

2) How much total revenue by the Top 50 customers collectively contribute to the business?

```sql
SELECT
    SUM(i.final_bill_amount)
    AS Total_Revenue,
    SUM(CASE WHEN tc.customer_id IS NOT NULL THEN i.final_bill_amount END)
    AS Revenue_by_Repeated_Customers,
    ROUND( SUM(CASE WHEN tc.customer_id IS NOT NULL THEN i.final_bill_amount END) * 100 /
    SUM(i.final_bill_amount), 2)
    AS Countribution_of_Repeated_Customers_in_Revenue
FROM invoices as i
LEFT JOIN top_50_customers as tc
ON tc.customer_id=i.customer_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Total_Revenue	Revenue_by_Repeated_Customers	Countribution_of_Repeated_Customers_in_Reven
46811413.00	2173180.00	4.64



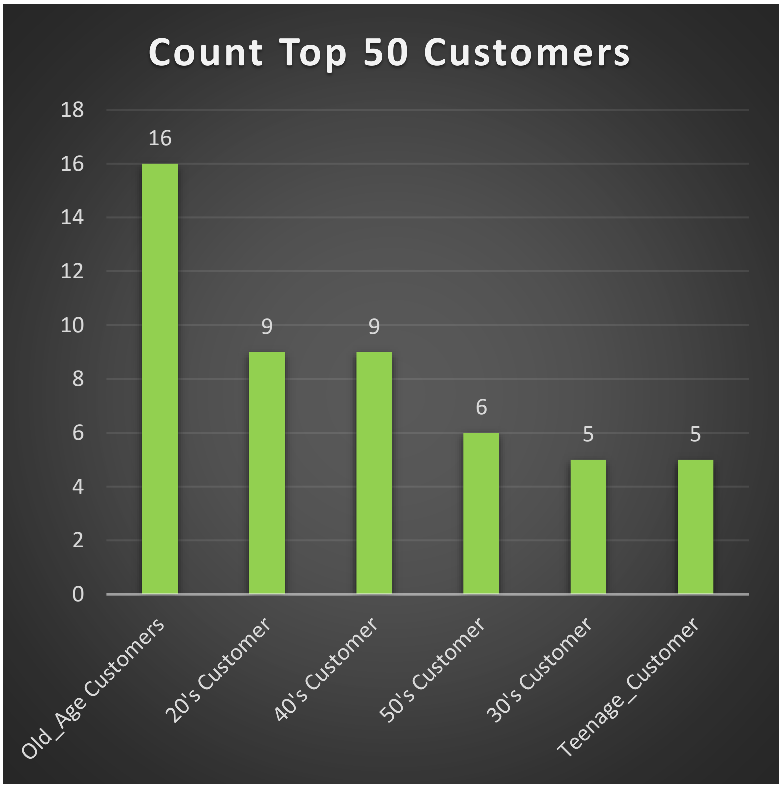
3) Age group segment of Top 50 Customers by revenue

3) Which age group dominates among our top 50 customers and does our highest value customer segment align with the overall age distribution of the store?

```sql
SELECT
CASE
    WHEN age < 20 THEN "Teenage_Customer"
    WHEN age BETWEEN 20 AND 30 THEN "20's Customer"
    WHEN age BETWEEN 31 AND 40 THEN "30's Customer"
    WHEN age BETWEEN 41 AND 50 THEN "40's Customer"
    WHEN age BETWEEN 51 AND 60 THEN "50's Customer"
    WHEN age > 60 THEN "Old_Age Customers"
  END AS Age_group,
  COUNT(DISTINCT tc.customer_id) AS COUNT_customers
FROM customers_details as c
INNER JOIN top_50_customers as tc
ON tc.customer_id=c.customer_id
GROUP BY Age_group
ORDER BY COUNT_customers DESC;
```

Result Grid | Filter Rows:

Age_group	COUNT_customers
Old_Age Customers	16
20's Customer	9
40's Customer	9
50's Customer	6
30's Customer	5
Teenage_Customer	5



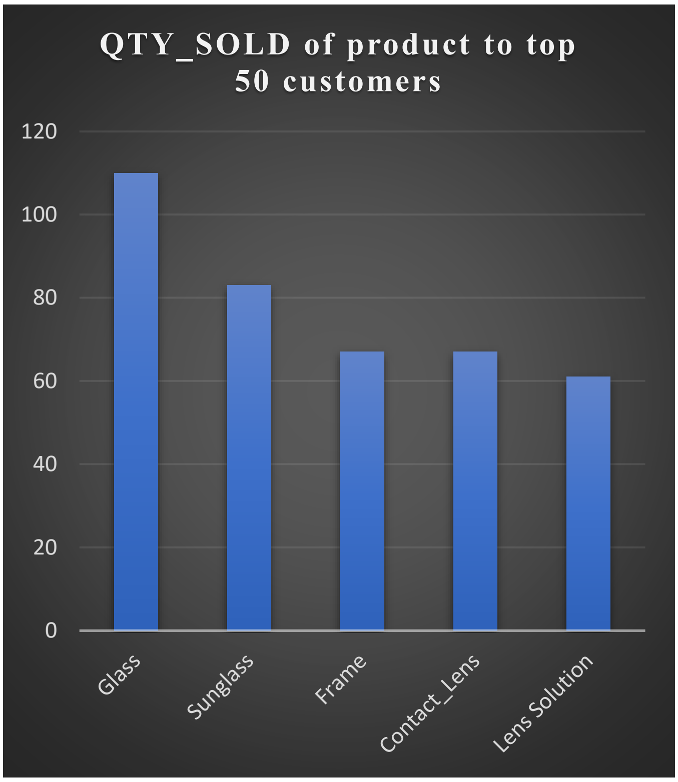
4) QTY_SOLD of each product to TOP 50 Customers

4) Which products are most frequently purchased by our top 50 customers, and do high-value customers show a different product preference compared to the average buyer?

```sql
SELECT
       p.product_name, COUNT(ii.item_id)
       AS total_units_sold
FROM top_50_customers as v
JOIN invoices i
ON v.customer_id = i.customer_id
JOIN invoice_items ii
ON i.invoice_id = ii.invoice_id
JOIN products p
ON ii.product_id = p.product_id
GROUP BY p.product_name
ORDER BY total_units_sold DESC;
```

Result Grid | Filter Rows:

product_name	total_units_sold
Glass	110
Sunglass	83
Frame	67
Contact_Lens	67
Lens Solution	61



5) Top Customers who has not purchased since last 180 days (appx 6 months)

5) Which of our top 50 customers have not made a purchase in over 180 days and are now at risk of being lost and what is the urgency of re-engaging them?

```sql
SELECT
    tc.customer_id,
    c.customer_full_name,
    c.mobile,
    MAX(i.invoice_date)
    AS last_purchase_date,
    DATEDIFF(CURDATE(),
    MAX(i.invoice_date))
    AS days_since_last_visit
FROM top_50_customers tc
JOIN customers_details c
ON tc.customer_id = c.customer_id
JOIN invoices i
ON tc.customer_id = i.customer_id
GROUP BY
     tc.customer_id,
     c.customer_full_name,
     c.mobile
HAVING days_since_last_visit > 180
ORDER BY days_since_last_visit DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	customer_full_name	mobile	last_purchase_date	days_since_last_visit
3293	Vedant Saxena	5480227836	2025-02-06 00:00:00	506
5487	Unni Borde	3865031997	2025-04-06 00:00:00	447
5465	Hemangini Doshi	4075937519	2025-05-04 00:00:00	419
478	Jhalak Tandon	5864182497	2025-05-13 00:00:00	410
4074	Isha Menon	8924518051	2025-06-06 00:00:00	386
7111	Vritti Acharya	8450100202	2025-06-16 00:00:00	376
2141	Tarak Garg	1670533401	2025-06-20 00:00:00	372
982	Ayaan Mann	9018662089	2025-06-27 00:00:00	365
3702	Meghana Pall	5368327615	2025-07-05 00:00:00	357
7450	Falan Banerjee	6780721218	2025-07-21 00:00:00	341
742	Deepa Bir	9987784616	2025-07-26 00:00:00	336
6226	Yug Manne	1654843051	2025-08-05 00:00:00	326
480	Mason Agrawal	4482577744	2025-08-10 00:00:00	321
5205	Eesha Dhawan	8420036932	2025-08-18 00:00:00	313
7021	Mohammed Sawhney	9434489662	2025-08-18 00:00:00	313
1532	Akshay Shanker	6047570873	2025-08-19 00:00:00	312
2489	Darika Sami	6520985027	2025-09-02 00:00:00	298
5586	Ekta Narayanan	0141383975	2025-09-03 00:00:00	297
1872	Yashvi Rout	8421374081	2025-09-05 00:00:00	295
2316	Yug Borde	6289378796	2025-09-09 00:00:00	291
6348	Om Sanghvi	4031571008	2025-09-13 00:00:00	287
699	Laksh Sarin	0000698951	2025-10-04 00:00:00	266
6091	Ayushman Mohanty	6788475484	2025-10-13 00:00:00	257
5866	Tanveer Choudhry	6670148839	2025-10-21 00:00:00	249
1670	Ria Sathe	7537099105	2025-10-27 00:00:00	243
7720	Vrishti Karnik	2729898132	2025-11-04 00:00:00	235

6) Identifying the % of repeated customers in the Top customers

6) At what point does the overlap between our top revenue generating customers and our repeated customers begin to break down and what does this reveal about the relationship between customer loyalty and revenue contribution?

```sql
WITH top_50_customers AS (
SELECT
    i.customer_id,
    SUM(i.final_bill_amount)
FROM invoices as i
GROUP BY i.customer_id
ORDER BY SUM(i.final_bill_amount) DESC
LIMIT 50
),
top_100_customers AS (
SELECT
     i.customer_id,
    SUM(i.final_bill_amount)
FROM invoices as i
GROUP BY i.customer_id
ORDER BY SUM(i.final_bill_amount) DESC
LIMIT 100
),
top_150_customers AS (
SELECT
     i.customer_id,
    SUM(i.final_bill_amount)
FROM invoices as i
GROUP BY i.customer_id
ORDER BY SUM(i.final_bill_amount) DESC
LIMIT 150
),
```

```sql
top_2500_customers AS (
SELECT
     i.customer_id,
    SUM(i.final_bill_amount)
FROM invoices as i
GROUP BY i.customer_id
ORDER BY SUM(i.final_bill_amount) DESC
LIMIT 2500
),
top_3000_customers AS (
SELECT
     i.customer_id,
    SUM(i.final_bill_amount)
FROM invoices as i
GROUP BY i.customer_id
ORDER BY SUM(i.final_bill_amount) DESC
LIMIT 3000
),
top_3721_customers AS (
SELECT
     i.customer_id,
    SUM(i.final_bill_amount)
FROM invoices as i
GROUP BY i.customer_id
ORDER BY SUM(i.final_bill_amount) DESC
LIMIT 3721
)
```

```sql
SELECT
        3000,
        COUNT(*),
        ROUND(COUNT(*) * 100 / 3000, 2)
FROM repeated_customers AS rc
INNER JOIN top_3000_customers AS tc3000
ON tc3000.customer_id = rc.customer_id

UNION ALL

SELECT
        3721,
        COUNT(*),
        ROUND(COUNT(*) * 100 / 3721, 2)
FROM repeated_customers AS rc
INNER JOIN top_3721_customers AS tc3721
ON tc3721.customer_id = rc.customer_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

TOP_Customers	Repeated_Customers	Percentage_of_Repeated_Customer_among_the_Top_50_Customers
50	50	100.00
100	100	100.00
150	149	99.33
200	196	98.00
300	287	95.67
400	361	90.25
500	434	86.80
600	503	83.83
700	560	80.00
800	610	76.25
900	659	73.22
1000	697	69.70
1500	859	57.27
2000	952	47.60
2500	952	38.08
3000	1039	34.63
3721	1051	28.25

Top Revenue Customers Analysis

Why This Analysis Matters

Understanding the store's highest-value customers is critical because a relatively small group of customers often contributes a disproportionately large share of total business revenue. Identifying these customers enables the business to implement personalized relationship management strategies that strengthen customer loyalty, increase customer lifetime value, and reduce the risk of losing high-value accounts.

Unlike the general customer base, these customers deserve proactive engagement through dedicated follow-ups, personalized recommendations, periodic eye-care reminders, priority service, and continuous relationship building. Retaining even a few of these customers can have a significant impact on long-term business profitability.

Insight 1 – A Small Group of Customers Generates Significant Business Revenue

The analysis identified the Top 50 customers based on total revenue generated. Collectively, these customers contributed approximately ₹23.71 lakh, representing 4.64% of the store's total revenue.

Considering that this revenue is generated by only 50 customers out of 3,721 buyer customers, this segment represents one of the business's highest-value customer groups.

Strategic Recommendation

The business should establish a High-Value Customer Management Program by assigning a dedicated relationship manager or implementing personalized customer engagement initiatives for these customers.

Recommended actions include:

- Regular follow-up calls and customer feedback collection
- Personalized product recommendations
- Priority appointment scheduling
- Periodic eye examination and prescription reminders
- Exclusive loyalty benefits and appreciation programs
- Building long-term trust through consultative service rather than transactional selling

Protecting this customer segment should be considered a strategic priority due to its substantial contribution to overall revenue.

Insight 2 – Senior Customers Dominate the Highest-Value Customer Segment

Consistent with previous demographic analyses, Senior Customers (60+) represent the largest proportion of the Top 50 revenue-generating customers, accounting for 16 customers within the top segment.

The remaining high-value customers are distributed across the working-age population, with customers in their 20s and 40s contributing nine customers each, followed by customers in their 50s and teenagers.

This finding further reinforces that senior customers consistently generate the highest business value across multiple dimensions, including revenue, repeat purchases, and referrals.

Strategic Recommendation

Customer retention initiatives should place particular emphasis on senior customers through personalized service, proactive relationship management, and regular vision care follow-ups. Given their recurring prescription needs, this segment offers the greatest opportunity for long-term customer lifetime value.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

TOP_Customers	Repeated_Customers	Percentage_of_Repeated_Customer_among_the_Top_50_Customers
50	50	100.00
100	100	100.00
150	149	99.33
200	196	98.00
300	287	95.67
400	361	90.25
500	434	86.80
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700	560	80.00
800	610	76.25
900	659	73.22
1000	697	69.70
1500	859	57.27
2000	952	47.60
2500	952	38.08
3000	1039	34.63
3721	1051	28.25

Insight 3 – Nearly Every High-Value Customer is a Repeat Customer

One of the most significant findings from the analysis is the exceptionally strong relationship between customer loyalty and revenue generation.

This demonstrates that virtually every high-value customer is also a repeat customer.

As the revenue threshold expands, the proportion of repeat customers gradually decreases; however, even among the Top 400 revenue-generating customers, more than 90% are repeat purchasers.

This finding provides compelling evidence that the store's highest revenue is generated not by first-time buyers, but by customers who continue returning over time.

Executive Business Finding

100% of the Top 50 and Top 100 highest-revenue customers are repeat customers, while over 90% of the Top 400 revenue-generating customers are repeat purchasers.

This finding confirms that long-term customer loyalty is the strongest predictor of high customer value. The store's most profitable customers are not first-time buyers—they are customers who repeatedly return and build an ongoing relationship with the business. Consequently, investments in customer retention are likely to deliver substantially higher long-term returns than an equivalent investment focused solely on acquiring new customers.

▪ Churn Analysis (Churn Customers)

```sql
CREATE VIEW churn_customers AS
SELECT
       rc.customer_id,
       c.customer_full_name,
       c.mobile,
       c.age,
       c.assigned_staff_id,
       DATE(MAX(i.invoice_date))
       AS last_visited_date,
       DATEDIFF( (SELECT MAX(invoice_date) FROM invoices),
       MAX(i.invoice_date))
       AS Number_of_days_from_last_visit
FROM repeated_customers as rc
INNER JOIN customers_details as c
ON rc.customer_id=c.customer_id
INNER JOIN invoices as i
ON rc.customer_id=i.customer_id
GROUP BY rc.customer_id
HAVING Number_of_days_from_last_visit > 180
ORDER BY Number_of_days_from_last_visit DESC;
```

```sql
CREATE VIEW non_repeated_churn_customers AS
WITH customers AS (
SELECT DISTINCT customer_id
FROM customers_details
),
customer_details AS (
SELECT c.customer_id, cc.age, cc.mobile,
       cc.customer_type, cc.referred_by_id,
       cc.assigned_staff_id
FROM customers as c
INNER JOIN customers_details as cc
ON c.customer_id=cc.customer_id
),
customers_with_repeated_customers AS (
SELECT cd.customer_id, rc.customer_id as repeated_customers,
       cd.age, cd.mobile,
       cd.customer_type, cd.referred_by_id,
       cd.assigned_staff_id
FROM customer_details as cd
LEFT JOIN repeated_customers as rc
ON cd.customer_id=rc.customer_id
),
-- extracting_the_non_repeated_customers
extracting_the_non_repeated_customers AS (
SELECT cwrc.customer_id, cwrc.repeated_customers,
       cwrc.age, cwrc.mobile,
       cwrc.customer_type, cwrc.referred_by_id,
       cwrc.assigned_staff_id
FROM customers_with_repeated_customers  as cwrc
```

```sql
FROM customers_with_repeated_customers  as cwrc
WHERE cwrc.repeated_customers IS NULL
),
buyer_non_repeated_customer_details AS (
SELECT enrc.customer_id, enrc.age, enrc.mobile,
       enrc.customer_type, enrc.referred_by_id,
       enrc.assigned_staff_id, i.invoice_id,
       i.invoice_date, i.total_amount_items,
       i.total_tax,
       i.final_bill_amount
FROM extracting_the_non_repeated_customers  as enrc
INNER JOIN invoices as i
ON enrc.customer_id=i.customer_id
)
SELECT
       bnrcd.customer_id, bnrcd.age, bnrcd.mobile,
       bnrcd.customer_type, bnrcd.referred_by_id,
       bnrcd.assigned_staff_id,
       DATE(MAX(bnrcd.invoice_date)) AS Last_visit_date,
       DATEDIFF( (SELECT MAX(invoice_date) FROM invoices),
       MAX(bnrcd.invoice_date))
       AS Number_of_days_from_last_visit
FROM buyer_non_repeated_customer_details as bnrcd
GROUP BY  bnrcd.customer_id, bnrcd.age, bnrcd.mobile,
       bnrcd.customer_type, bnrcd.referred_by_id,
       bnrcd.assigned_staff_id
HAVING Number_of_days_from_last_visit > 180
ORDER BY Number_of_days_from_last_visit DESC;
```

Before diving into the churn analysis queries, a dedicated VIEW named Churn customers was created to store all the churned customer records in a reusable virtual table. The VIEW identifies every repeated customer who has not made a purchase in more than 180 days classifying them as churned based on the difference between the store's most recent invoice date and each customer's last purchase date.

NOTE - Assumption: *Since the dataset is historical, customer inactivity and churn are calculated relative to the latest transaction date in the dataset (2026-01-11) rather than the system's current date CURDATE(). This avoids overstating inactivity due to the absence of data beyond the dataset period.*

Business Questions Answer about the Churning Customers are as follows:

1) How many of our repeated customers those who have previously demonstrated loyalty by purchasing more than once have not returned in over 180 days and can now be classified as churned, & what percentage of our 1,051 repeated customers does this represent?

2) How much revenue did these churned customers generate before they stopped visiting and what is the total revenue the business is at risk of losing permanently?

3) Which age group has the highest number of churned customers and does a particular generation show a stronger tendency to disengage from the store over time?

4) Which churned customers were among our top revenue generators and who should be prioritized first for re-engagement outreach?

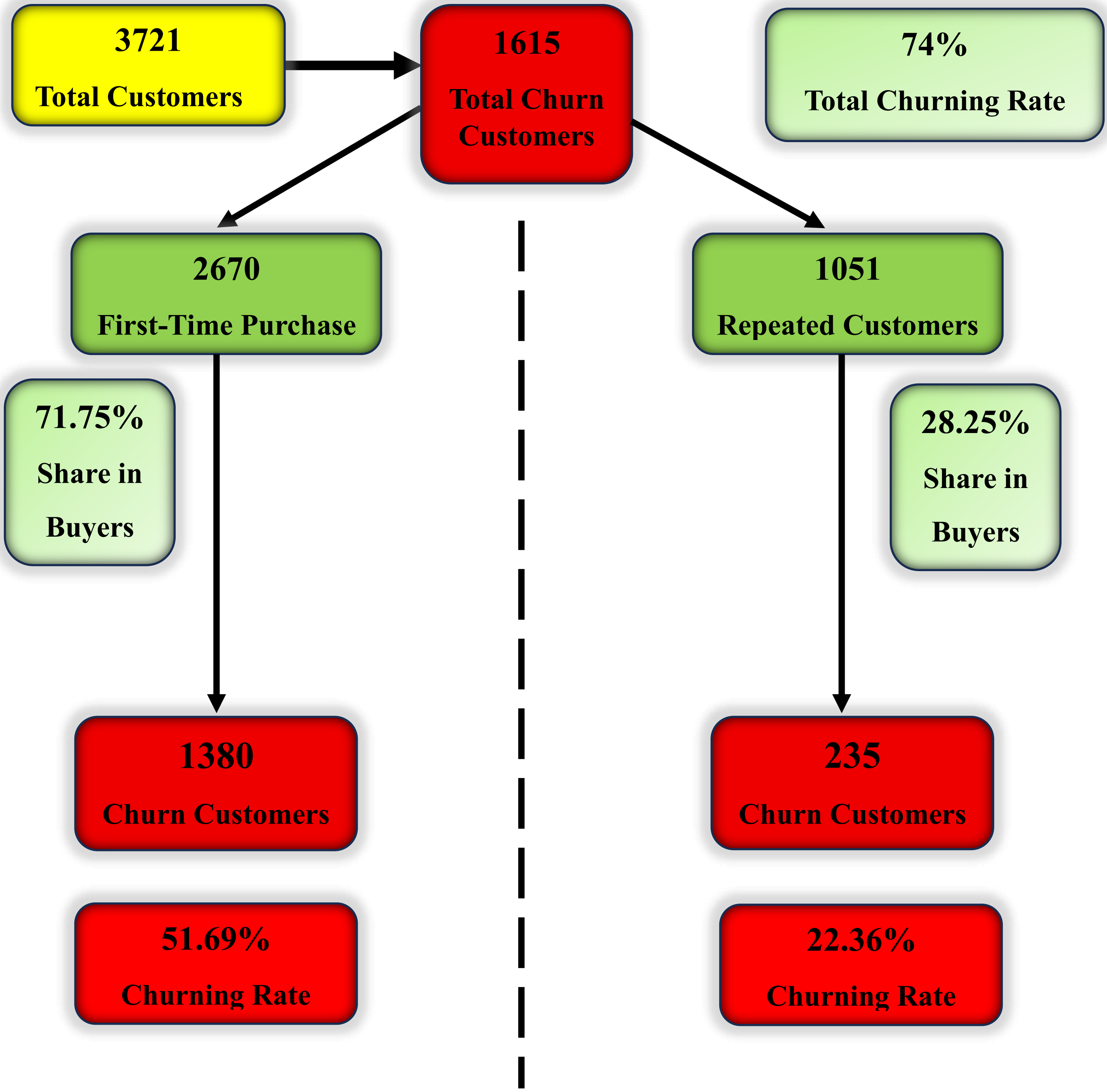
5) How many repeated customers are currently in the "At Risk" zone having not visited in 90 to 180 days and can still be recovered before they churn completely?

6) What is the average number of days since the last purchase across all churned customers and how long ago did the business begin losing these customers?

7) Which staff members have the highest number of churned customers under their assignment and does staff performance play a role in customer retention?

8) Which vision type is most commonly associated with churned customers and does the type of eye condition influence long term loyalty?

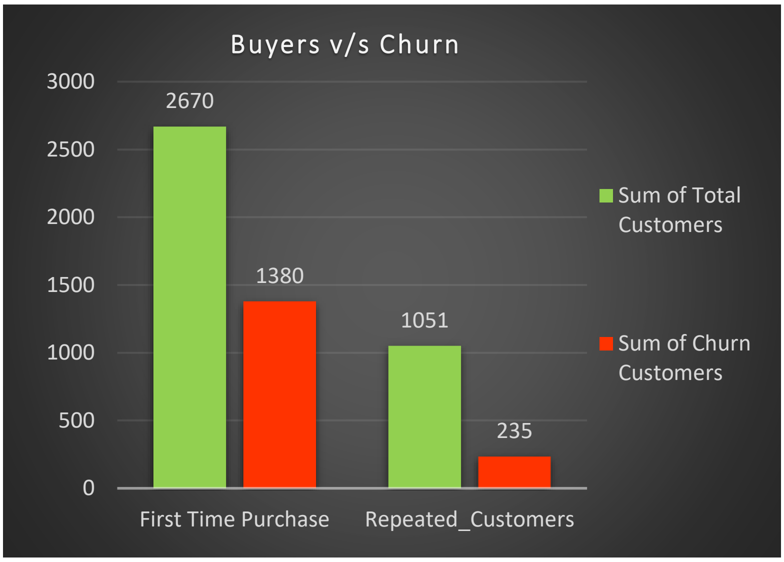
Dashboard of Churning Customers (Churn Analysis)

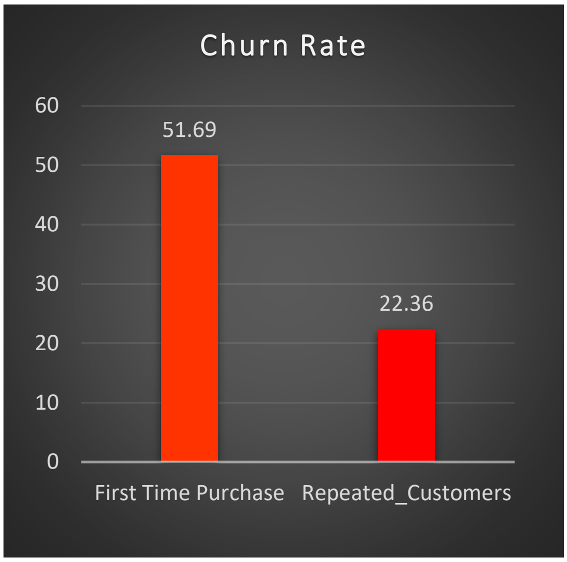


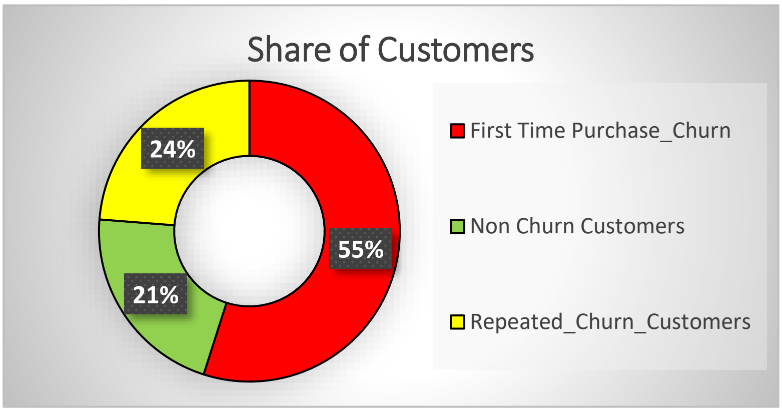
SQL QUERY OUTPUT:

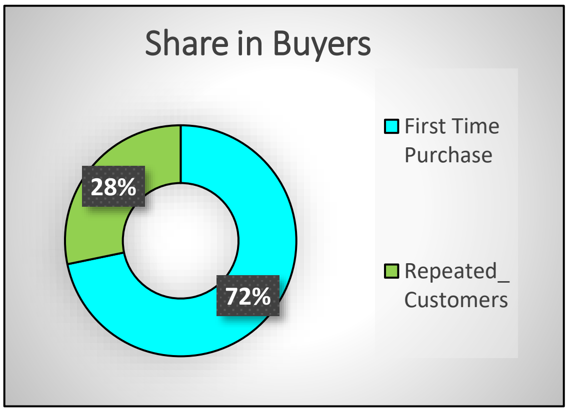
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

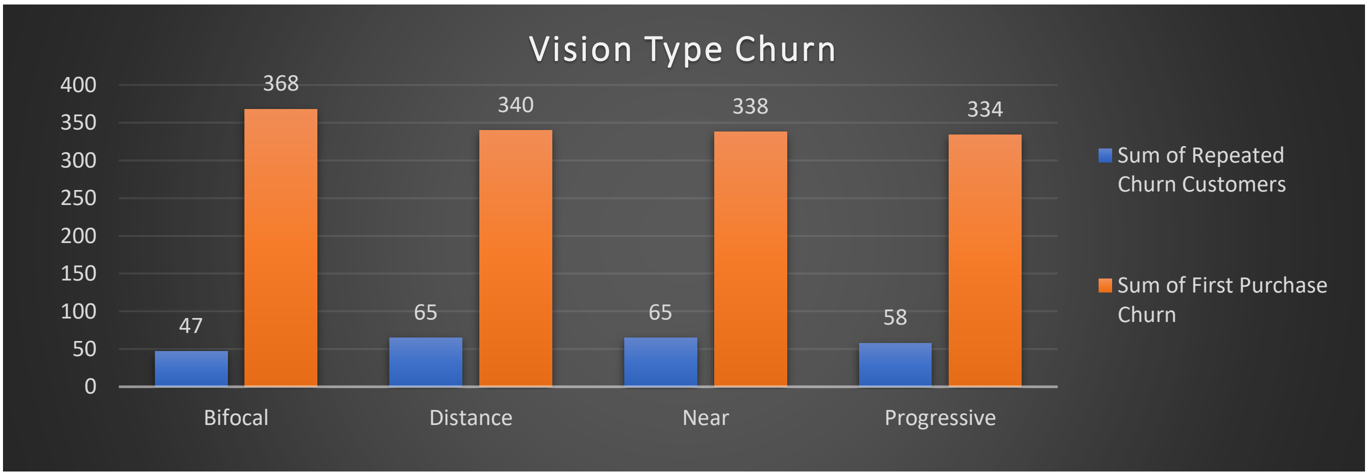
Churn_Customer_Type	Total_Customers	Share_in_Buyers	Churn_Customers	Churn_Rate
First Time Purchase	2670	71.75	1380	51.69
Repeated_Customers	1051	28.25	235	22.36
Total	3721	100.00	1615	74.05

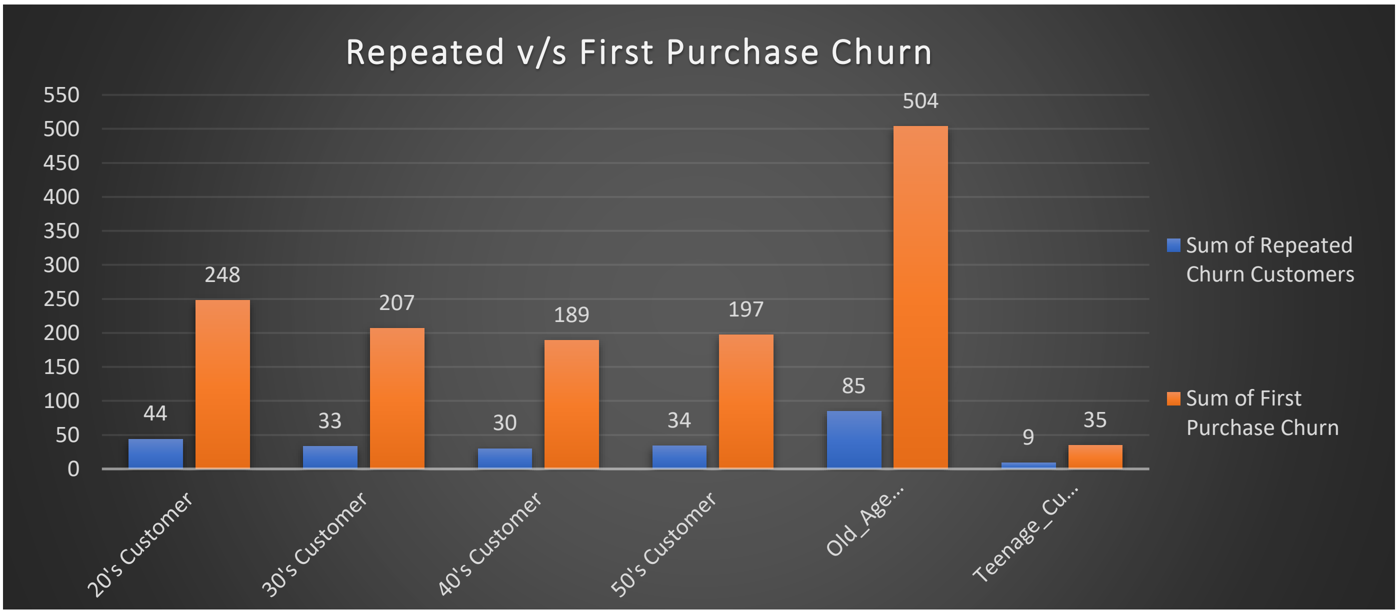

Buyers v/s Churn
3000
2500
2000
1500
1000
500
0
2670
1380
1051
235
First Time Purchase
Repeated_Customers
Sum of Total Customers
Sum of Churn Customers

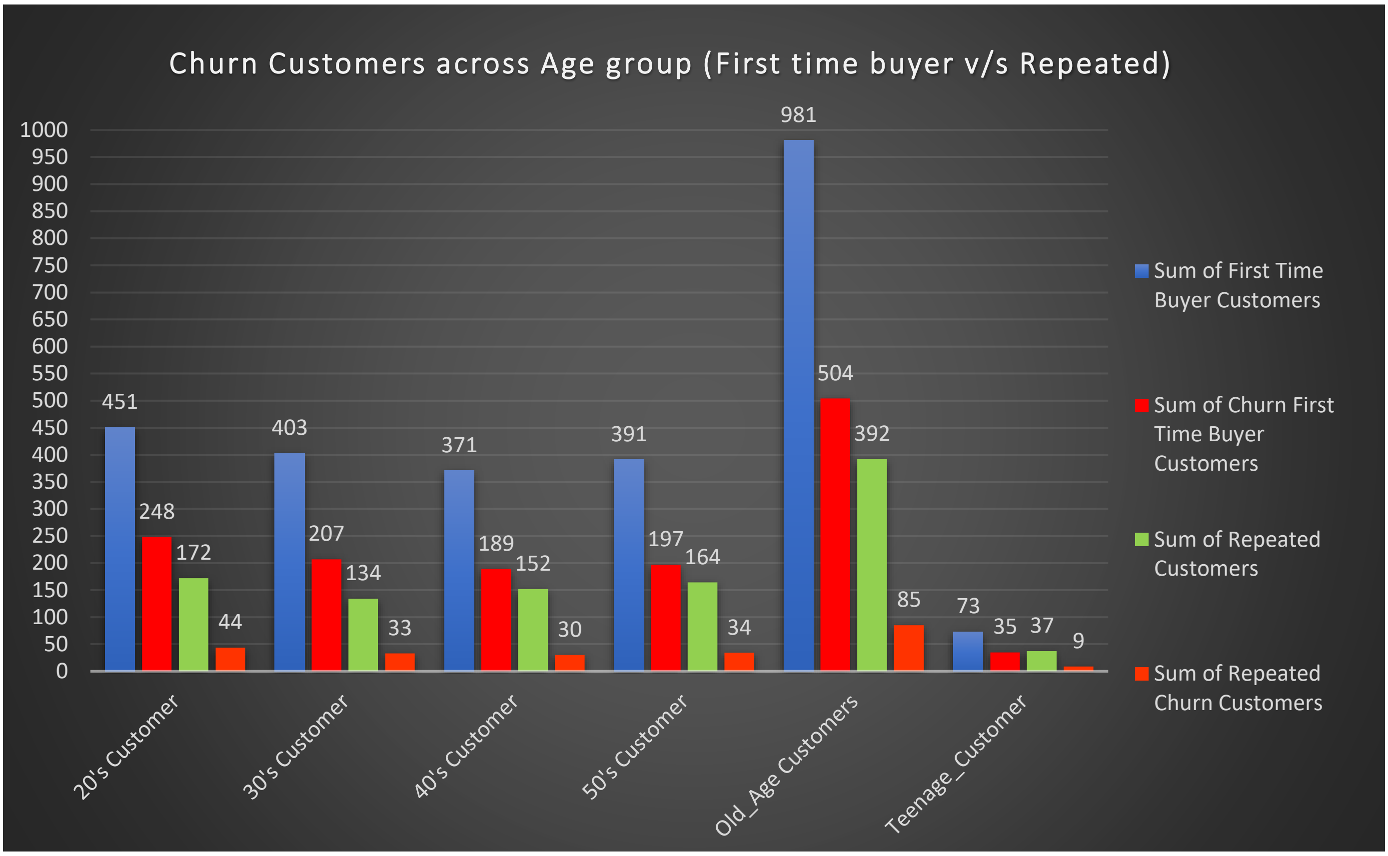

Churn Rate
60
50
40
30
20
10
0
51.69
22.36
First Time Purchase
Repeated_Customers

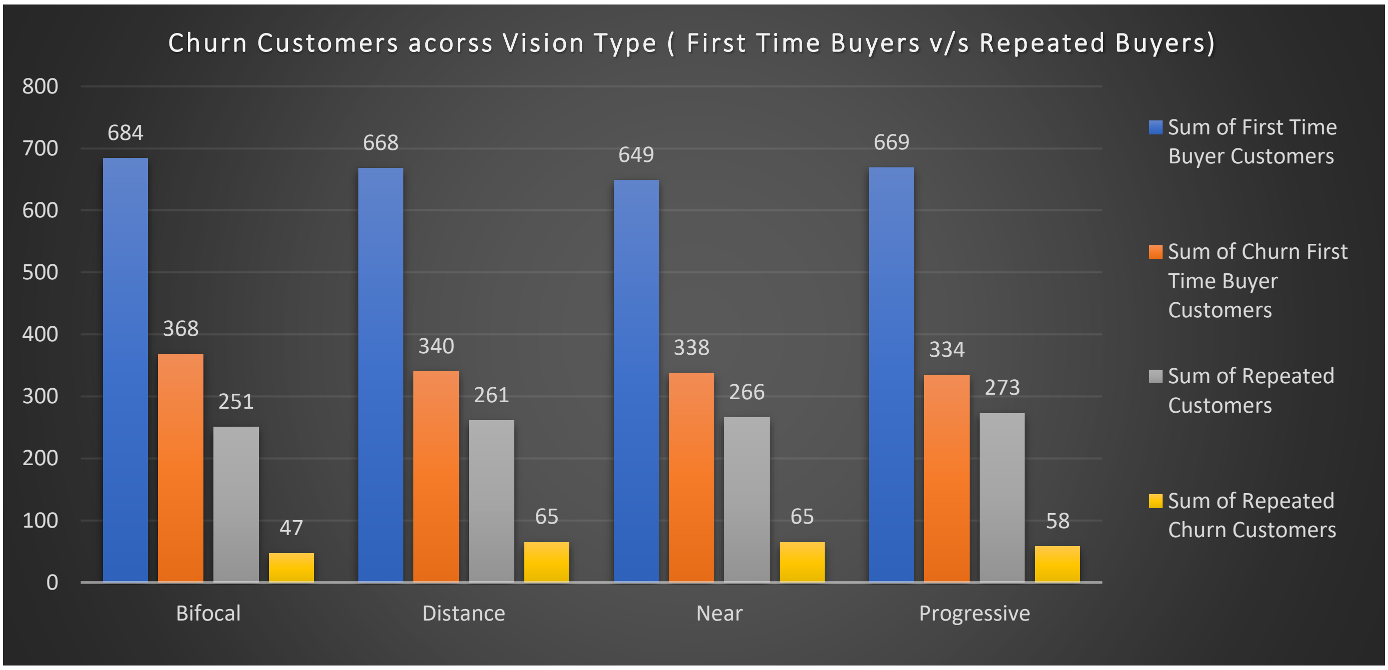

Share of Customers
24%
55%
21%
First Time Purchase_Churn
Non Churn Customers
Repeated_Churn_Customers


Share in Buyers
28%
72%
First Time Purchase
Repeated_ Customers


Vision Type Churn
400
350
300
250
200
150
100
50
0
47
368
65
340
65
338
58
334
Bifocal
Distance
Near
Progressive
Sum of Repeated Churn Customers
Sum of First Purchase Churn


Repeated v/s First Purchase Churn
550
500
450
400
350
300
250
200
150
100
50
0
44
248
33
207
30
189
34
197
85
504
9
35
20's Customer
30's Customer
40's Customer
50's Customer
Old_Age...
Teenage_Cu...
Sum of Repeated Churn Customers
Sum of First Purchase Churn





Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Age_group	First_time_buyers	Churn_Customers	Churn_Rate	repeated_customers	Churn_Customers	Churn_Rate
NULL	2670	1380	51.69	1051	235	22.36
Old_Age Customers	981	504	51.38	392	85	21.68
20's Customer	451	248	54.99	172	44	25.58
30's Customer	403	207	51.36	134	33	24.63
50's Customer	391	197	50.38	164	34	20.73
40's Customer	371	189	50.94	152	30	19.74
Teenage_Customer	73	35	47.95	37	9	24.32

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

vision_type	First_time_buyers	Churn_Customers	Churn_Rate	repeated_customers	Churn_Customers	Churn_Rate
NULL	2670	1380	51.69	1051	235	22.36
Bifocal	684	368	53.80	251	47	18.73
Progressive	669	334	49.93	273	58	21.25
Distance	668	340	50.90	261	65	24.90
Near	649	338	52.08	266	65	24.44

Churn Customer Analysis

Why this Analysis Matters

Customer churn is one of the most critical business metrics because acquiring a new customer is significantly more expensive than retaining an existing one. In this project, repeat customers have already been identified as one of the store's most valuable customer segments, contributing nearly half of the total business revenue despite representing only a small proportion of buyers.

Therefore, understanding which customers are leaving the business, why they are leaving, and which customers are at risk of churning enables management to take proactive actions before valuable customer relationships are permanently lost.

The objective of this analysis is to identify churn patterns across customer demographics, vision types, revenue contribution, customer value, and staff performance so that targeted retention strategies can be implemented.

Key Insight 1 – Most Churn Comes from First-Time Buyers

Out of 3,721 buyer customers, the analysis identifies 1,615 churned customers, resulting in an overall churn rate of 43.4% *(don't use the summed 74.05%; use 1615/3721 instead)*. Among these:

- 1,380 are First-Time Buyers.
- 235 are Repeated Customers.

This means approximately 85% of all churned customers are first-time buyers, while only 15% are repeated customers.

Business Insight:

The largest customer leakage occurs immediately after the customer's first purchase.

Many customers purchase once but never return, indicating that the business performs well in customer acquisition but has significant room for improvement in converting first-time buyers into loyal customers.

Strategic Recommendation

Management should strengthen the post-purchase customer journey by implementing:

- Follow-up calls or WhatsApp messages
- Customer satisfaction surveys
- Eye check-up reminder campaigns
- Personalized product recommendations
- Loyalty enrollment after the first purchase

The objective should be to convert more first-time buyers into repeat customers.

Key Insight 2 – Repeated Customers Churn Less but Their Loss is More Costly

Although repeated customers account for only 28.25% of total buyers (1,051 customers), only 235 of them have churned, resulting in a churn rate of 22.36%.

This churn rate is significantly lower than the 51.69% observed among first-time buyers.

However, repeated customer churn represents a much greater business risk.

Previous analysis showed that:

- Repeated customers represent only 22% of buyers
- Yet they generate 47% of total business revenue

Therefore, every repeated customer lost represents the loss of multiple future purchases rather than a single transaction.

Business Insight

Repeated customers demonstrate substantially higher loyalty than first-time buyers.

However, because of their high lifetime value, losing even a small proportion of repeated customers can significantly reduce long-term profitability.

Strategic Recommendation

Repeated customers should receive dedicated retention initiatives, including:

- Personalized follow-ups
- Annual eye examination reminders
- Prescription renewal campaigns
- VIP loyalty benefits
- Relationship-based customer service

Protecting this customer segment should remain one of the business's highest strategic priorities.

Key Insight 3 – Senior Customers Record the Highest Churn Volume Across Both Customer Types

When churn is analysed across age groups, Senior Customers (60+) consistently record the highest number of churned customers in both categories.

This finding aligns with previous demographic analysis, where senior customers also represented the largest customer base.

Business Insight

The higher churn volume among senior customers is primarily driven by their larger population rather than poor loyalty.

Despite having the highest number of churned customers, senior customers also remain:

- The largest revenue contributors
- The largest repeat customer segment
- The largest referral source

Strategic Recommendation

Rather than treating senior customer churn as a weakness, management should view it as the largest retention opportunity.

Improving retention within this already large customer base can produce significant long-term revenue growth.

Key Insight 4 – First-Time Buyer Churn is Consistently High Across Every Age Group

Across every age segment, first-time buyer churn is substantially higher than repeated customer churn.

Business Insight

This consistent gap indicates that the primary business challenge is not retaining loyal customers it is converting first-time buyers into repeat customers.

Once customers return for a second purchase, their likelihood of remaining engaged with the business increases considerably.

Strategic Recommendation

The business should redesign its first-time customer experience by focusing on:

- Better onboarding
- After-sales engagement
- Educational content
- Follow-up reminders
- Building trust immediately after the first purchase

Key Insight 5 – Churn is Evenly Distributed Across Vision Types

Churn analysis across vision types shows relatively similar customer counts for:

- Bifocal
- Distance
- Near Vision
- Progressive

No single vision category dominates customer churn.

This indicates that churn is not driven by a particular vision condition but is instead influenced by overall customer relationship management.

Business Insight

The business serves a well-diversified vision portfolio without excessive dependence on any single vision segment.

Key Insight 6 – Progressive Customers Require Special Retention Attention

Although Progressive Vision customers do not record the highest churn volume, they deserve the greatest strategic attention.

Previous analysis showed that Progressive lenses generate:

- The highest revenue
- Higher profit margins
- Greater customization requirements
- Strong alignment with senior customers

Because Progressive dispensing requires precise eye testing, accurate PD measurement, expert fitting, and skilled consultation, even one poor customer experience can permanently lose a high-value customer.

Strategic Recommendation

The business should continue investing in:

- Experienced optometrists
- Precision diagnostic equipment
- Staff training
- Accurate lens fitting
- Premium customer consultation

Protecting Progressive customers will preserve one of the store's highest-margin revenue streams.

Overall Business Conclusion

The churn analysis reveals two distinct business challenges:

First-Time Buyer Churn

- Represents the largest source of customer loss.
- Requires improvements in post-purchase engagement and customer conversion.
- Every additional repeat customer created today strengthens future revenue.

Repeated Customer Churn

- Occurs at a much lower rate.
- However, it carries disproportionately high financial impact because repeated customers generate 47% of total business revenue.
- Retaining these customers should remain one of the business's highest strategic priorities.

Executive Comparison (Highlight Box)

Critical Business Finding

- 71.75% of buyers are first-time purchasers, yet 51.69% of them churn after their initial purchase.
- 28.25% of buyers become repeated customers, and only 22.36% of them churn.
- Repeated customers contribute 47% of total business revenue, making them nearly twice as valuable as their population share suggests.

Management implication:

1. The business must pursue a dual strategy reduce first-time buyer churn to create more repeat customers, and aggressively retain existing repeat customers because they are the primary drivers of long-term profitability.
2. Protect Senior Customers, who remain the largest source of recurring revenue and referrals.
3. Strengthen Progressive Lens Customer Experience, where both revenue potential and service sensitivity are highest.

The analysis clearly shows that sustainable business growth will depend not only on acquiring new customers, but on successfully converting first-time buyers into repeat customers and protecting the high-value customers who already generate the majority of long-term revenue.

Additional Churn Analysis

Beyond demographic and vision-type analysis, additional SQL analyses were performed to identify customers requiring immediate management attention.

These analyses include:

1. Churned Repeat Customers

- Number of previously loyal customers who have now churned
- Percentage of churn among the 1,051 repeat customers

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2. Revenue at Risk

- Total historical revenue generated by churned customers
- Estimated revenue permanently at risk if these customers are not recovered

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3. High-Value Churn Customers

- Identification of churned customers who previously generated the highest revenue
- Priority list for personalized re-engagement campaigns

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4. Customers Currently "At Risk"

- Customers inactive between 90–180 days
- Customers who can still be recovered before becoming fully churned

This provides management with an early-warning system rather than waiting until customers are permanently lost.

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5. Average Customer Inactivity

- Average number of days since the last purchase across all churned customers

This helps management understand how long customers remain inactive before the relationship is effectively lost.

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6. Staff-wise Churn Analysis

- Number of churned customers assigned to each staff member

This analysis helps determine whether customer retention differences are associated with sales staff performance, follow-up quality, or relationship management practices.

Overall Business Conclusion

The churn analysis indicates that the business currently maintains a relatively healthy repeated customer base, with an overall churn rate of **22.36%**, which is comparatively very high, since repeated customers based is the one which has contributed 47% revenue of the business, hence losing this 22% of the customers is like something losing a very high portion of the revenue. However, the analysis highlights clear opportunities to improve long-term customer lifetime value through targeted retention initiatives.

The most important strategic priorities are:

1. **Protect Senior Customers**, as they generate the highest revenue, repeat purchases, referrals, and lifetime value.
2. **Closely monitor Progressive Lens customers**, where service quality directly influences customer retention and profitability.
3. **Implement proactive re-engagement campaigns** for customers in the 90–180 day "At Risk" window before they become permanently inactive.
4. **Prioritize recovery of high-value churned customers**, as recovering an existing profitable customer is significantly more cost-effective than acquiring a new one.
5. **Increase engagement with Teenage Customers** through fashion-oriented marketing and digital channels to reduce their relatively high churn rate.

Overall, the findings reinforce a key business principle identified throughout this project: sustainable growth depends not only on attracting new customers, but on retaining valuable existing customers, preventing churn, and maximizing customer lifetime value.

Additional Churn Analysis:

1) Extracting the Churn Customers details & % in repeated customers

1) How many of our repeated customers those who have previously demonstrated loyalty by purchasing more than once have not returned in over 180 days and can now be classified as churned, and what percentage of our 1,051 repeated customers does this represent?

```sql
WITH churning_customers AS (
SELECT
      rc.customer_id,
      DATE(MAX(i.invoice_date)) AS Last_Visit_Date,
      DATEDIFF(CURDATE(), MAX(i.invoice_date))
      AS Days_from_last_visit
FROM repeated_customers as rc
INNER JOIN invoices as i
ON rc.customer_id=i.customer_id
GROUP BY rc.customer_id
HAVING Days_from_last_visit > 180
ORDER BY Days_from_last_visit DESC
),
churning_customers_details AS (
SELECT
       cc.customer_id, c.customer_full_name,
       c.mobile, c.age, cc.Last_Visit_Date,
       cc.Days_from_last_visit
FROM churning_customers as cc
INNER JOIN customers_details as c
ON cc.customer_id=c.customer_id
)
SELECT
       ccd.customer_id, ccd.customer_full_name,
       ccd.mobile, ccd.age, ccd.Last_Visit_Date,
       ccd.Days_from_last_visit
FROM churning_customers_details as ccd;
```

```sql
WITH churning_customers AS (
SELECT
      rc.customer_id,
      DATEDIFF( (SELECT MAX(invoice_date) FROM invoices ),
      MAX(i.invoice_date)) AS Number_of_days_from_last_visits
FROM repeated_customers as rc
INNER JOIN invoices as i
ON rc.customer_id=i.customer_id
GROUP BY rc.customer_id
HAVING Number_of_days_from_last_visits > 180
ORDER BY Number_of_days_from_last_visits DESC
)
SELECT
      COUNT( DISTINCT rcc.customer_id)
      AS Total_Repeated_Customers,
      COUNT( DISTINCT CASE WHEN cc.customer_id IS NOT NULL THEN cc.customer_id END)
      AS Churning_repeated_customers,
      ROUND( COUNT( DISTINCT CASE WHEN cc.customer_id IS NOT NULL THEN cc.customer_id END)* 100 /
      COUNT( DISTINCT rcc.customer_id), 2) AS Percentage_of_Churn_Customers
FROM repeated_customers as rcc
LEFT JOIN churning_customers as cc
ON cc.customer_id=rcc.customer_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	customer_full_name	mobile	age	Last_Visit_Date	Days_from_last_visit
7325	Saumya Karpe	0568512814	49	2025-01-26	520
3293	Vedant Saxena	5480227836	22	2025-02-06	509
4169	Hitesh Korpal	1624156656	42	2025-02-06	509
1166	Quincy Bala	7314923403	51	2025-02-07	508
3944	Eesha Banik	2033560086	38	2025-02-10	505
3986	Maanas Brar	1643364188	22	2025-02-11	504
6266	Mugdha Datta	1866781344	34	2025-02-13	502
1809	Leena Dugal	9621641944	76	2025-02-16	499
1111	Krish Bail	0183544535	60	2025-02-19	496
6616	Kala Vora	2127629587	45	2025-02-19	496
1629	Niharika Nath	6059400213	75	2025-02-20	495
6708	Caleb Konda	9203467054	65	2025-02-22	493
7005	Chasmum Wali	0166012578	53	2025-02-22	493
2325	Anvi Tara	3461214400	30	2025-02-24	491
7329	Indrajit Devan	1076196510	75	2025-02-25	490
4664	Jagrati Narula	7546971956	51	2025-02-26	489
344	Kiaan Madan	5929300620	23	2025-02-27	488
2764	Chakradev Das	3541300349	62	2025-02-27	488
4500	Dominic Kothari	1735730331	53	2025-02-27	488
1867	Madhav Palan	2414597827	30	2025-03-01	486
7522	Amruta Minhas	7940878896	33	2025-03-07	480

Result 6 | Result 7

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Total_Repeated_Customers	Churning_repeated_customers	Percentage_of_Churn_Customers
1051	235	22.36

2) Total revenue the business is at risk of losing permanently

2) How much revenue did these churned customers generate before they stopped visiting and what is the total revenue the business is at risk of losing permanently?

```sql
SELECT
      SUM(i.final_bill_amount)
      AS Revenue_from_Churn_Customers,
      ROUND( SUM(i.final_bill_amount)*100/
      (SELECT SUM(final_bill_amount) FROM invoices ), 2)
      AS Percentage_of_revenue_at_lost
FROM churn_customers as cc
INNER JOIN invoices as i
ON cc.customer_id=i.customer_id;
```

Result Grid | Filter Rows: | Export: | Wr

Revenue_from_Churn_Customers	Percentage_of_revenue_at_lost
4779880.00	10.21

3) Count of Churned Customers in each Age Group

3) Which age group has the highest number of churned customers and does a particular generation show a stronger tendency to disengage from the store over time?

```sql
SELECT
      CASE
      WHEN cc.age < 21 THEN "Teenage_Customers"
      WHEN cc.age BETWEEN 21 AND 30 THEN "20's Customers"
      WHEN cc.age BETWEEN 31 AND 40 THEN "30's Customers"
      WHEN cc.age BETWEEN 41 AND 50 THEN "40's Customers"
      WHEN cc.age BETWEEN 51 AND 60 THEN "50's Customers"
      WHEN cc.age > 60 THEN "Old_Age_Customers"
   END AS Age_Group,
   COUNT(DISTINCT cc.customer_id) AS COUNT_Customers
FROM churn_customers as cc
GROUP BY Age_Group WITH ROLLUP
ORDER BY COUNT_Customers DESC;
```

Result Grid | Filter Rows:

Age_Group	COUNT_Customers
NULL	235
Old_Age_Customers	85
20's Customers	36
50's Customers	34
30's Customers	33
40's Customers	30
Teenage_Customers	17

4) Top Revenue generating Churn Customers

4) Which churned customers were among our top revenue generators and who should be prioritized first for re-engagement outreach?

```sql
SELECT
      cc.customer_id,
      cc.customer_full_name,
      cc.mobile, cc.age,
      SUM(i.final_bill_amount)
      AS Revenue
FROM churn_customers as cc
INNER JOIN invoices as i
ON cc.customer_id=i.customer_id
GROUP BY cc.customer_id, cc.customer_full_name,
         cc.mobile, cc.age
ORDER BY Revenue DESC;
```

Result Grid | Filter Rows: | Export: | Wrap

customer_id	customer_full_name	mobile	age	Revenue
3293	Vedant Saxena	5480227836	22	47956.00
2141	Tarak Garg	1670533401	23	42550.00
7111	Vritti Acharya	8450100202	35	42134.00
5487	Unni Borde	3865031997	82	41777.00
5465	Hemangini Doshi	4075937519	18	41068.00
982	Ayaan Mann	9018662089	69	39833.00
3702	Meghana Pall	5368327615	29	39563.00
478	Jhalak Tandon	5864182497	23	38887.00
4074	Isha Menon	8924518051	18	38518.00
4500	Dominic Kothari	1735730331	53	38430.00
5666	Omisha Keer	6298702456	50	38315.00
344	Kiaan Madan	5929300620	23	37826.00
2809	Orinder Sankaran	4379917915	55	37719.00

Result 11 ×

5) Customers "AT RISK Zone" of getting Churn (90 to 180 Days)

5) How many repeated customers are currently in the "At Risk" zone having not visited in 90 to 180 days and can still be recovered before they churn completely?

```sql
WITH at_risk_customers AS (
SELECT
       i.customer_id,
       DATEDIFF( (SELECT MAX(invoice_date) FROM invoices),
       MAX(i.invoice_date) )
       AS Number_of_days_from_last_visit
FROM invoices as i
GROUP BY i.customer_id
HAVING Number_of_days_from_last_visit BETWEEN 90 AND 180
ORDER BY Number_of_days_from_last_visit DESC
),
at_risk_customers_details AS (
SELECT
      arc.customer_id, c.customer_full_name,
      c.mobile, c.age,
      arc.Number_of_days_from_last_visit
FROM customers_details as c
INNER JOIN at_risk_customers as arc
ON arc.customer_id=c.customer_id
)
SELECT arcd.customer_id, arcd.customer_full_name,
       arcd.mobile, arcd.age,
       arcd.Number_of_days_from_last_visit
FROM at_risk_customers_details as arcd;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	customer_full_name	mobile	age	Number_of_days_from_last_visit
205	Leela Ghosh	4666327309	25	180
3494	Nikita Ravel	9273115352	76	180
4507	Christopher Prasad	6018976926	84	180
6824	Arunima Atwal	9524762839	31	180
7730	Girindra Mahal	0046592511	32	180
1237	Chandran Magar	2030964277	68	179
1777	Meera Kannan	8045864144	65	179
2393	Saksham Wadhwa	6016130067	41	179
2815	Triya Gulati	7459102014	29	179
4493	Advik Prasad	4558652297	72	179
7466	Caleb Barman	4955725308	66	179
479	Yashica Shah	6103731956	34	178
1833	Arunima Sampath	6223898537	41	178
2862	Oni Palan	1098790386	27	178
3287	Jonathan Raghavan	5385009733	52	178
4895	Jackson Kar	5021228343	83	178
4899	Darsh Mitra	5060164446	59	178
4935	Chanchal Jhaveri	7569313495	76	178

Result 13 ×

6) Average Days of all the Churn Customers from last Visit

6) What is the average number of days since the last purchase across all churned customers and how long ago did the business begin losing these customers?

```sql
SELECT
	COUNT(DISTINCT cc.customer_id)
	AS Total_Churned_Customers,
	AVG(cc.Number_of_days_from_last_visit)
	AS Average_days_since_last_purchase,
	MAX(cc.Number_of_days_from_last_visit)
	AS Longest_Absence,
	MIN(cc.Number_of_days_from_last_visit)
	AS Shortest_Absence,
	MIN(cc.last_visited_date)
	AS Earliest_Churn_date,
	MAX(cc.last_visited_date)
	AS Most_recent_Churn_date
FROM churn_customers as cc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Total_Churned_Customers	Average_days_since_last_purchase	Longest_Absence	Shortest_Absence	Earliest_Churn_date	Most_recent_Churn_date
235	239.6128	350	181	2025-01-26	2025-07-14

7) Staff with the Highest number of Churn Customers

7) Which staff members have the highest number of churned customers under their assignment and does staff performance play a role in customer retention?

```sql
WITH staff_revenue AS (
SELECT
	i.staff_id, s.staff_full_name,
	SUM(i.final_bill_amount) AS staff_revenue
FROM invoices as i
INNER JOIN staff as s
ON i.staff_id=s.staff_id
GROUP BY i.staff_id, s.staff_full_name
ORDER BY staff_revenue DESC
)
SELECT
	cc.assigned_staff_id, sr.staff_full_name,
	sr.staff_revenue,
	COUNT(DISTINCT cc.customer_id)
	AS COUNT_Churn_Customers
FROM churn_customers as cc
INNER JOIN staff_revenue as sr
ON sr.staff_id=cc.assigned_staff_id
GROUP BY cc.assigned_staff_id, sr.staff_full_name,
	  sr.staff_revenue
ORDER BY COUNT_Churn_Customers DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

assigned_staff_id	staff_full_name	staff_revenue	COUNT_Churn_Customers
12	Naksh Gandhi	2151305.00	16
16	Pratyush Kata	2401868.00	15
19	Anita Vala	2146680.00	15
1	Neel Varty	2398491.00	14
5	Dhriti Mohan	2478569.00	14
20	Utkarsh Hayre	2500938.00	14
3	Luke Kata	2411549.00	13
9	Neha Lalla	2197589.00	13
13	Indali Keer	2323801.00	13
2	Anjali Amble	2114982.00	12
4	Vinaya Lalla	2343254.00	12
15	Netra Lal	2471046.00	12
14	Gautam Choud...	2682251.00	11
17	Anmol Gole	2549710.00	11
6	Sai Keer	2373953.00	10
8	Mohini Srinivasan	2288597.00	10
11	Tara Swamy	2479416.00	9
7	Robert Sawhney	2325696.00	8
10	Neelima Sibal	1836366.00	7
18	Owen Narayanan	2335352.00	6

8) Count of Churn Customers based on Vision Type

8) Which vision type is most commonly associated with churned customers and does the type of eye condition influence long term loyalty?

```
SELECT
       p.vision_type,
       COUNT(DISTINCT cc.customer_id)
       AS Churn_Customers
FROM churn_customers as cc
INNER JOIN prescriptions as p
ON cc.customer_id=p.customer_id
GROUP BY p.vision_type
ORDER BY Churn_Customers DESC;
```

Result Grid | Filter Rows:

vision_type	Churn_Customers
Near	65
Distance	65
Progressive	58
Bifocal	47

5.3) Staff Analysis

Business Questions Answer about the Churning Customers are as follows:

1) Which staff members contribute the most and least to overall store revenue, and is the business overly dependent on a small group of high-performing employees?

2) Which staff members are most effective at converting assigned customers into buyers, and who performs above or below the overall store conversion benchmark?

3) Which staff members are most successful at retaining customers and converting first-time buyers into repeat customers, indicating stronger long-term customer relationships?

OR

which staff members have the largest share of one-time customers, indicating weaker customer retention?

4) Do staff members maintain a balanced product sales portfolio, or are they overly dependent on a limited number of product categories that could expose the business to revenue concentration risk?

5) Which staff members generate the highest revenue per customer handled, demonstrating greater sales efficiency rather than simply serving a larger number of customers?

6) Considering revenue generation, customer conversion, customer loyalty, product-selling behaviour, and sales efficiency, who are the strongest and weakest performing staff members across the complete customer journey?

1) Evaluate Staff Performance based on the Revenue Contribution by Staff

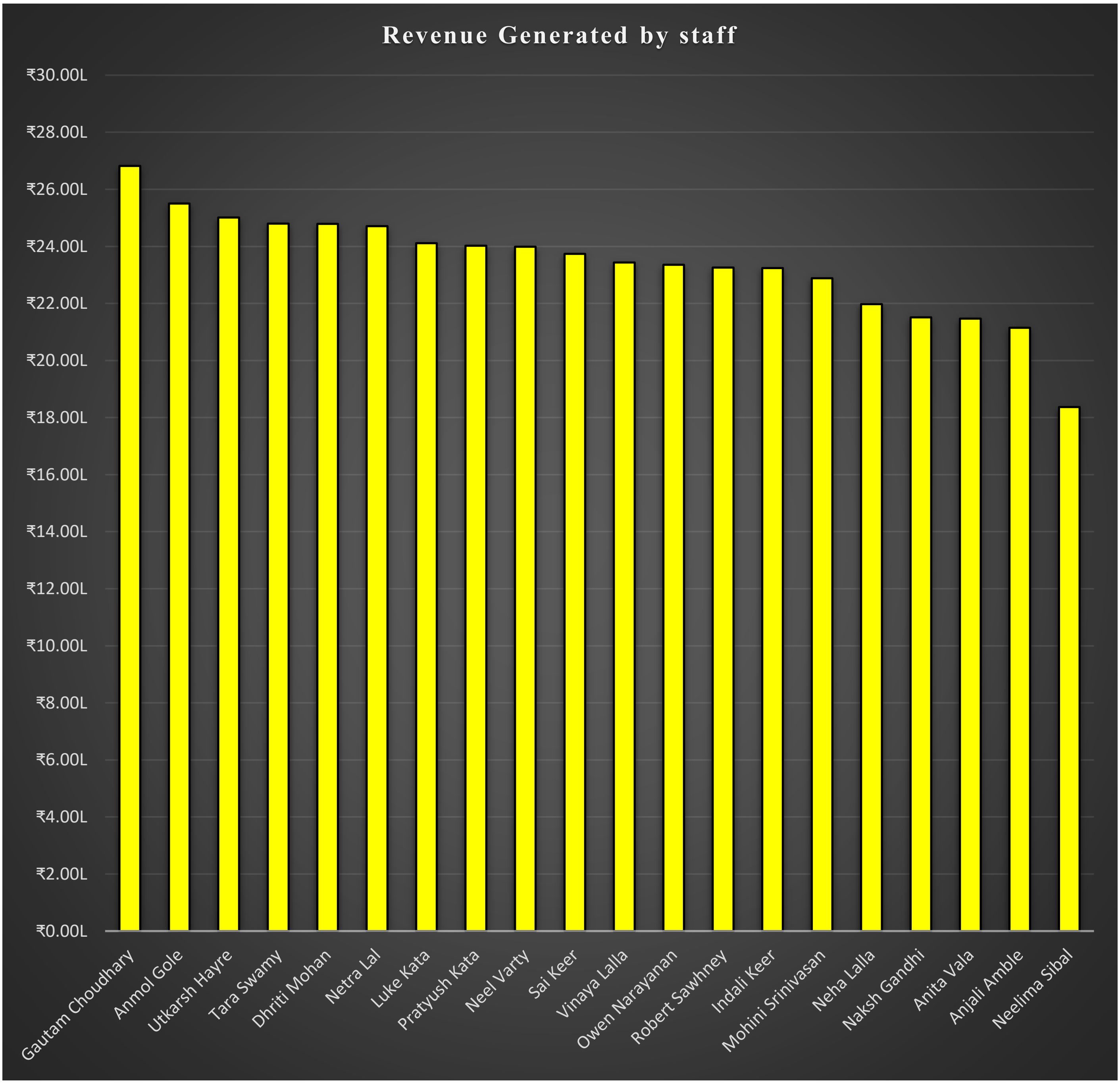
1) How wide is the performance gap between our highest and lowest revenue generating staff, and is the business overly dependent on a small group of top performers?

```sql
SELECT

     i.staff_id, s.staff_full_name,
     SUM(i.final_bill_amount) AS Revenue_by_Staff,
     ROUND(SUM(i.final_bill_amount) * 100 / 46811413.00, 2)
     AS Percentage_of_Revenue
FROM invoices as i
INNER JOIN staff as s
ON s.staff_id=i.staff_id
GROUP BY i.staff_id, s.staff_full_name
ORDER BY Revenue_by_Staff DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Con

staff_id	staff_full_name	Revenue_by_Staff	Percentage_of_Revenue
14	Gautam Choudhary	2682251.00	5.73
17	Anmol Gole	2549710.00	5.45
20	Utkarsh Hayre	2500938.00	5.34
11	Tara Swamy	2479416.00	5.30
5	Dhriti Mohan	2478569.00	5.29
15	Netra Lal	2471046.00	5.28
3	Luke Kata	2411549.00	5.15
16	Pratyush Kata	2401868.00	5.13
1	Neel Varty	2398491.00	5.12
6	Sai Keer	2373953.00	5.07
4	Vinaya Lalla	2343254.00	5.01
18	Owen Narayanan	2335352.00	4.99
7	Robert Sawhney	2325696.00	4.97
13	Indali Keer	2323801.00	4.96
8	Mohini Srinivasan	2288597.00	4.89
9	Neha Lalla	2197589.00	4.69
12	Naksh Gandhi	2151305.00	4.60
19	Anita Vala	2146680.00	4.59
2	Anjali Amble	2114982.00	4.52
10	Neelima Sibal	1836366.00	3.92



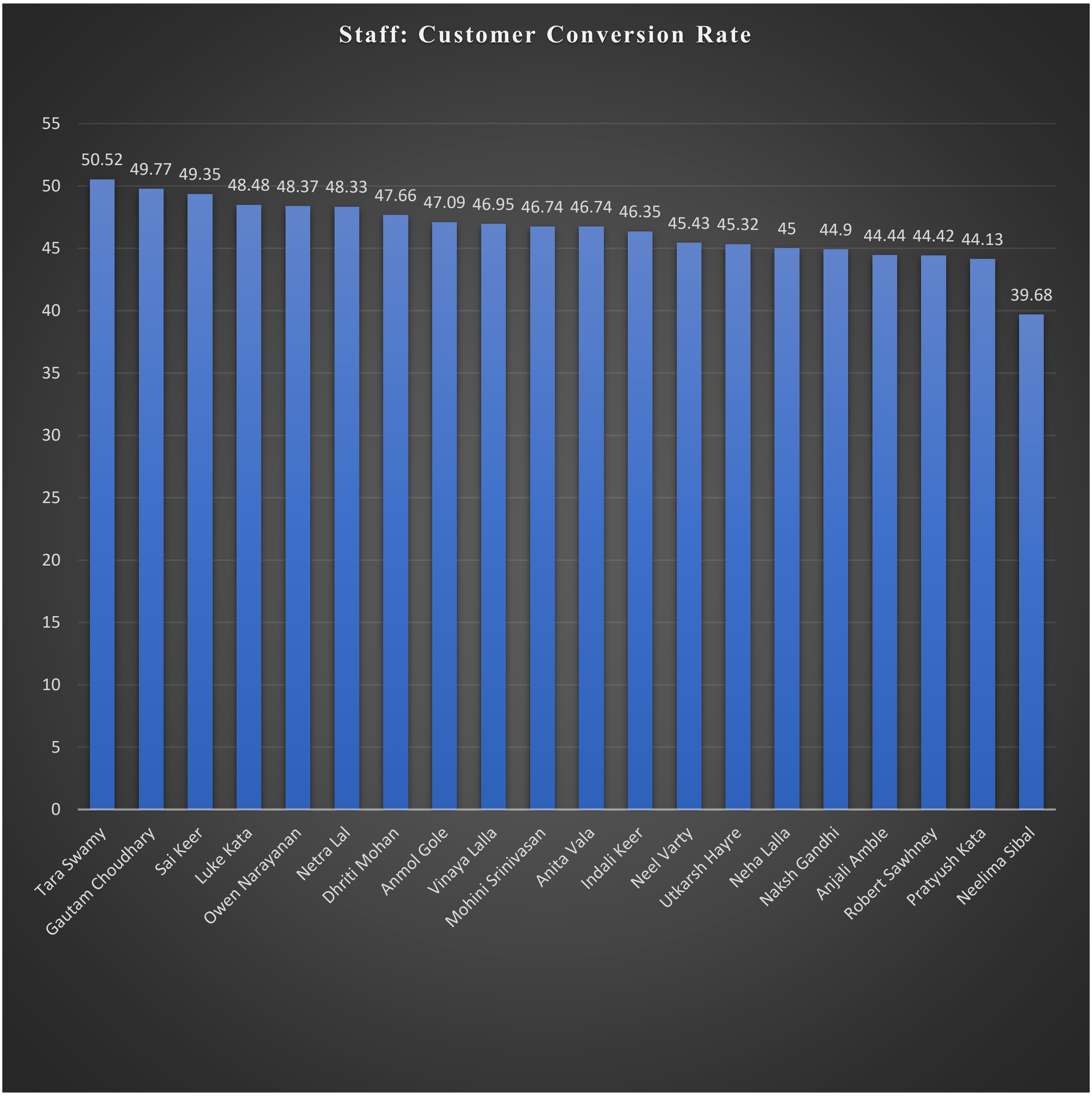
2) Staff Customer Conversion Rate

2) Which staff members deviate most significantly from the store-wide conversion rate benchmark, and are they flagged as above or below expectation?

```sql
SELECT
    c.assigned_staff_id, s.staff_full_name,
    COUNT(DISTINCT c.customer_id) AS Visited_Customers,
    COUNT(DISTINCT i.customer_id) AS Buyer_Customers,
    ROUND( COUNT(DISTINCT i.customer_id) * 100 /
    COUNT(DISTINCT c.customer_id), 2 )
    AS Customer_Conversion_Rate
FROM customers_details as c
INNER JOIN staff as s
ON c.assigned_staff_id=s.staff_id
LEFT JOIN invoices as i
ON c.customer_id=i.customer_id
GROUP BY c.assigned_staff_id, s.staff_full_name
ORDER BY Customer_Conversion_Rate DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

assigned_staff_id	staff_full_name	Visited_Customers	Buyer_Customers	Customer_Conversion_Rate
11	Tara Swamy	388	196	50.52
14	Gautam Choudhary	434	216	49.77
6	Sai Keer	387	191	49.35
3	Luke Kata	396	192	48.48
18	Owen Narayanan	399	193	48.37
15	Netra Lal	420	203	48.33
5	Dhriti Mohan	428	204	47.66
17	Anmol Gole	412	194	47.09
4	Vinaya Lalla	394	185	46.95
8	Mohini Srinivasan	383	179	46.74
19	Anita Vala	383	179	46.74
13	Indali Keer	397	184	46.35
1	Neel Varty	383	174	45.43
20	Utkarsh Hayre	406	184	45.32
9	Neha Lalla	400	180	45.00
12	Naksh Gandhi	392	176	44.90
2	Anjali Amble	396	176	44.44
7	Robert Sawhney	403	179	44.42
16	Pratyush Kata	426	188	44.13
10	Neelima Sibal	373	148	39.68



3) Identifying the Staff who builds long term relationship with customers

3) What percentage of each staff member's customer base eventually becomes a loyal, repeated customer identifying who builds long-term relationships versus one-time transactions?

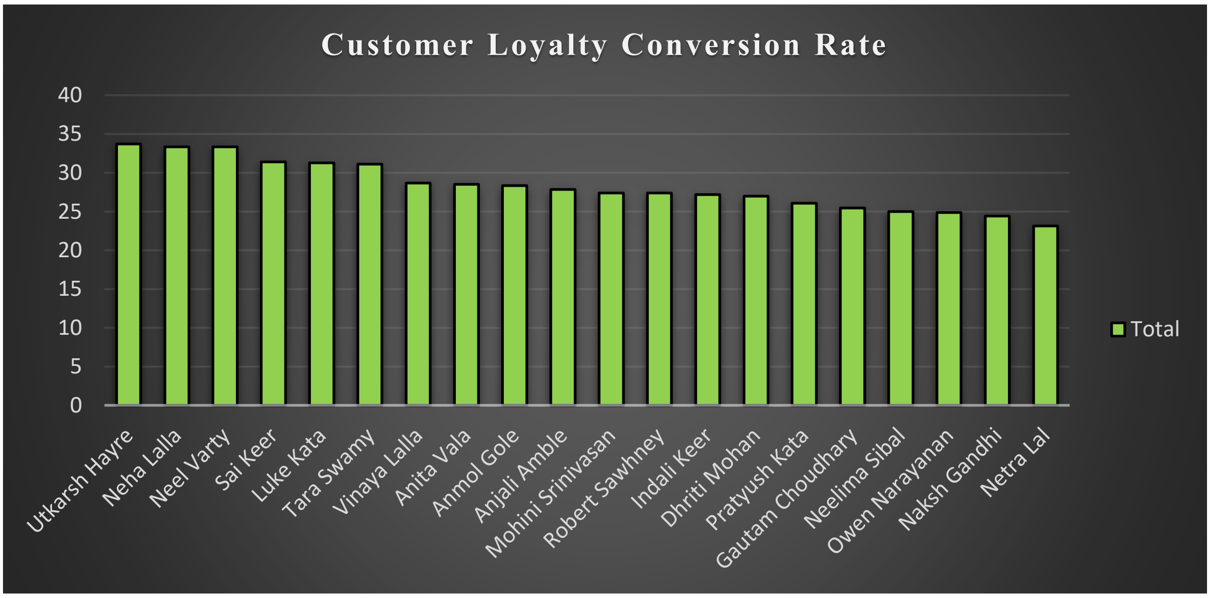
OR

which staff members have the largest share of one-time customers, indicating weaker customer

```sql
WITH staff_customers AS (
SELECT
      DISTINCT i.customer_id,
      i.staff_id, s.staff_full_name
FROM staff as s
INNER JOIN invoices as i
ON s.staff_id=i.staff_id
),
staff_customers_loyalty AS (
SELECT
      sc.staff_id, sc.staff_full_name,
      sc.customer_id,
      rc.customer_id AS repeated_customers_id
FROM staff_customers as sc
LEFT JOIN repeated_customers as rc
ON rc.customer_id=sc.customer_id
)
SELECT
      scl.staff_id, scl.staff_full_name,
      COUNT( DISTINCT scl.customer_id)
      AS Buyer_Customers,
      COUNT( DISTINCT scl.repeated_customers_id)
      AS Loyal_Customers_Base,
      ROUND( COUNT( DISTINCT scl.repeated_customers_id) * 100 /
      COUNT( DISTINCT scl.customer_id), 2) AS Loyalty_Conversion_Rate
FROM staff_customers_loyalty as scl
GROUP BY scl.staff_id, scl.staff_full_name
ORDER BY Loyalty_Conversion_Rate DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

staff_id	staff_full_name	Buyer_Customers	Repeated_Customers_Base	Non_Repeated_Customers	Loyalty_Conversion_Rate
20	Utkarsh Hayre	184	62	122	33.70
1	Neel Varty	174	58	116	33.33
9	Neha Lalla	180	60	120	33.33
6	Sai Keer	191	60	131	31.41
3	Luke Kata	192	60	132	31.25
11	Tara Swamy	196	61	135	31.12
4	Vinaya Lalla	185	53	132	28.65
19	Anita Vala	179	51	128	28.49
17	Anmol Gole	194	55	139	28.35
2	Anjali Amble	176	49	127	27.84
7	Robert Sawhney	179	49	130	27.37
8	Mohini Srinivasan	179	49	130	27.37
13	Indali Keer	184	50	134	27.17
5	Dhriti Mohan	204	55	149	26.96
16	Pratyush Kata	188	49	139	26.06
14	Gautam Choudhary	216	55	161	25.46
10	Neelima Sibal	148	37	111	25.00
18	Owen Narayanan	193	48	145	24.87
12	Naksh Gandhi	176	43	133	24.43
15	Netra Lal	203	47	156	23.15



4) Staff Products Sales Portfolio

4) Do staff members maintain a balanced product sales portfolio, or are they overly dependent on a limited number of product categories that could expose the business to revenue concentration risk?

```
SELECT
    s.staff_id, s.staff_full_name,
    SUM(CASE WHEN pd.product_id=1 THEN ii.quantity END) AS Frame,
    SUM(CASE WHEN pd.product_id=2 THEN ii.quantity END) AS Glass,
    SUM(CASE WHEN pd.product_id=3 THEN ii.quantity END) AS Sunglass,
    SUM(CASE WHEN pd.product_id=4 THEN ii.quantity END) AS Contact_Lens,
    SUM(CASE WHEN pd.product_id=6 THEN ii.quantity END) AS Lens_Solution
FROM staff as s
INNER JOIN customers_details as c
ON s.staff_id=assigned_staff_id
INNER JOIN invoices as i
ON c.customer_id=i.customer_id
INNER JOIN invoice_items as ii
ON i.invoice_id=ii.invoice_id
INNER JOIN products as pd
ON ii.product_id=pd.product_id
GROUP BY s.staff_id, s.staff_full_name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

staff_id	staff_full_name	Frame	Glass	Sunglass	Contact_Lens	Lens_Solution
1	Neel Varty	90	212	96	92	110
7	Robert Sawhney	97	220	87	98	97
18	Owen Narayanan	110	198	100	101	91
8	Mohini Srinivasan	80	172	107	108	103
15	Netra Lal	107	224	108	96	105
12	Naksh Gandhi	102	204	90	97	80
3	Luke Kata	117	216	93	94	116
2	Anjali Amble	90	178	109	92	104
5	Dhriti Mohan	121	174	101	114	123
13	Indali Keer	88	206	106	95	88
6	Sai Keer	98	198	116	106	95
20	Utkarsh Hayre	103	212	117	91	97
10	Neelima Sibal	79	172	76	68	79
16	Pratyush Kata	108	204	102	102	103
4	Vinaya Lalla	73	192	100	94	128
11	Tara Swamy	105	192	115	107	109
19	Anita Vala	97	176	91	99	97
17	Anmol Gole	110	222	100	106	109
14	Gautam Choudhary	113	220	123	112	115
9	Neha Lalla	96	188	103	96	89

---

5) Staff Sales efficiency

5) Which staff members generate the highest revenue per customer handled, demonstrating greater sales efficiency rather than simply serving a larger number of customers?

```
WITH customers_invoices AS (
SELECT
      i.customer_id, i.staff_id,
      s.staff_full_name,
      i.final_bill_amount
FROM invoices as i
INNER JOIN staff as s
ON i.staff_id=s.staff_id
)
SELECT
       ci.staff_id, ci.staff_full_name,
       SUM(ci.final_bill_amount) AS Revenue,
       ROUND( SUM(ci.final_bill_amount) /
       COUNT(DISTINCT ci.customer_id), 2)
       AS Revenue_per_Customers
FROM customers_invoices as ci
GROUP BY ci.staff_id, ci.staff_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

staff_id	staff_full_name	Revenue	Revenue_per_Customers
1	Neel Varty	2398491.00	13784.43
2	Anjali Amble	2114982.00	12016.94
3	Luke Kata	2411549.00	12560.15
4	Vinaya Lalla	2343254.00	12666.24
5	Dhriti Mohan	2478569.00	12149.85
6	Sai Keer	2373953.00	12429.07
7	Robert Sawhney	2325696.00	12992.72
8	Mohini Srinivasan	2288597.00	12785.46
9	Neha Lalla	2197589.00	12208.83
10	Neelima Sibal	1836366.00	12407.88
11	Tara Swamy	2479416.00	12650.08
12	Naksh Gandhi	2151305.00	12223.32
13	Indali Keer	2323801.00	12629.35
14	Gautam Choudhary	2682251.00	12417.83
15	Netra Lal	2471046.00	12172.64
16	Pratyush Kata	2401868.00	12775.89
17	Anmol Gole	2549710.00	13142.84
18	Owen Narayanan	2335352.00	12100.27
19	Anita Vala	2146680.00	11992.63
20	Utkarsh Hayre	2500938.00	13592.05

6) Staff Performance Summary

6) Considering revenue generation, customer conversion, customer loyalty, product-selling behaviour, and sales efficiency, who are the strongest and weakest performing staff members across the complete customer journey?

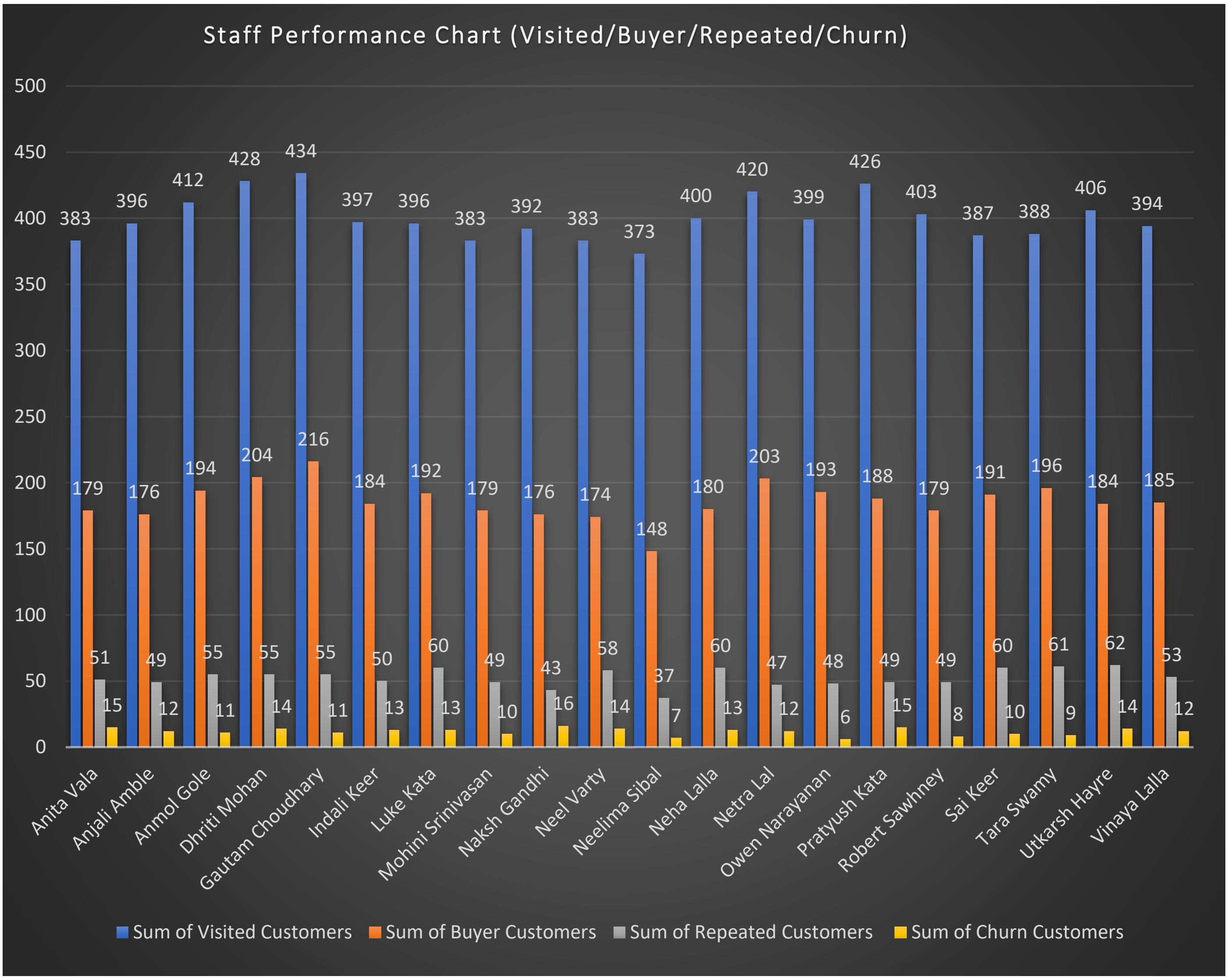
```sql
WITH customers AS (
SELECT
      DISTINCT c.customer_id AS all_customers,
      c.assigned_staff_id
FROM customers_details as c
),
customers_staff AS (
SELECT
      cc.all_customers, cc.assigned_staff_id,
      s.staff_full_name
FROM staff as s
INNER JOIN customers as cc
ON cc.assigned_staff_id=s.staff_id
),
customers_invoices AS (
SELECT
      DISTINCT i.customer_id AS buyer_customers
FROM invoices as i
),
```

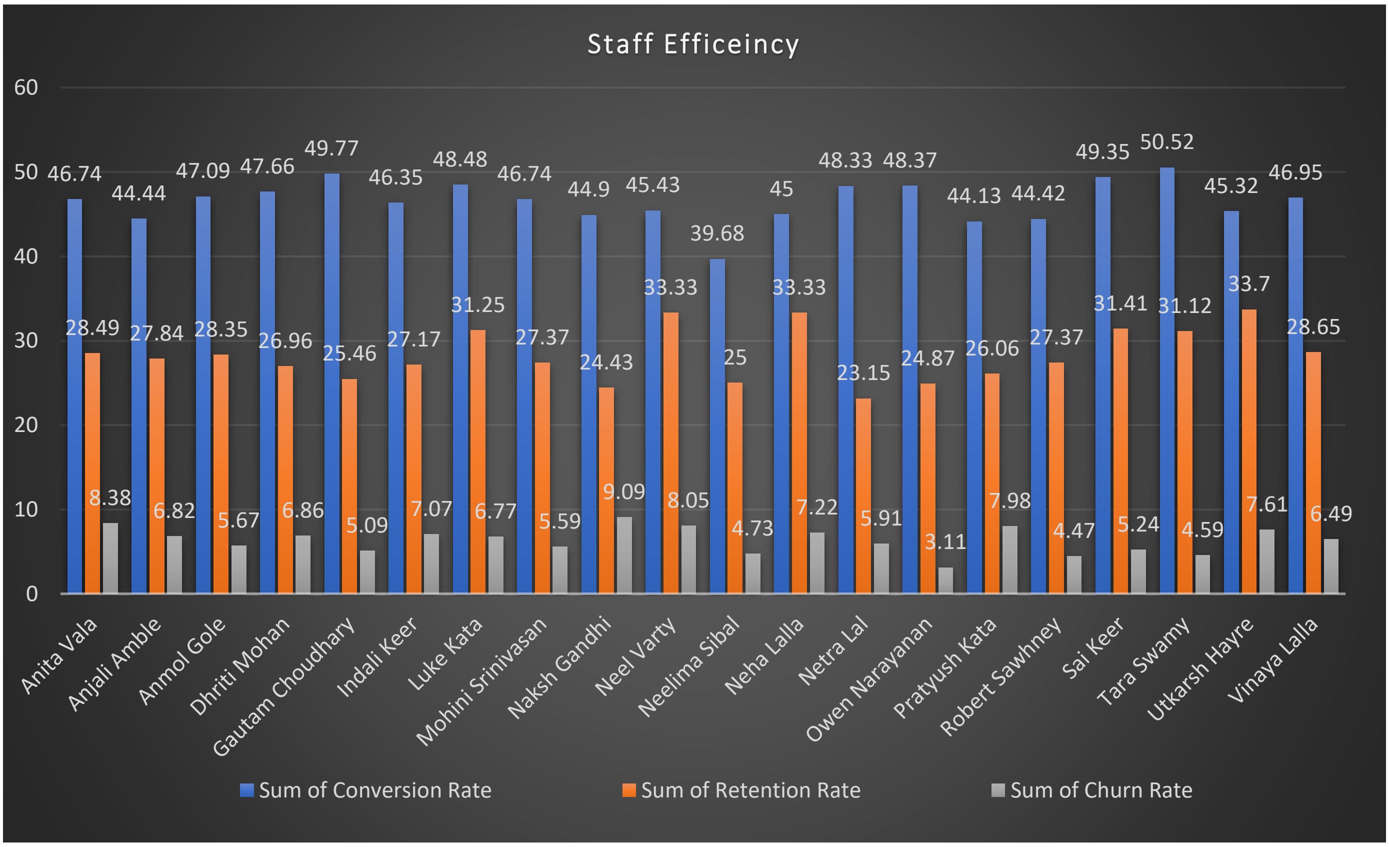
```sql
customers_staff_invoices AS (
SELECT
      cs.all_customers, ci.buyer_customers,
      cs.assigned_staff_id, cs.staff_full_name
FROM customers_staff as cs
LEFT JOIN customers_invoices as ci
ON cs.all_customers=ci.buyer_customers
),
customers_staff_invoices_with_repeated_customers AS (
SELECT
      csi.all_customers, csi.buyer_customers,
      rc.customer_id AS repeated_customers,
      csi.assigned_staff_id, csi.staff_full_name
FROM customers_staff_invoices as csi
LEFT JOIN repeated_customers as rc
ON rc.customer_id=csi.buyer_customers
)
SELECT
      csiwrc.assigned_staff_id, csiwrc.staff_full_name,
      COUNT(csiwrc.all_customers) AS Assigned_Customers,
      COUNT(csiwrc.buyer_customers) AS Buyer_Customers,
      COUNT(csiwrc.repeated_customers) AS Repeated_Customers,
      ROUND( COUNT(csiwrc.buyer_customers) * 100 /
      COUNT(csiwrc.all_customers), 2) AS Customer_Conversion_Rate,
      ROUND( COUNT(csiwrc.repeated_customers) * 100 /
      COUNT(csiwrc.buyer_customers), 2) AS Loyalty_Conversion_Rate
FROM customers_staff_invoices_with_repeated_customers as csiwrc
GROUP BY csiwrc.assigned_staff_id, csiwrc.staff_full_name;
```

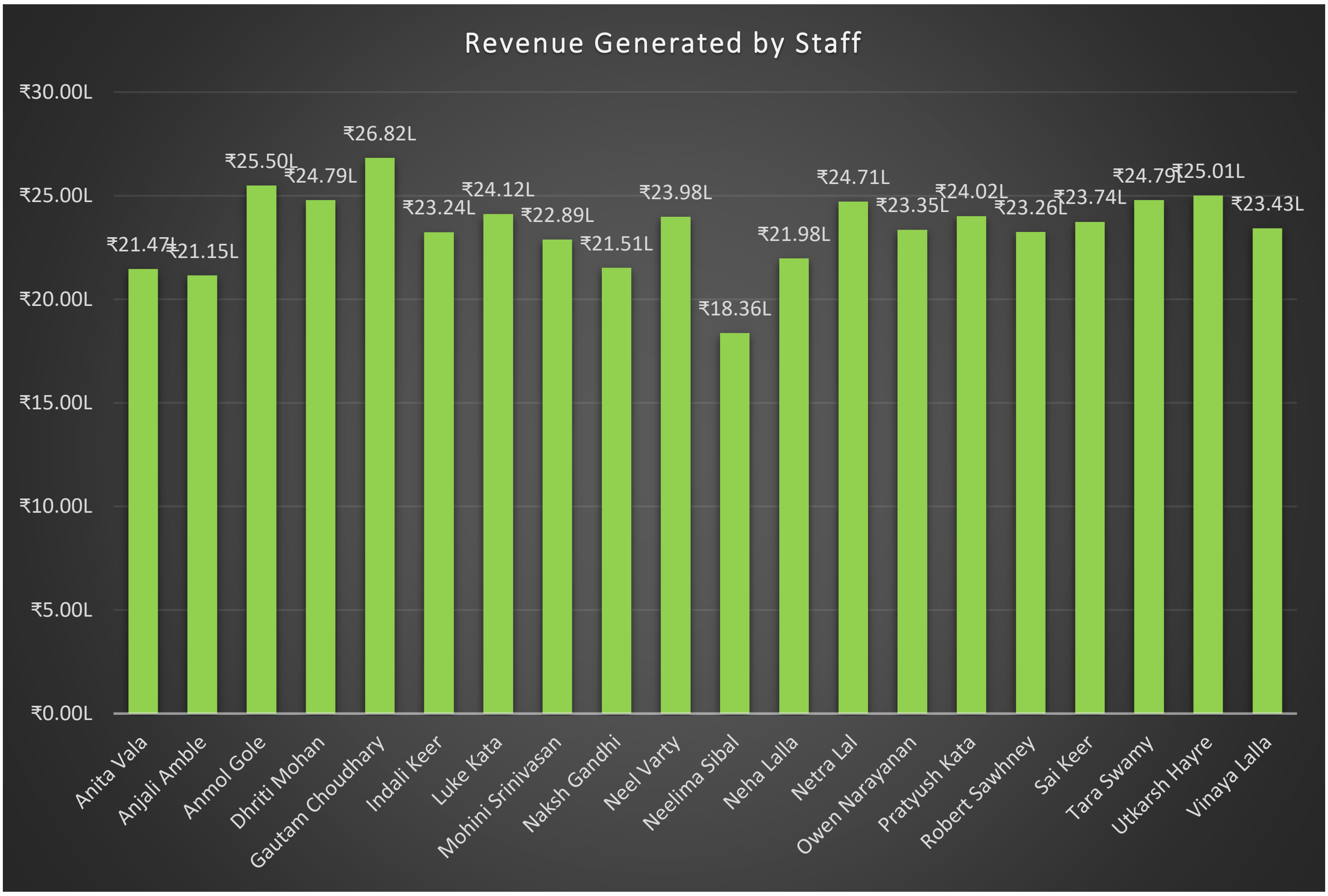
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

assigned_staff_id	staff_full_name	Assigned_Customers	Buyer_Customers	Repeated_Customers	Customer_Conversion_Rate	Loyalty_Conversion_Rate
1	Neel Varty	383	174	58	45.43	33.33
2	Anjali Amble	396	176	49	44.44	27.84
3	Luke Kata	396	192	60	48.48	31.25
4	Vinaya Lalla	394	185	53	46.95	28.65
5	Dhriti Mohan	428	204	55	47.66	26.96
6	Sai Keer	387	191	60	49.35	31.41
7	Robert Sawhney	403	179	49	44.42	27.37
8	Mohini Srinivasan	383	179	49	46.74	27.37
9	Neha Lalla	400	180	60	45.00	33.33
10	Neelima Sibal	373	148	37	39.68	25.00
11	Tara Swamy	388	196	61	50.52	31.12
12	Naksh Gandhi	392	176	43	44.90	24.43
13	Indali Keer	397	184	50	46.35	27.17
14	Gautam Choudhary	434	216	55	49.77	25.46
15	Netra Lal	420	203	47	48.33	23.15
16	Pratyush Kata	426	188	49	44.13	26.06
17	Anmol Gole	412	194	55	47.09	28.35
18	Owen Narayanan	399	193	48	48.37	24.87
19	Anita Vala	383	179	51	46.74	28.49
20	Utkarsh Hayre	406	184	62	45.32	33.70

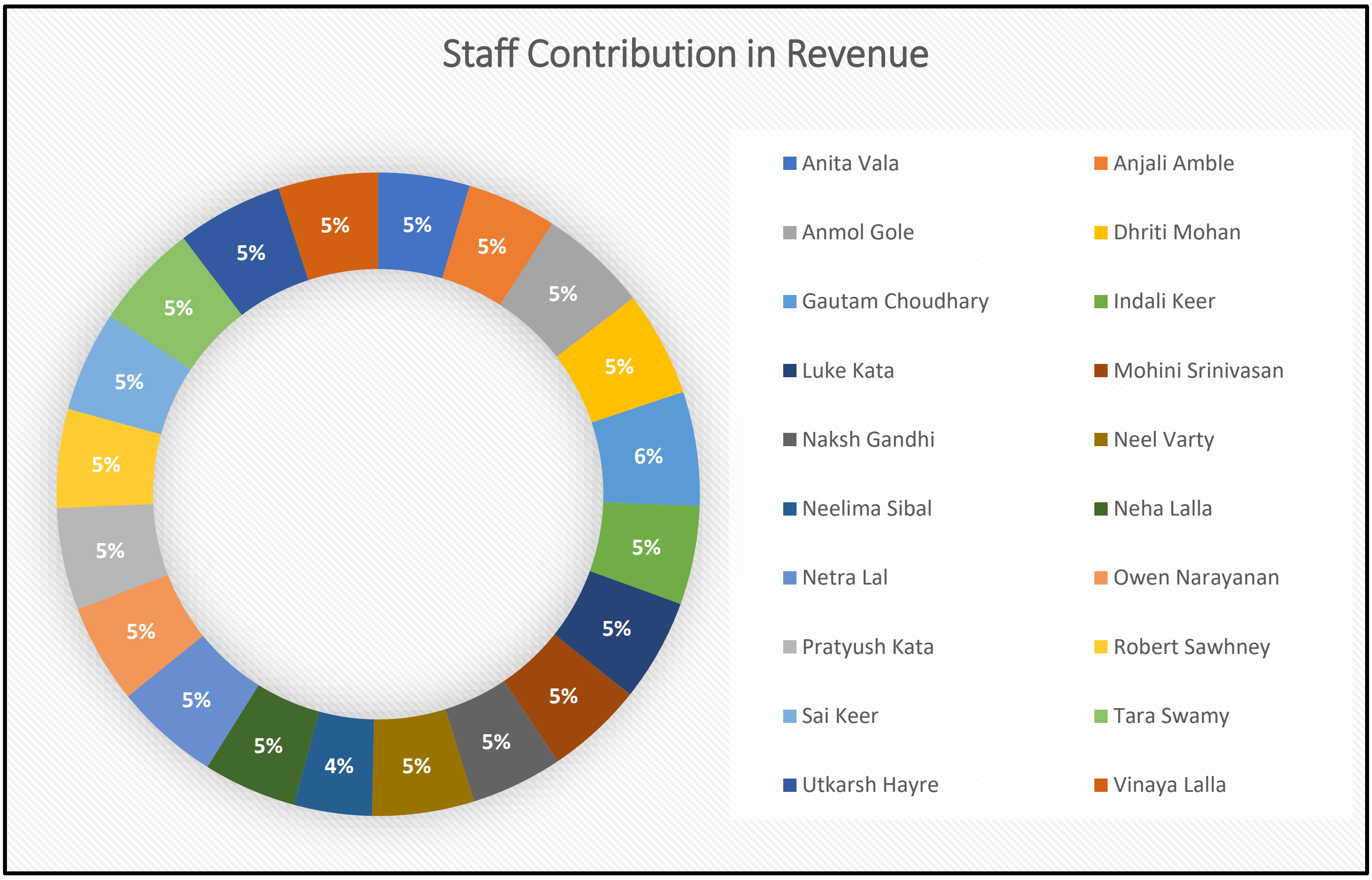
Dashboard Staff Analysis







Revenue Generated by Staff
₹30.00L
₹25.00L
₹20.00L
₹15.00L
₹10.00L
₹5.00L
₹0.00L
₹21.47L
₹21.15L
₹25.50L
₹24.79L
₹26.82L
₹23.24L
₹24.12L
₹22.89L
₹21.51L
₹23.98L
₹18.36L
₹21.98L
₹24.71L
₹23.35L
₹24.02L
₹23.26L
₹23.74L
₹24.79L
₹25.01L
₹23.43L
Anita Vala
Anjali Amble
Anmol Gole
Dhriti Mohan
Gautam Choudhary
Indali Keer
Luke Kata
Mohini Srinivasan
Naksh Gandhi
Neel Varty
Neelima Sibal
Neha Lalla
Netra Lal
Owen Narayanan
Pratyush Kata
Robert Sawhney
Sai Keer
Tara Swamy
Utkarsh Hayre
Vinaya Lalla



Staff Contribution in Revenue
5%
5%
5%
5%
6%
5%
5%
5%
5%
5%
4%
5%
5%
5%
5%
5%
5%
5%
5%
5%
Anita Vala
Anjali Amble
Anmol Gole
Dhriti Mohan
Gautam Choudhary
Indali Keer
Luke Kata
Mohini Srinivasan
Naksh Gandhi
Neel Varty
Neelima Sibal
Neha Lalla
Netra Lal
Owen Narayanan
Pratyush Kata
Robert Sawhney
Sai Keer
Tara Swamy
Utkarsh Hayre
Vinaya Lalla

Staff Performance Analysis

Executive Summary

The staff analysis reveals a well-balanced workforce where business volume is distributed relatively evenly across employees. No single employee dominates the business, reducing operational dependency on one individual.

However, noticeable differences exist in conversion, retention, and churn performance, indicating that customer handling quality varies among staff members despite similar workloads.

Overall, the findings suggest that improving staff efficiency presents a greater opportunity for revenue growth than simply increasing customer traffic.

Why this Analysis Matters

Evaluating staff performance helps the business understand whether customer outcomes are driven by individual employee performance or by overall operational processes. By analysing revenue generation, customer handling, conversion rate, retention rate, and churn rate across all staff members, management can identify performance gaps, recognize operational strengths, and determine whether future improvements should focus on employees or broader business strategies.

Key Insight – Staff Performance is Consistent Across the Workforce

Unlike many retail businesses where employee performance varies significantly, the analysis reveals that staff performance is remarkably consistent across the organization.

Across all key performance indicators—including revenue generated, customer handling volume, conversion rate, retention rate, and churn rate—only minor variations exist between staff members. No employee consistently outperforms or underperforms the rest across all business metrics.

This indicates that the store has successfully established standardized operating procedures and customer service practices, resulting in relatively uniform performance throughout the workforce.

The findings suggest that:

- Revenue contribution is evenly distributed across staff members, with no excessive dependence on a single employee.
- Customer handling volume is balanced, indicating a fair allocation of customer interactions.
- Conversion, retention, and churn rates remain within a narrow range across the team, demonstrating consistent sales and customer service performance.
- Overall staff performance appears stable and standardized rather than highly dependent on individual performers.

Strategic Recommendation

Since staff performance is already well-balanced, the business should prioritize maintaining operational consistency rather than restructuring employee responsibilities or focusing solely on individual performance improvements.

Management should continue to:

- Maintain standardized training and customer service practices across all staff members.
- Regularly monitor key performance indicators to identify any future performance deviations.
- Encourage knowledge sharing and best-practice exchange among employees.
- Continue investing in staff development to sustain service quality and customer experience.

More importantly, the analysis indicates that the business's future growth is unlikely to be constrained by staff performance. Instead, greater opportunities for improvement exist in broader business functions such as:

- Customer acquisition strategies
- Customer retention initiatives
- Referral generation programs
- Customer Relationship Management (CRM)
- Marketing and customer engagement activities

Strengthening these business processes is expected to generate a greater impact on long-term revenue growth than attempting to optimize an already consistent workforce.

Overall Business Conclusion

The staff analysis demonstrates that employee performance is well-balanced across the organization, reflecting effective operational processes, standardized service quality, and consistent customer handling practices. Rather than identifying significant differences between individual employees, the analysis highlights that the business's next stage of growth should focus on enhancing customer acquisition, improving retention strategies, strengthening referral programs, and building long-term customer relationships. These areas offer significantly greater potential for sustainable business growth than further optimizing an already stable and consistently performing workforce.

• Referrer & Referrals Analysis

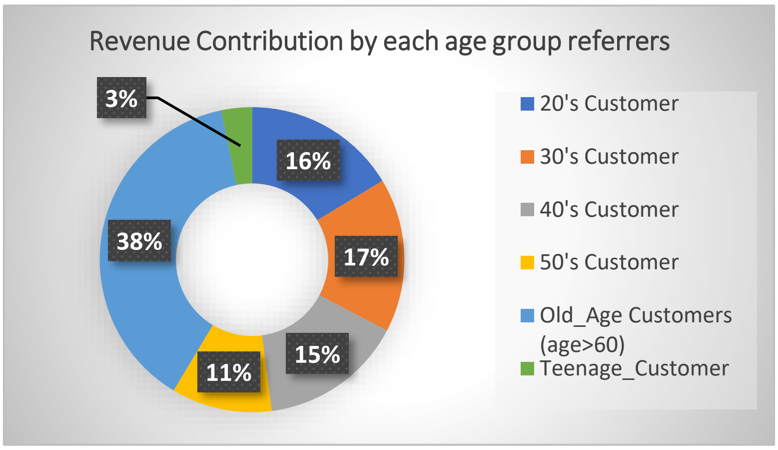
8000 Walk-in-visitors → **721** Referrers → **810** Referrals → **381** Buyer Referrals

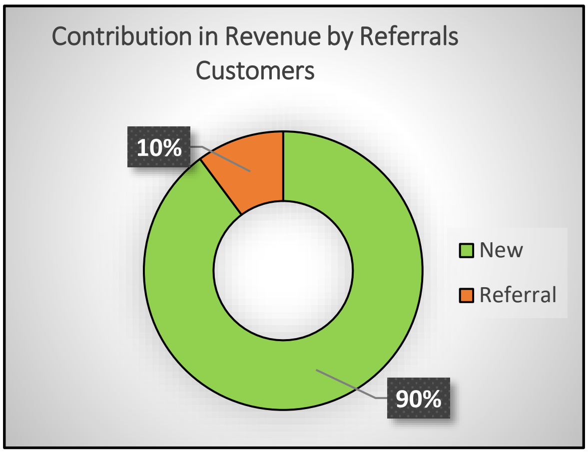
47% Referrals Conversion Rate

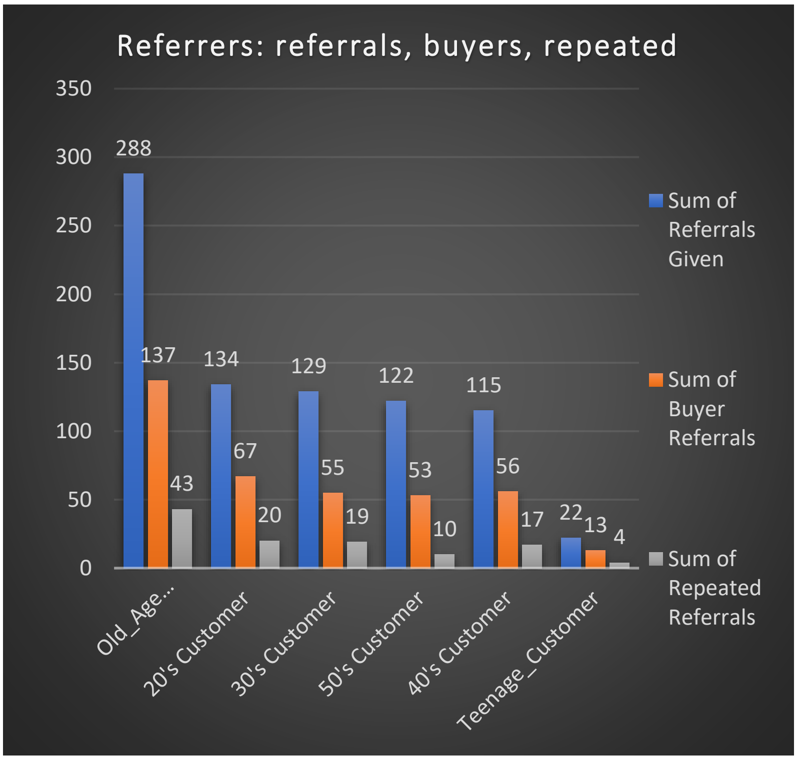
10% Contribution in Revenue

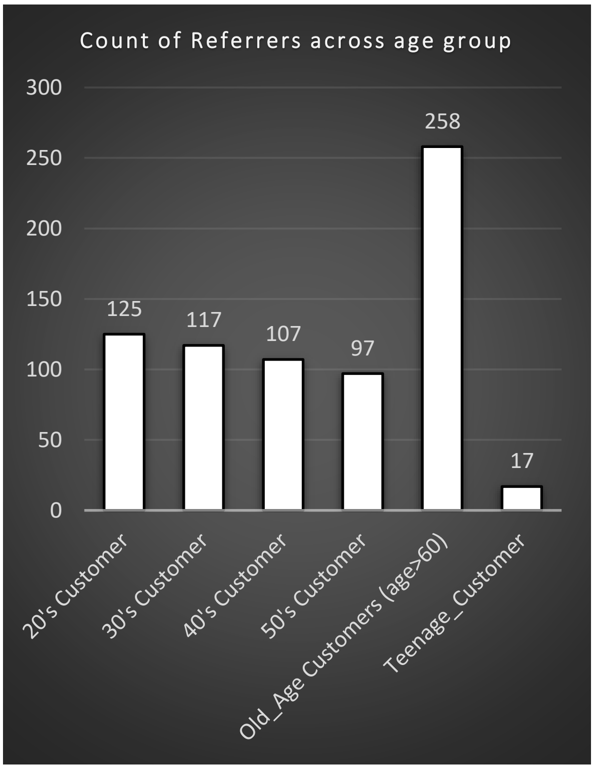
47.63L Rs Revenue by Referrals

Referrers Performance Dashboard









Problem Statement:

"The store has an organic referral program operating without any formal structure or tracking. The business does not know which age groups are most likely to refer customers, whether referred customers from different age groups convert and stay loyal at different rates, and whether the age of the referrer influences the quality of the referral they give. This analysis investigates referral behaviour across age segments to identify where the referral program is strongest and where it needs attention."

Referrer Analysis

- About the Referrers

Referral customers represent one of the lowest-cost customer acquisition channels because they require virtually no marketing expenditure while bringing new customers through word-of-mouth. For a local optical store operating without a formal marketing budget or digital advertising presence, referrals represent one of the most natural and cost-free ways where the new customers discover the business.

This section it investigates the entire referral ecosystem from asking who gives referrals, whether those referrals actually buy, how much revenue they generate, and most importantly uncovering a phenomenon that completely challenges the conventional assumption about how word-of-mouth works.

- Scale of Referrals program

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Total_Visited_Customers	Total_Referrers	Percentage_of_referrers	Total_Referrals	Buyer_referrals	Referral_Conversion_rate	Revenue_from_referrals	Contribution_of_referrals_in_Revenue	Repeated_Referral	Referral_Retention_Rate
8000	721	9.01	810	381	47.04	4763973.00	10.18	113	29.66

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Visited_Customers	Referrers_Customers	Percetage_of_Referrers_Customers	Buyer_Customers	Buyer_referrals	Percentage_of_referral_customers_in_buyers	Referral_Conversion_Rate
8000	721	9.01	3721	381	10.24	4.76

Business Insights:

The analysis reveals that 721 customers (9.01% of all 8,000 visitors) acted as referrers by recommending our store to friends or family. While this represents less than one-tenth of the overall customer base, these referrers generated 810 referral customers (10.13% of the 8000 visitors), demonstrating the significant impact that a relatively small group of satisfied customers can have on business growth.

Out of these 810 referred customers, 381 completed a purchase, resulting in a Referral Conversion Rate of 47.04%. This indicates that nearly one out of every two referred visitors became a paying customer, which is a strong acquisition performance for a completely organic customer source. These 381 referral buyers account for approximately 10.24% of all buyers (3,721) in the business.

Financially, the impact is even more significant. Referral customers generated approximately ₹47.64 lakh in revenue, contributing 10.18% of the store's total revenue. Considering that only 9% of customers became referrers, generating over 10% of total business revenue highlights the high commercial value of referral-based customer acquisition.

However, customer retention remains an area requiring attention. Out of the 381 referral buyers, only 113 became repeat customers, resulting in a Referral Retention Rate of 29.66%. While the initial conversion is strong, retaining these acquired customers presents a clear opportunity for improvement.

Strategic Business Recommendation:

Hence, we must focus more on the referral network functions as a zero-cost customer acquisition channel for the business. Existing customers act as brand advocates by recommending the store to friends and family, increasing awareness without requiring investment in advertising or promotional campaigns.

Rather than investing solely in paid advertising, management should prioritize increasing the number of customers who are willing to recommend the business.

The most sustainable way to achieve this is by consistently delivering an exceptional customer experience throughout the sales journey. Customers are far more likely to refer friends and family when they genuinely trust the business and feel that their interests are prioritized over short-term sales.

If the business can increase the proportion of customers who become active referrers from the current 9% to even 15–20%, referral-driven revenue could increase substantially without proportionally increasing marketing expenditure.

While referral visitors still need to be converted through effective sales and customer service, they represent pre-qualified prospects who arrive with higher initial trust. Strengthening and incentivizing this referral ecosystem has the potential to increase customer acquisition while maintaining a significantly lower Customer Acquisition Cost (CAC) than traditional marketing methods.

Now the question arrives: Who are this Referrers? Whom they Refers (basically referrals)?

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Referrers_Age_group	COUNT_Referrers	Referrals	Rate_of_Referrals	Buyer_Referrals	Referral_Buyer_Rate	Retain_Buyer_Referrals	Buyer_Referrals_Retention_Rate	Revenue_Collected
20's Customer	125	134	1.07	67	50.00	20	29.85	778770.00
30's Customer	117	129	1.10	55	42.64	19	34.55	778895.00
40's Customer	107	115	1.07	56	48.70	17	30.36	726073.00
50's Customer	97	122	1.26	53	43.44	10	18.87	508640.00
Old_Age Customers (age>60)	258	288	1.12	137	47.57	43	31.39	1816498.00
Teenage_Customer	17	22	1.29	13	59.09	4	30.77	155097.00
NULL	721	810	1.12	381	47.04	113	29.66	4763973.00

Insight 1: Senior Customers are the strongest advocates of the business

Customers aged 60+ emerge as the most valuable advocates within the referral ecosystem.

Compared with every other demographic, this segment contributes:

- Highest number of Referrers
- Highest number of Referral Customers
- Highest number of Retained Referral Customers
- Highest Revenue generated through referrals

This indicates that senior customers are significantly more willing to recommend the store to friends and family, making them the strongest source of organic customer acquisition.

Although their referral-to-buyer conversion rate is slightly below some younger age groups, this is more reflective of the sales conversion process after the referral rather than the referral behaviour itself.

Therefore, the business should continue nurturing this segment through exceptional post-purchase service, relationship management and trust-building initiatives, as they generate substantial long-term value beyond their own purchases.

Insight 2: Middle-aged customer segments demonstrate stable referral behaviour

Customers aged 20–50 display remarkably consistent referral performance across nearly every business metric.

The number of referrals generated, buyer conversion, retained referrals and revenue contribution remain relatively balanced across these age groups, indicating a stable and predictable referral pattern.

This suggests that referral behaviour is not heavily concentrated within a single working-age demographic but is instead broadly distributed. Because these segments represent a large share of the customer base, even small improvements in referral conversion or retention could translate into meaningful revenue growth.Rather than treating these customers differently, the business should apply scalable referral programs and standardized customer experience improvements across all working-age groups.

Insight 3 Teenage customers represent an underpenetrated but high-quality acquisition opportunity

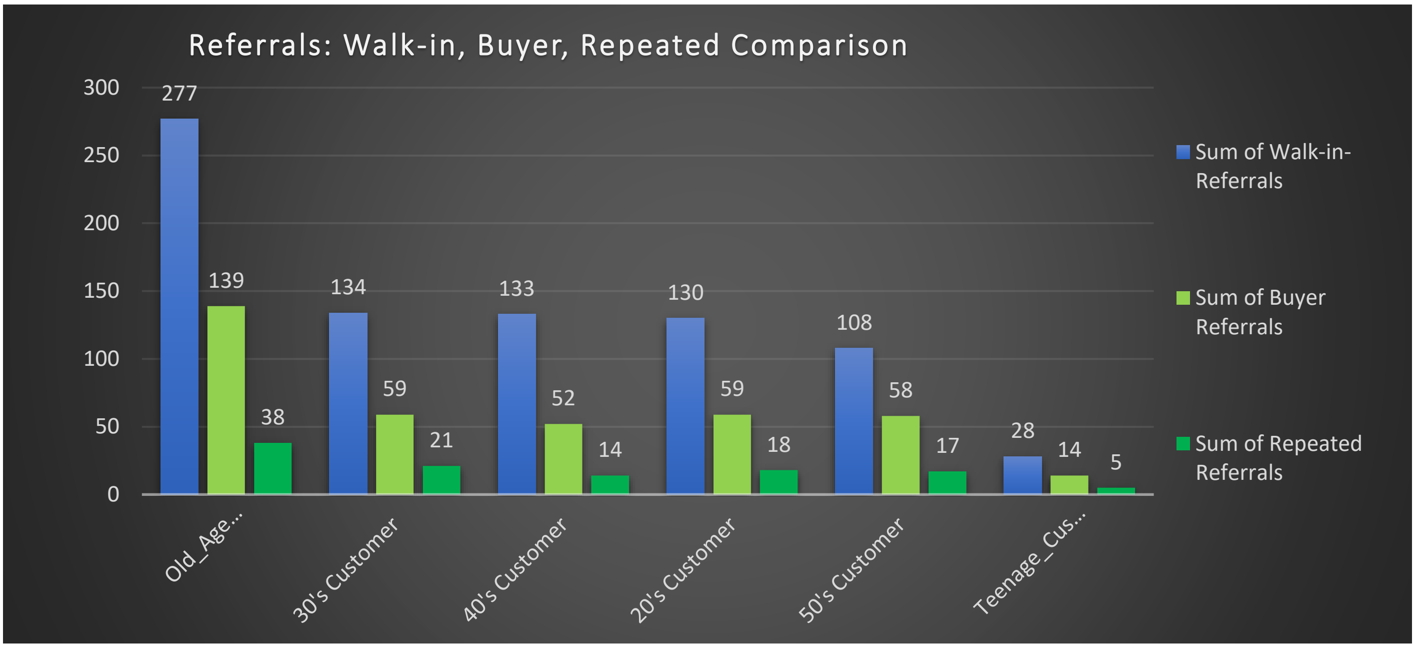
Although teenagers account for the smallest share of referrers and referral customers, they exhibit one of the strongest behavioural profiles once acquired. Compared with other age groups, teenage referrals demonstrate:

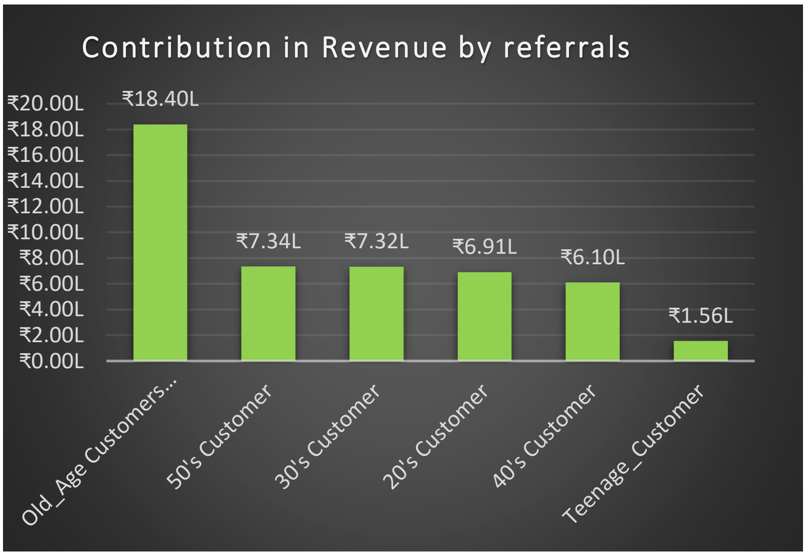
  - Highest Referral-to-Buyer Conversion Rate
  - Lower customer churn
  - Strong retention performance

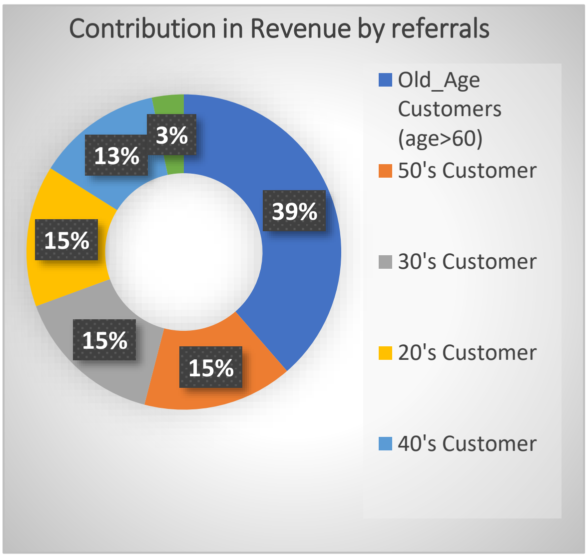
This suggests that the business is not facing a conversion problem within this segment. Instead, the challenge appears to be customer acquisition. In other words,

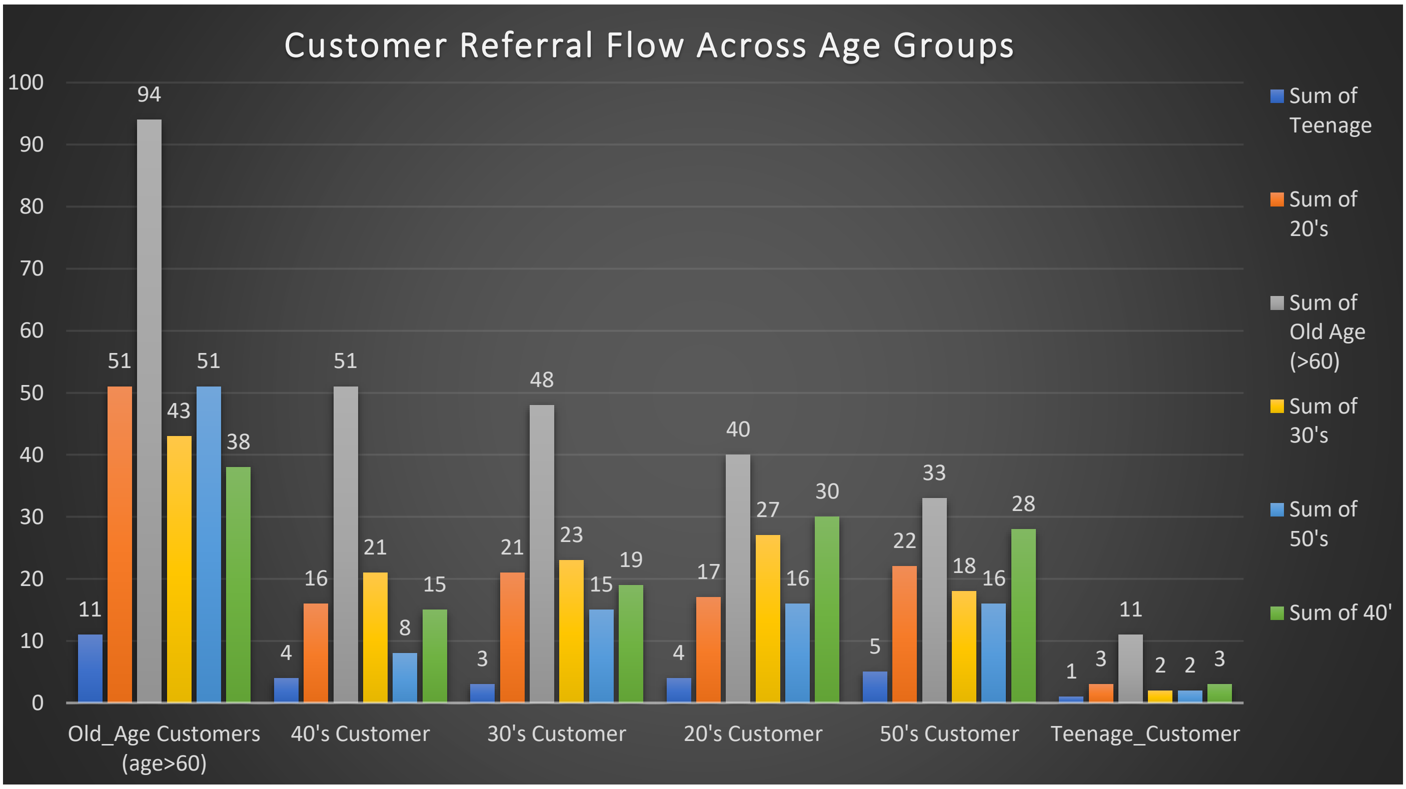
The business converts teenage customers effectively once they visit the store, but relatively few teenagers enter the acquisition funnel. These highlights an opportunity to expand marketing efforts targeted toward students and young adults through digital marketing, school partnerships, social media campaigns and youth-oriented referral incentives.

Referrals Performance Dashboard









Referral Network Analysis by Customer Age Group

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Referrers_Age_group	Teenage_Customers	20s_Customers	30s_Customers	40s_Customers	50s_Customers	Old_Age_Customers
NULL	28	130	134	133	108	277
Old_Age Customers (age>60)	11	51	43	38	51	94
50's Customer	5	22	18	28	16	33
20's Customer	4	17	27	30	16	40
40's Customer	4	16	21	15	8	51
30's Customer	3	21	23	19	15	48
Teenage_Customer	1	3	2	3	2	11

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Age_group_Referrals	Count_Referral_Customers	Buyer_Referrals	Referral_Conversion_Rate	repeated_referrals	Referrals_Retention_Rate	Revenue_Collected
20's Customer	130	59	45.38	18	30.51	691298.00
30's Customer	134	59	44.03	21	35.59	732491.00
40's Customer	133	52	39.10	14	26.92	610299.00
50's Customer	108	58	53.70	17	29.31	734382.00
Old_Age Customers (age>60)	277	139	50.18	38	27.34	1839787.00
Teenage_Customer	28	14	50.00	5	35.71	155716.00
NULL	810	381	47.04	113	29.66	4763973.00

Key Insight 1: Senior Customers are the Primary Referral Drivers

Customers aged **60+** dominate nearly every referral metric; are as follows:

- Generated the highest number of referrals.
- Referred the highest number of fellow senior customers.
- Produced the highest number of buyer referrals.
- Generated the highest number of retained referral customers.
- Contributed the largest referral revenue (₹18.4 Lakhs).

This reinforces an important business finding observed throughout previous analyses:

Senior customers are not only the store's largest customer segment, but also its strongest brand advocates. Their influence extends beyond their own purchases by continuously bringing new customers into the business.

Key Insight 2: Teenage Referrals Are Small in Volume but High in Quality

Teenage customers represent only a small portion of the referral ecosystem. However, once acquired they demonstrate exceptionally strong performance.

Compared with every other segment, teenage referrals exhibit:

- One of the highest Referral Conversion Rates (~50%)
- The highest Referral Retention Rate (~35.7%)
- Low levels of customer churn

This suggests the business is not facing a conversion problem among teenage referrals. Instead, the challenge lies in customer acquisition. If more teenage customers could be attracted into the referral funnel through school campaigns, digital marketing or social media engagement, this segment could become a meaningful source of long-term growth.

Key Insight 3: Senior Referral Customers Need Better Retention

Although senior customers generate the largest referral revenue, they do not perform equally well in long-term retention. Their referral retention rate remains below several younger segments.

This observation aligns with findings from the earlier Churn Analysis, where older customers also exhibited relatively higher churn behaviour. While the current data cannot determine the exact reason, it indicates that a significant proportion of senior referral customers purchase only once and fail to return.

Recommended actions include:

- Structured follow-up calls after purchase
- Customer satisfaction surveys
- After-sales support
- Reminder campaigns for eye checkups
- Personalized relationship management for senior customers

Improving retention within this high-volume segment is likely to produce a greater business impact than attempting to increase referrals alone.

Insight 4: Senior Customers Receive Referrals Across Nearly Every Age Group

One of the most interesting patterns observed in the referral network is that customers from almost every age segment predominantly refer senior customers (60+) to the store.

This indicates that senior customers are not only referring others themselves, but are also the primary recipients of referrals from family members and acquaintances across multiple age groups.

This pattern suggests that the business has developed a strong reputation and trust in serving senior customers. Families often play an active role in recommending optical care for their parents and grandparents, making senior customers the central segment within the referral ecosystem.

Strategic Business Recommendation:

Context: From an optical retail perspective, this segment naturally requires periodic vision correction due to age-related conditions such as presbyopia and progressive changes in eyesight. As a result, many customers in this age group typically revisit an optical store every 2–3 years for updated prescriptions, replacement lenses or new eyewear. This recurring demand makes customer retention significantly more valuable.

Therefore, the business should prioritize:

- Strengthening customer relationships through personalized service.
- Building long-term trust by recommending products based on customer needs rather than short-term sales incentives.
- Conducting post-purchase follow-ups and collecting customer feedback.
- Scheduling periodic eye-checkup and prescription reminder campaigns to encourage repeat visits.

Protecting and retaining this customer segment should be considered a strategic priority, as today's first-time senior customers are the most likely to become tomorrow's repeat customers, brand advocates and referral generators.

Note: However, the analysis also reveals that a noticeable proportion of senior referral customers do not return after their initial purchase. Since this segment contributes the largest share of referral-driven business, even a small improvement in retention could generate substantial long-term revenue growth.

- Identifying the reasons behind old age customer churn and implementing corrective actions.
- Protect this high-value customer segment.
- Prioritize the retention of this strategic customer segment.
- Safeguarding this segment should be a key business objective.
- Improving retention within this segment offers one of the highest-return opportunities for sustainable revenue growth.

Referral Customers vs Regular Customers

Type_Customers	Visited_Customers	Buyer_Customers	Conversion_rate_of_non_referral_customers	Repeated_Customers	Retention_rate	Average_Revenue_per_invoice
Regular_Customers	7190	3340	46.45	938	28.08	9375.13
Referral_Customers	810	381	47.04	113	29.66	9250.43

Business Insight:

At first glance, referral customers appear to perform slightly better than regular customers across key business metrics. However, the difference is relatively small.

Now the Question Arrives what's next? What decisions we should make further?

"Based on the analyses performed in this project, two strategic opportunities appear to offer the greatest potential for future business growth:

1. **Expanding Customer acquisition among younger demographics**

The analysis consistently shows that teenage customers exhibit:

- One of the highest conversion rates
- Strong retention performance
- Lower customer churn

The current limitation is not conversion but customer volume. Very few teenage customers enter the business, despite performing exceptionally well once acquired.

2. **Improving the retention of senior customers (Age above 60)**

The analysis also identifies senior customers as the most strategically valuable customer segment.

They contribute the highest:

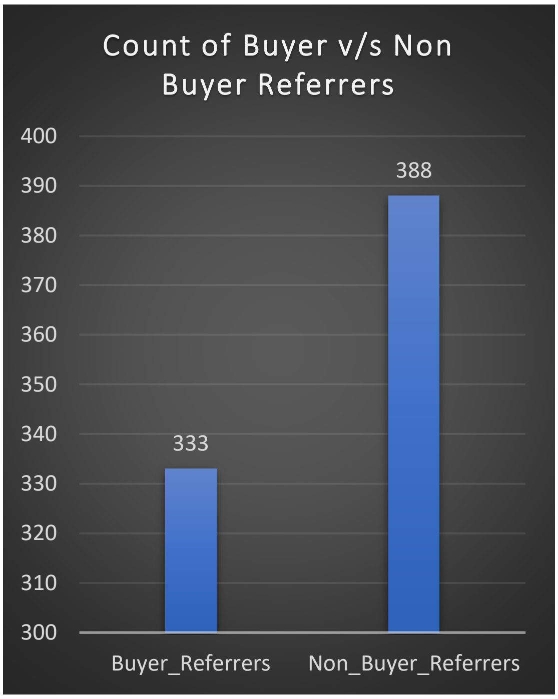
- Revenue
- Repeat customers
- Referral generation
- Referral revenue
- Long-term business value

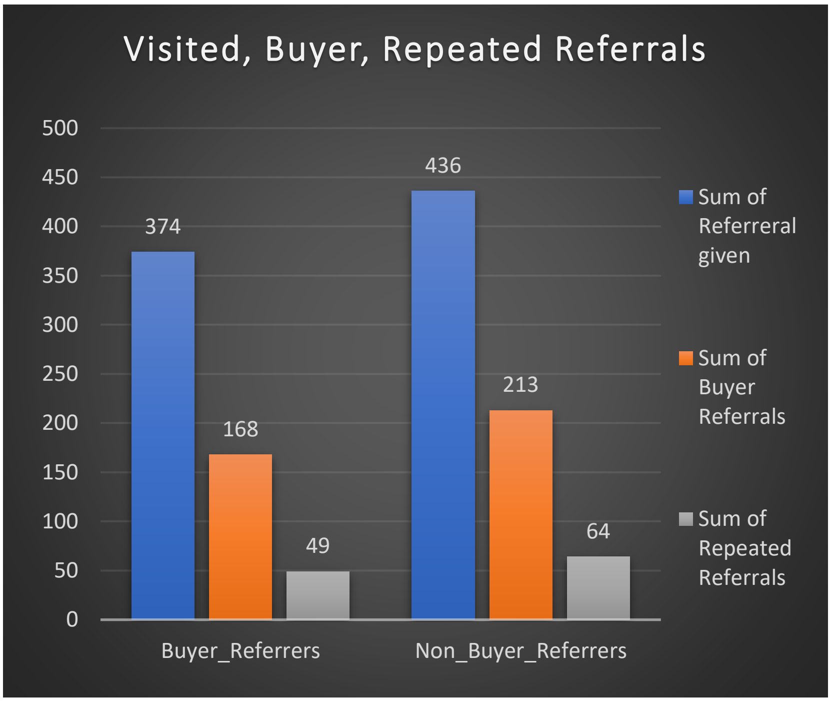
Key Insight – Non-Buyer Referrers Are an Unexpected Growth Driver

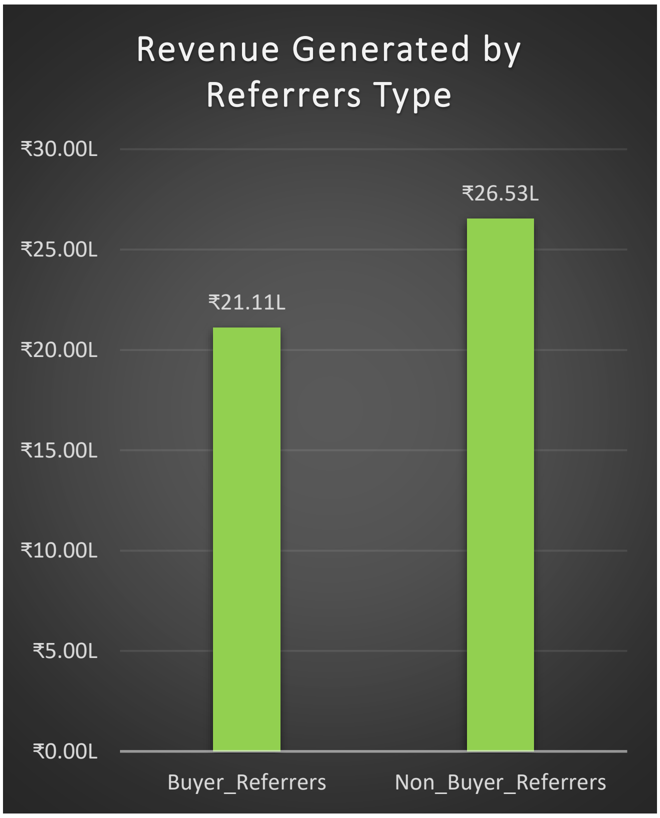
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

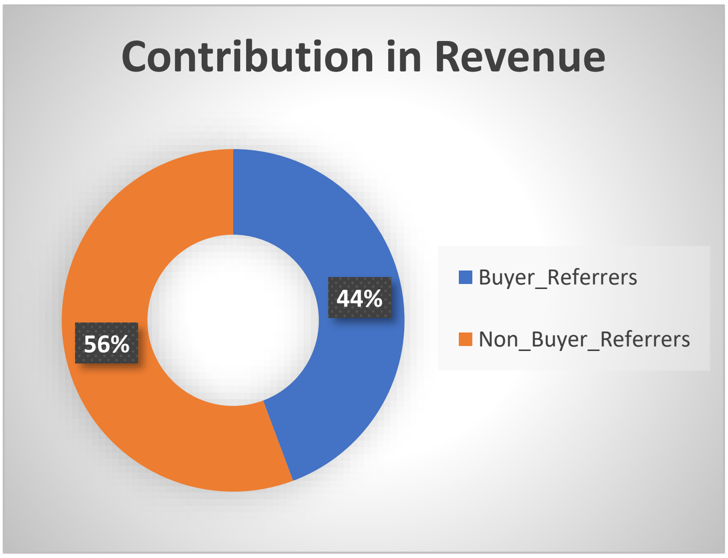
Type_of_Referrers	Buyer_Referrers	Total_Referral_given	Referrals_Rate	Buyer_Referrals	Referrals_Conversion_Rate	Repeated_referrals	Referrals_retention_rate	Revenue_from_referrals	Contribution_in_Revenue
Buyer_Referrers	333	374	1.12	168	44.92	49	29.17	2110932.00	4.51
Non_Buyer_Referrers	388	436	1.12	213	48.85	64	30.05	2653041.00	5.67
Total	721	810	1.12	381	47.04	113	29.66	4763973.00	10.18

Dashboard of Buyer and Non-Buyer Referrers Performance









Insights from buyers and non-buyer referrers

- One of the most surprising findings from this analysis is that customers who have never purchased from the store are recommending the business almost as actively as existing buyers.
- Although the business has a larger number of Non-Buyer Referrers (388) than Buyer Referrers (333), both groups generate referrals at an almost identical rate of 1.12 referrals per referrer. More importantly, the quality of these referrals remains remarkably similar throughout the customer journey.
- Referrals generated by non-buyers achieved a 48.85% conversion rate, compared to 44.92% for buyer referrers, while referral retention rates remained almost identical (30.05% vs. 29.17%). Consequently, both groups contributed nearly the same business value, with buyer referrers generating approximately ₹21.1 lakh in revenue and non-buyer referrers contributing approximately ₹26.5 lakh.
- Collectively, the referral ecosystem generated ₹47.64 lakh, accounting for 10.18% of total business revenue without any direct marketing expenditure.

The most important takeaway from this analysis is not the revenue itself, but the question it raises:

"Why are customers who have never purchased from us still recommending our business?"

This behaviour suggests that purchasing is not the only factor influencing word-of-mouth referrals. Several business HYPOTHESIS / Explanations are possible:

- The customer may have had a positive consultation or store experience despite not making a purchase.
- Price-sensitive visitors may have chosen not to buy immediately but still trusted the store enough to recommend it to friends or family.
- Existing reputation, service quality, staff behaviour, or brand trust may be creating positive impressions independent of sales.

This represents an opportunity for further investigation through customer feedback or surveys. Understanding why non-buyers become advocates could help the business strengthen referral acquisition while also improving conversion among these visitors.

Referrer Analysis Section Summary

The Referrer Analysis demonstrates that the store's referral ecosystem is a significant contributor to long-term business growth. Although only 721 customers (approximately 9% of the total customer base) acted as referrers, they generated 810 referral customers, of which 381 became buyers, contributing ₹47.64 lakh, or approximately 10.18% of the store's total revenue. This highlights referrals as one of the most valuable acquisition channels, achieved without direct marketing expenditure.

The analysis also revealed that buyer and non-buyer referrers exhibit nearly identical referral behaviour, generating approximately 1.12 referrals per referrer on average. Surprisingly, non-buyer referrers performed almost as well as buyer referrers in terms of referral generation, conversion, retention, and revenue contribution. This suggests that customer advocacy is driven not only by purchases but also by the overall customer experience and perception of the business.

Demographic analysis further showed that customers above 60 years of age form the backbone of the referral network. They contribute the highest number of referrers, referrals, repeat referral customers, and referral revenue. Given that this customer segment naturally revisits optical stores every 2–3 years due to age-related prescription changes, retaining these customers should be treated as a strategic priority. At the same time, teenage customers, despite their relatively small volume, demonstrated the highest referral conversion and retention rates, indicating a promising opportunity for future customer acquisition through digital marketing and targeted engagement.

Overall, the findings suggest that the referral program is already performing effectively and represents a highly cost-efficient growth channel. Future business efforts should focus on strengthening customer experience, increasing referral participation, improving referral customer retention, and expanding acquisition among younger customer segments while continuing to nurture the highly valuable senior customer base.

Final Business Recommendations

- **Strengthen customer experience** by delivering accurate prescriptions, transparent consultations, and exceptional after-sales support, as satisfied customers become the business's most effective marketers.
- **Retain senior customers** through periodic follow-ups, reminder campaigns, and proactive eye-checkup programs, recognizing their high lifetime value and recurring vision care needs.
- **Expand referral participation** by encouraging satisfied customers to recommend friends and family through structured referral initiatives.
- **Target younger demographics** using social media and digital marketing, leveraging their strong conversion and retention characteristics to build the next generation of loyal customers.
- **Investigate churn among senior referral customers** by collecting feedback from one-time buyers and implementing corrective actions to improve long-term retention

SQL Queries used for Referrer & Referrals Analysis:

```sql
SELECT
      COUNT(DISTINCT c.customer_id)
      AS Total_Visited_Customers,
      COUNT(DISTINCT CASE WHEN c.referred_by_id IS NOT NULL THEN c.customer_id END)
      AS Referral_Customers,
      ROUND( COUNT(DISTINCT CASE WHEN c.referred_by_id IS NOT NULL THEN c.customer_id END) * 100 /
      COUNT(DISTINCT c.customer_id), 2)
      AS Percentage_of_referraL_Customers_in_total_visited_customers
FROM customers_details as c;
```

```sql
WITH unique_referrers AS (
SELECT DISTINCT c.referred_by_id
FROM customers_details as c
),
unique_referrers_referrals AS (
SELECT ur.referred_by_id, c.customer_id as all_customers_id
FROM customers_details as c
INNER JOIN unique_referrers as ur
ON ur.referred_by_id=c.referred_by_id
),
ur_referral_buyers AS (
SELECT urr.referred_by_id, urr.all_customers_id,
       i.customer_id as buyer_customer_id,
       i.final_bill_amount
FROM unique_referrers_referrals as urr
LEFT JOIN invoices as i
ON urr.all_customers_id=i.customer_id
),
urrb_repeated_referrals AS (
SELECT
       urrb.referred_by_id,
       urrb.all_customers_id,
       urrb.buyer_customer_id,
       urrb.final_bill_amount,
       rc.customer_id as repeated_customers_id
FROM ur_referral_buyers as urrb
LEFT JOIN repeated_customers as rc
ON rc.customer_id=urrb.all_customers_id
)
```

```sql
SELECT
      COUNT(DISTINCT urrbrr.referred_by_id)
      AS Total_Referrers,
      COUNT(DISTINCT urrbrr.all_customers_id)
      AS Total_Referrals,
      COUNT(DISTINCT urrbrr.buyer_customer_id)
      AS Buyer_referrals,
      ROUND( COUNT(DISTINCT urrbrr.buyer_customer_id)*100/
      COUNT(DISTINCT urrbrr.all_customers_id), 2)
      AS Referral_Conversion_rate,
      SUM(urrbrr.final_bill_amount)
      AS Revenue_from_referrals,
      ROUND( SUM(urrbrr.final_bill_amount)*100/
      (SELECT SUM(final_bill_amount) FROM invoices), 2)
      AS Contribution_of_referrals_in_Revenue,
      COUNT(DISTINCT urrbrr.repeated_customers_id)
      AS Repeated_Referral,
      ROUND(COUNT(DISTINCT urrbrr.repeated_customers_id)*100/
      COUNT(DISTINCT urrbrr.buyer_customer_id), 2)
      AS Referral_Retention_Rate
FROM urrb_repeated_referrals as urrbrr;
```

```sql
WITH referrers AS (
SELECT
      c.referred_by_id
FROM customers_details as c
GROUP BY c.referred_by_id
)
SELECT
CASE
    WHEN c.age < 20 THEN "Teenage_Customer"
    WHEN c.age BETWEEN 20 AND 30 THEN "20's Customer"
    WHEN c.age BETWEEN 31 AND 40 THEN "30's Customer"
    WHEN c.age BETWEEN 41 AND 50 THEN "40's Customer"
    WHEN c.age BETWEEN 51 AND 60 THEN "50's Customer"
    WHEN c.age > 60 THEN "Old_Age Customers (age>60)"
  END AS Referrer_Age_group,
COUNT(DISTINCT referrers.referred_by_id ) AS COUNT_Referrals
FROM customers_details as c
INNER JOIN referrers
ON referrers.referred_by_id=c.customer_id
GROUP BY Referrer_Age_group
ORDER BY COUNT_Referrals DESC;
```

```sql
SELECT
    COUNT( DISTINCT c.customer_id ) AS Visited_Customers,
    COUNT(DISTINCT i.customer_id) AS Buyer_Customers,
    COUNT( DISTINCT CASE WHEN c.referred_by_id IS NOT NULL
    AND
    i.customer_id IS NOT NULL THEN i.customer_id END)
    AS Buyer_referrals,
    ROUND( COUNT( DISTINCT CASE WHEN c.referred_by_id IS NOT NULL
    AND
    i.customer_id IS NOT NULL THEN i.customer_id END) * 100 /
    COUNT( DISTINCT i.customer_id ), 2)
    AS Percentage_of_referral_customers_in_buyers,
    ROUND( (COUNT( DISTINCT CASE WHEN c.referred_by_id IS NOT NULL
    AND
    i.customer_id IS NOT NULL THEN i.customer_id END))*100/
    COUNT( DISTINCT c.customer_id ), 2)
    AS Referral_Conversion_Rate
FROM customers_details as c
LEFT JOIN invoices as i
ON c.customer_id=i.customer_id;
```

```sql
SELECT
    'Regular_Customers' AS Type_Customers,
     COUNT(DISTINCT c.customer_id) AS Visited_Customers,
     COUNT(DISTINCT i.customer_id) AS Buyer_Customers,
     ROUND( COUNT(DISTINCT i.customer_id) * 100 /
     COUNT(DISTINCT c.customer_id), 2)
     AS Conversion_rate_of_non_referral_customers,
     COUNT( DISTINCT rc.customer_id) AS Repeated_Customers,
     ROUND( COUNT( DISTINCT rc.customer_id) *100 /
     COUNT(DISTINCT i.customer_id), 2) AS Retention_rate,
     ROUND(AVG(i.final_bill_amount), 2)
     AS Average_Revenue_per_invoice
FROM customers_details as c
LEFT JOIN invoices as i
On c.customer_id=i.customer_id
LEFT JOIN repeated_customers as rc
ON rc.customer_id=c.customer_id
WHERE c.referred_by_id IS NULL
UNION ALL
SELECT
     'Referral_Customers',
     COUNT(DISTINCT c.customer_id) AS Visited_Customers,
     COUNT(DISTINCT i.customer_id) AS Buyer_Customers,
     ROUND( COUNT(DISTINCT i.customer_id) * 100 /
     COUNT(DISTINCT c.customer_id), 2)
     AS Conversion_rate_of_non_referral_customers,
     COUNT(DISTINCT rc.customer_id) AS Repeated_referral_customers,
     ROUND( COUNT(DISTINCT rc.customer_id)*100/
     COUNT(DISTINCT i.customer_id), 2)
     AS Retention_rate,
     ROUND(AVG(i.final_bill_amount), 2) AS Average_Revenue_per_invoice
FROM customers_details as c
LEFT JOIN invoices as i
On c.customer_id=i.customer_id
LEFT JOIN repeated_customers as rc
ON rc.customer_id=c.customer_id
WHERE c.referred_by_id IS NOT NULL;
```

```sql
SELECT
    COUNT( DISTINCT c.customer_id)
    AS Count_Referral_Customers,
    CASE
    WHEN c.age < 20 THEN "Teenage_Customer"
    WHEN c.age BETWEEN 20 AND 30 THEN "20's Customer"
    WHEN c.age BETWEEN 31 AND 40 THEN "30's Customer"
    WHEN c.age BETWEEN 41 AND 50 THEN "40's Customer"
    WHEN c.age BETWEEN 51 AND 60 THEN "50's Customer"
    WHEN c.age > 60 THEN "Old_Age Customers (age>60)"
  END AS Age_group
FROM customers_details as c
WHERE c.referred_by_id IS NOT NULL
GROUP BY Age_group WITH ROLLUP;
```

SQL Queries Output of Referrer & Referrals Analysis

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Total_Referrers	Total_Referrals	Buyer_referrals	Referral_Conversion_rate	Revenue_from_referrals	Contribution_of_referrals_in_Revenue	Repeated_Referral	Referral_Retention_Rate
721	810	381	47.04	4763973.00	10.18	113	29.66

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Total_Visited_Customers	Referral_Customers	Percentage_of_referraL_Customers_in_total_visit
8000	810	10.13

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Visited_Customers	Buyer_Customers	Buyer_referrals	Percentage_of_referral_customers_in_buyers	Referral_Conversion_Rate
8000	3721	381	10.24	4.76

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Referrers_Age_group	COUNT_Referrers	Referrals	Buyer_Referrals	Referral_Buyer_Rate	Retain_Buyer_Referrals	Buyer_Referrals_Retention_Rate
20's Customer	125	134	67	50.00	20	29.85
30's Customer	117	129	55	42.64	19	34.55
40's Customer	107	115	56	48.70	17	30.36
50's Customer	97	122	53	43.44	10	18.87
Old_Age Customers (age>60)	258	288	137	47.57	43	31.39
Teenage_Customer	17	22	13	59.09	4	30.77
NULL	721	810	381	47.04	113	29.66

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Age_group_Referrals	Count_Referral_Customers	Buyer_Referrals	Referral_Conversion_Rate	repeated_referrals	Referrals_Retention_Rate
20's Customer	130	59	45.38	18	30.51
30's Customer	134	59	44.03	21	35.59
40's Customer	133	52	39.10	14	26.92
50's Customer	108	58	53.70	17	29.31
Old_Age Customers (age>60)	277	139	50.18	38	27.34
Teenage_Customer	28	14	50.00	5	35.71
NULL	810	381	47.04	113	29.66

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Referrers_Age_group	Teenage_Customers	20s_Customers	30s_Customers	40s_Customers	50s_Customers	Old_Age_Customers
NULL	28	130	134	133	108	277
Old_Age Customers (age>60)	11	51	43	38	51	94
50's Customer	5	22	18	28	16	33
20's Customer	4	17	27	30	16	40
40's Customer	4	16	21	15	8	51
30's Customer	3	21	23	19	15	48
Teenage_Customer	1	3	2	3	2	11

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Type_Customers	Visited_Customers	Buyer_Customers	Conversion_rate_of_non_referral_customers	Repeated_Customers	Retention_rate	Average_Revenue_per_invoice
Regular_Customers	7190	3340	46.45	938	28.08	9375.13
Referral_Customers	810	381	47.04	113	29.66	9250.43

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Type_of_Referrers	Buyer_Referrers	Total_Referral_given	Buyer_Referrals	Referrals_Conversion_Rate	Repeated_referrals	Referrals_retention_rate	Revenue_from_referrals	Contribution_in_Revenue
Buyer_Referrers	333	374	168	44.92	49	29.17	2110932.00	4.51
Non_Buyer_Referrers	388	436	213	48.85	64	30.05	2653041.00	5.67
Total	721	810	381	47.04	113	29.66	4763973.00	10.18

7.5) Product Analysis

Product Analysis View

Before beginning the product analysis, a dedicated SQL VIEW named **Product Analysis Data** was created to consolidate all product-related information into a single reusable analytical dataset.

The view integrates customer demographics, vision type, staff assignment, purchase history, repeat customer status, repeated customer churn status, Top 50 customer flag, invoice details, product information, quantity sold, and final product price by joining multiple normalized tables into one centralized view.

This approach eliminates the need to repeatedly write long and complex JOIN operations for every product-related analysis, making subsequent SQL queries significantly shorter, easier to maintain, and more efficient.

The view serves as the foundation for all product analytics performed in this project, including:

- Product-wise sales analysis
- Brand performance analysis
- Product type analysis
- Revenue analysis
- Quantity sold analysis
- Vision type vs product preference analysis
- Age group vs product preference analysis
- Repeat customer purchasing behaviour
- Churn customer product analysis
- Top customer product purchasing patterns

By creating this reusable analytical view, the product analysis becomes more modular, scalable, and easier to extend for future business questions without rewriting the complete data preparation process each time.

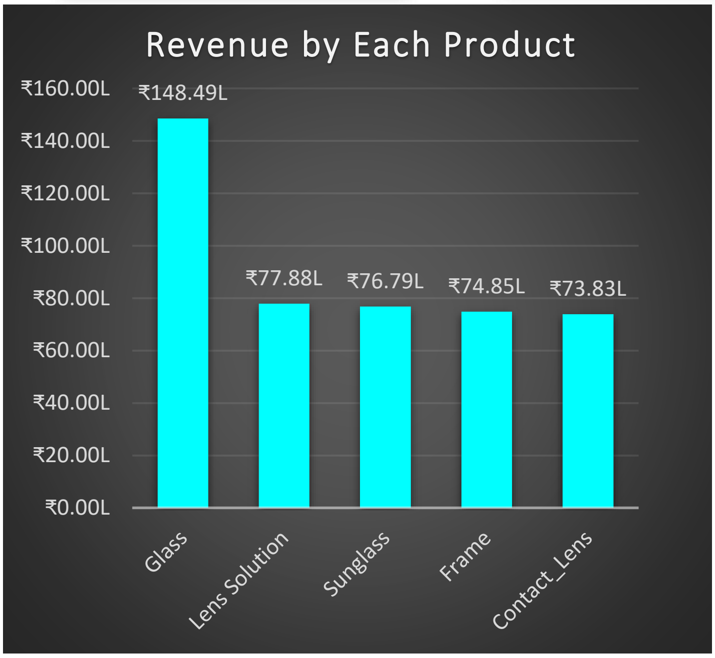
```sql
CREATE VIEW Product_Analysis_data AS
WITH customers AS (
SELECT DISTINCT customer_id
FROM customers_details
),
customers_detail AS (
SELECT
        c.customer_id, cd.age, cd.mobile, cd.customer_type,
        cd.assigned_staff_id, cd.referred_by_id
FROM customers as c
INNER JOIN customers_details as cd
ON cd.customer_id=c.customer_id
),
cd_vision AS (
SELECT
        cd.customer_id, cd.age, cd.mobile, cd.customer_type,
        cd.assigned_staff_id, cd.referred_by_id,
        p.vision_type
FROM customers_detail as cd
INNER JOIN prescriptions as p
ON p.customer_id=cd.customer_id
),
cdv_staff AS (
SELECT
        cdv.customer_id, cdv.age, cdv.mobile, cdv.customer_type,
        cdv.assigned_staff_id, cdv.referred_by_id,
        cdv.vision_type, s.staff_full_name
FROM cd_vision as cdv
INNER JOIN staff as s
ON s.staff_id=cdv.assigned_staff_id
),
```

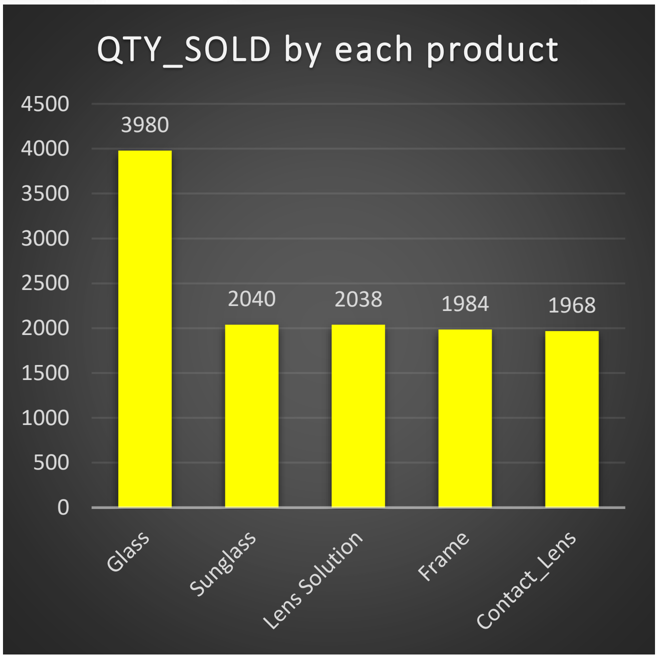
```sql
cdvs_repeated AS (
SELECT
        cdvs.customer_id, cdvs.age, cdvs.mobile, cdvs.customer_type,
        cdvs.assigned_staff_id, cdvs.referred_by_id,
        cdvs.vision_type, cdvs.staff_full_name,
        rc.customer_id as repeated_customers
FROM cdv_staff as cdvs
LEFT JOIN repeated_customers as rc
ON rc.customer_id=cdvs.customer_id
),
cdvsr_churn AS (
SELECT
        cdvsr.customer_id, cdvsr.age, cdvsr.mobile, cdvsr.customer_type,
        cdvsr.assigned_staff_id, cdvsr.referred_by_id,
        cdvsr.vision_type, cdvsr.staff_full_name,
        cdvsr.repeated_customers,
        rcc.customer_id as repeated_churn_customers
FROM cdvs_repeated as cdvsr
LEFT JOIN repeated_churn_customers as rcc
ON cdvsr.customer_id=rcc.customer_id
),
cdvsrc_top AS (
SELECT
        cdvsrc.customer_id, cdvsrc.age, cdvsrc.mobile, cdvsrc.customer_type,
        cdvsrc.assigned_staff_id, cdvsrc.referred_by_id,
        cdvsrc.vision_type, cdvsrc.staff_full_name,
        cdvsrc.repeated_customers,
        cdvsrc.repeated_churn_customers,
        tc.customer_id as top_50_customers
FROM cdvsr_churn as cdvsrc
LEFT JOIN top_50_customers as tc
ON tc.customer_id=cdvsrc.customer_id
),
```

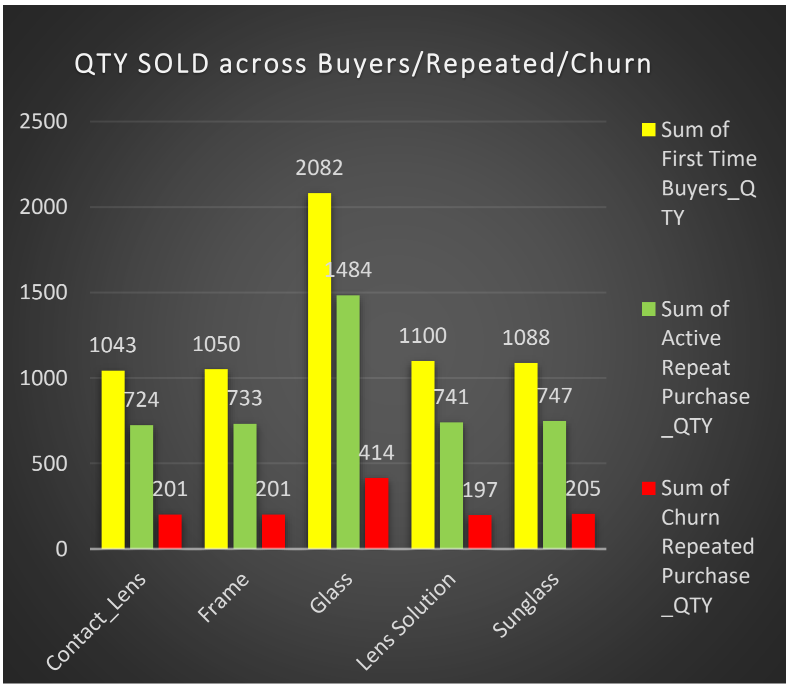
```sql
cdvsrct_invoices AS (
SELECT
        cdvsrct.customer_id, cdvsrct.age, cdvsrct.mobile, cdvsrct.customer_type,
        cdvsrct.assigned_staff_id, cdvsrct.referred_by_id,
        cdvsrct.vision_type, cdvsrct.staff_full_name,
        cdvsrct.repeated_customers,
        cdvsrct.repeated_churn_customers,
        cdvsrct.top_50_customers,
        i.invoice_id
FROM cdvsrc_top as cdvsrct
INNER JOIN invoices as i
ON i.customer_id=cdvsrct.customer_id
),
cdvsrcti_invoice_items AS (
SELECT
        cdvsrcti.customer_id, cdvsrcti.age, cdvsrcti.mobile, cdvsrcti.customer_type,
        cdvsrcti.assigned_staff_id, cdvsrcti.referred_by_id,
        cdvsrcti.vision_type, cdvsrcti.staff_full_name,
        cdvsrcti.repeated_customers,
        cdvsrcti.repeated_churn_customers,
        cdvsrcti.top_50_customers,
        cdvsrcti.invoice_id, ii.item_id, ii.product_id, ii.brand_id, ii.product_type_id,
        ii.quantity, ii.final_product_price
FROM cdvsrct_invoices as cdvsrcti
INNER JOIN invoice_items as ii
ON ii.invoice_id=cdvsrcti.invoice_id
)
SELECT
        cdvsrctiii.customer_id, cdvsrctiii.age, cdvsrctiii.mobile, cdvsrctiii.customer_type,
        cdvsrctiii.assigned_staff_id, cdvsrctiii.referred_by_id,
        cdvsrctiii.vision_type, cdvsrctiii.staff_full_name,
        cdvsrctiii.repeated_customers,
        cdvsrctiii.repeated_churn_customers,
        cdvsrctiii.top_50_customers,
        cdvsrctiii.invoice_id, cdvsrctiii.item_id,
        cdvsrctiii.product_id, cdvsrctiii.brand_id,
        cdvsrctiii.product_type_id,
        cdvsrctiii.quantity, cdvsrctiii.final_product_price
FROM cdvsrcti_invoice_items as cdvsrctiii;
```

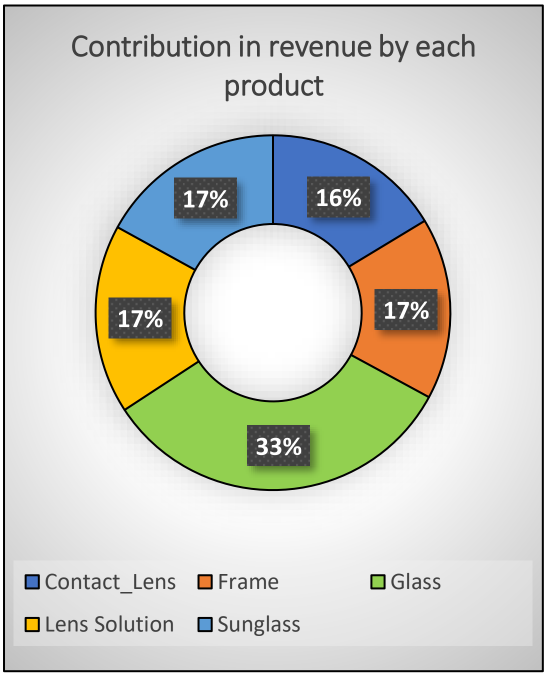
Dashboard of Product Analysis

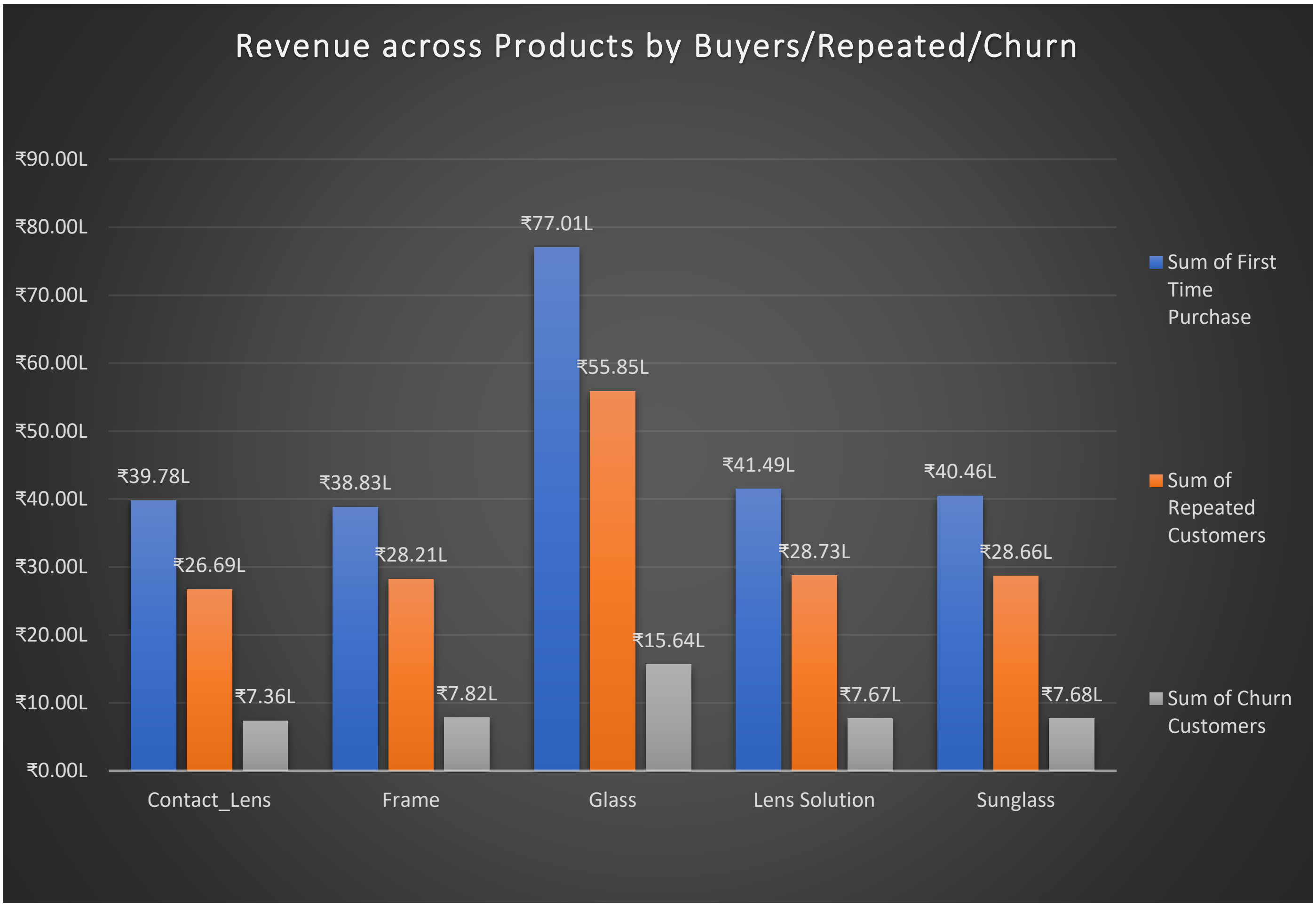
5	12,010	4.51 Cr (Rs)
Total Products	Total QTY Sold	Total Revenue
Frames	1984	74.85 L
Glass	3980	1.48Cr
Sunglasses	2040	76.79L
Contact Lens	1968	73.83L
Lens Solution	2038	77.88L

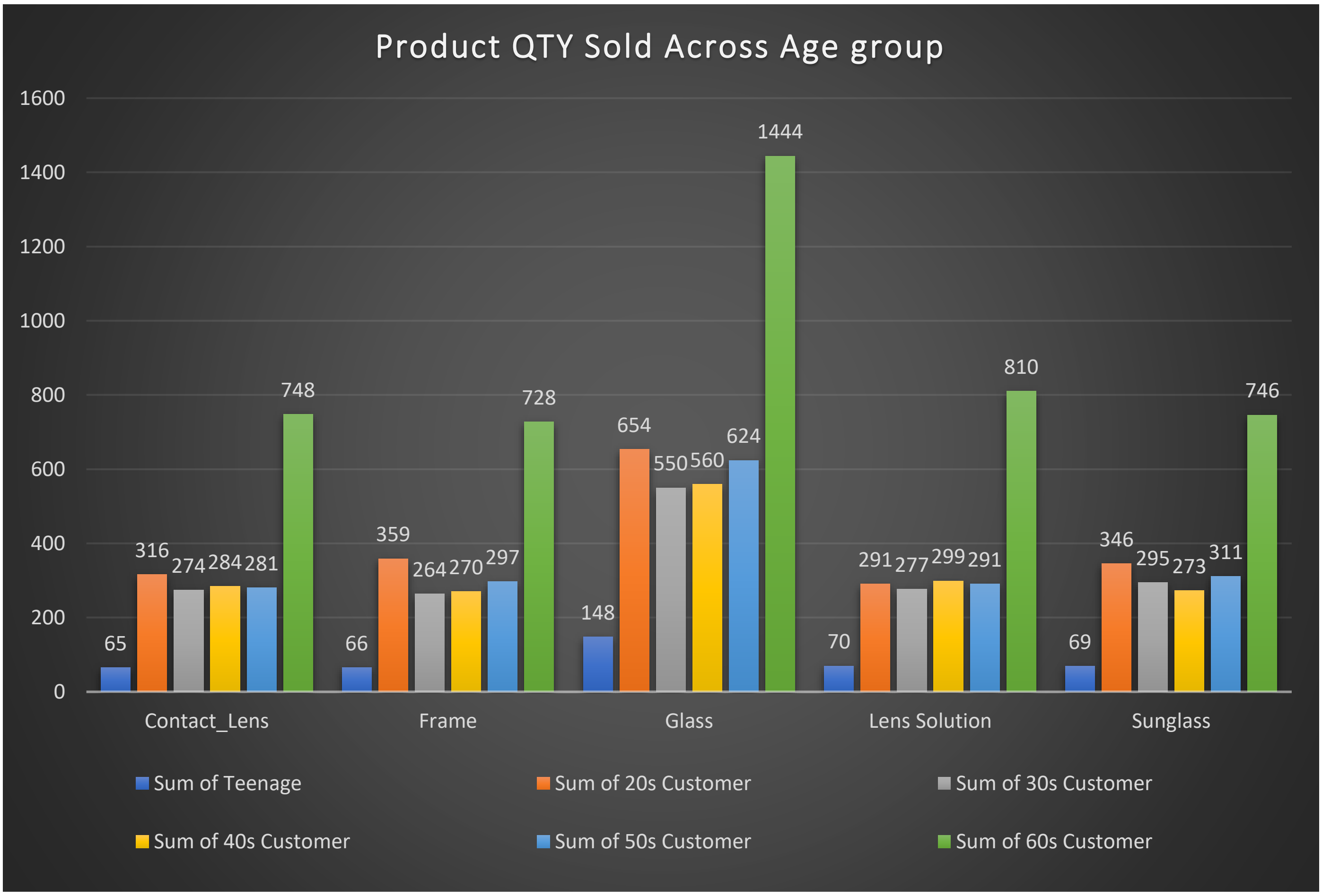









Revenue across Products by Buyers/Repeated/Churn
₹90.00L
₹80.00L
₹70.00L
₹60.00L
₹50.00L
₹40.00L
₹30.00L
₹20.00L
₹10.00L
₹0.00L
₹39.78L
₹26.69L
₹7.36L
₹38.83L
₹28.21L
₹7.82L
₹77.01L
₹55.85L
₹15.64L
₹41.49L
₹28.73L
₹7.67L
₹40.46L
₹28.66L
₹7.68L
Contact_Lens
Frame
Glass
Lens Solution
Sunglass
Sum of First Time Purchase
Sum of Repeated Customers
Sum of Churn Customers


Product QTY Sold Across Age group
1600
1400
1200
1000
800
600
400
200
0
65
316
274
284
281
748
66
359
264
270
297
728
148
654
550
560
624
1444
70
291
277
299
291
810
69
346
295
273
311
746
Contact_Lens
Frame
Glass
Lens Solution
Sunglass
Sum of Teenage
Sum of 20s Customer
Sum of 30s Customer
Sum of 40s Customer
Sum of 50s Customer
Sum of 60s Customer

Why Product Analysis Matters

Product and Brand Analysis plays a critical role in retail decision-making as it helps management understand which products and brands drive the highest revenue, sales volume, repeat purchases, and customer demand. These insights enable the business to optimize inventory planning, identify high-margin product categories, strengthen supplier relationships, improve cross-selling opportunities, and understand purchasing preferences across different customer segments. Ultimately, this analysis supports better merchandising decisions while maximizing profitability and reducing inventory-related risks.

Product Analysis

Key Findings & Business Insights

1. Glasses are the Primary Revenue Driver of the Business

The analysis clearly identifies Glasses as the business's flagship product category.

Not only does it record the highest quantity sold (3,980 units), but it also contributes approximately ₹1.48 Crore in revenue, accounting for nearly 33% of the company's total product revenue, almost twice the contribution of every other product category.

This indicates that the business is fundamentally driven by prescription eyewear rather than accessories or secondary products.

Furthermore, cross-analysis with customer demographics shows that Old Age Customers (>60 years) are the largest purchasers of glasses, which aligns perfectly with previous customer segmentation findings where this age group also demonstrated:

- Highest customer volume
- Highest repeat customer base
- Highest revenue contribution

This consistency across multiple analyses validates that senior customers represent the company's core customer segment.

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2. Progressive Lenses Present the Largest Cross-Selling Opportunity

Although the project analysis products at a higher level, the findings naturally indicate another important business opportunity.

Senior customers frequently replace their prescription lenses due to changing eye conditions every few years. This creates a recurring opportunity to recommend:

- Progressive lenses
- Premium anti-glare coatings
- Blue-cut lenses

- Premium spectacle frames

Among these, Progressive Lenses deserve special attention because they:

- Generate significantly higher profit margins
- Meet the needs of aging customers
- Increase average bill value
- Improve long-term customer satisfaction

Since senior customers already form the largest repeat customer segment, recommending premium lens upgrades can substantially increase customer lifetime value.

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3. Cross-Selling Frames Can Significantly Increase Revenue

The analysis also reveals an overlooked business opportunity.

Many customers purchase only replacement lenses while continuing to use their existing spectacle frames.

This creates an ideal opportunity for staff to recommend new premium frames during every prescription replacement.

Selling a new frame together with prescription lenses can substantially increase average order value while improving customer satisfaction through updated styles and improved comfort.

Considering spectacle frames generally carry retail margins ranging between **100–200%**, increasing frame attachment rates can become one of the most profitable sales strategies for the business.

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4. Churn Among Glass Buyers Represents a Significant Revenue Risk

One of the most important findings from the churn analysis is that a considerable number of **Glass buyers** have already entered the churn segment.

These customers collectively represent approximately **₹15 Lakhs** of historical revenue that is now at risk of being permanently lost if re-engagement efforts are not initiated.

Since prescription customers typically require periodic eye examinations and lens replacements, proactive follow-ups through:

- Eye check-up reminders
- WhatsApp campaigns
- Phone calls
- Personalized offers
- Frame upgrade recommendations

We can recover a meaningful portion of this revenue before these customers permanently shift to competitors.

5. Product Portfolio is Well Balanced

Apart from Glasses, the remaining product categories—including:

- Contact Lenses
- Spectacle Frames
- Sunglasses
- Lens Solutions

all contribute relatively similar sales volumes and revenue.

Their contribution differs only marginally, indicating that the business is not overly dependent on any single secondary product category.

This balanced product mix reduces operational risk while ensuring stable revenue generation from multiple product lines.

Product Brand Analysis

5
Total Products

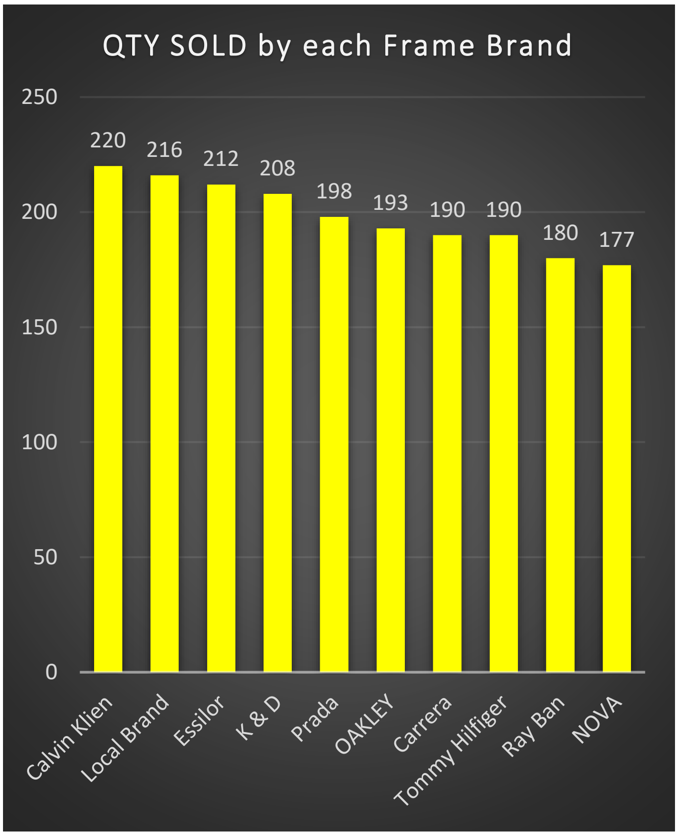
25
Total Brands

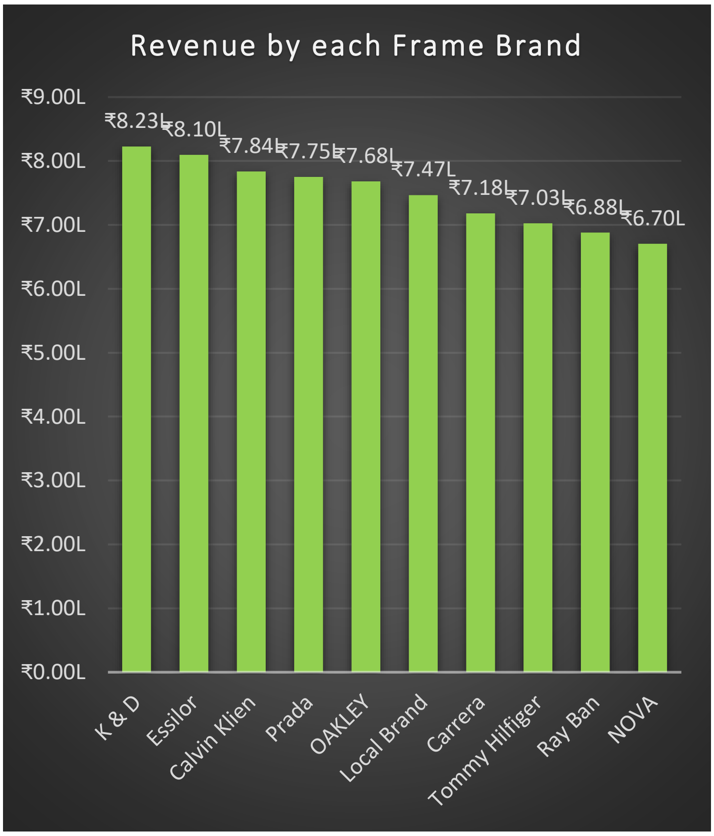
201
Total Variety of Products

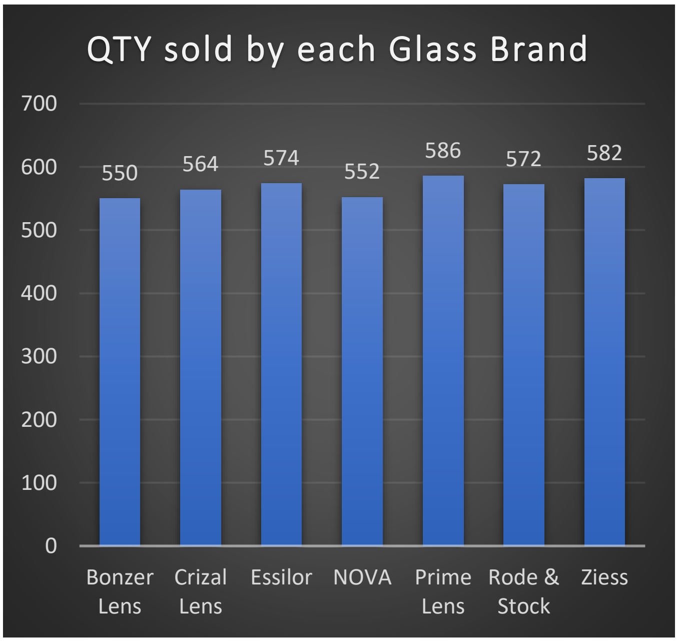
Ray Ban | **Prada** | **Prime** | **Bonzer** | **Renu Fresh**

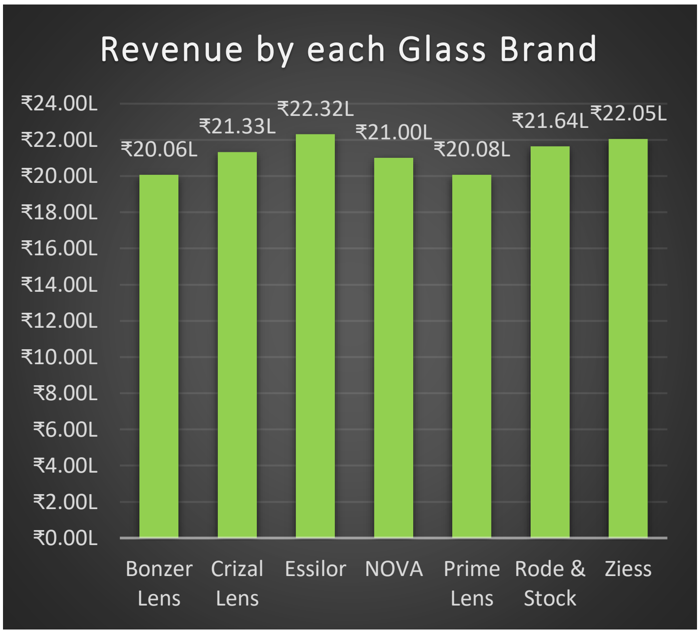
Essilor | **Acme** | **Crizal** | **Local Brand** | **Aqua Soft Bio**

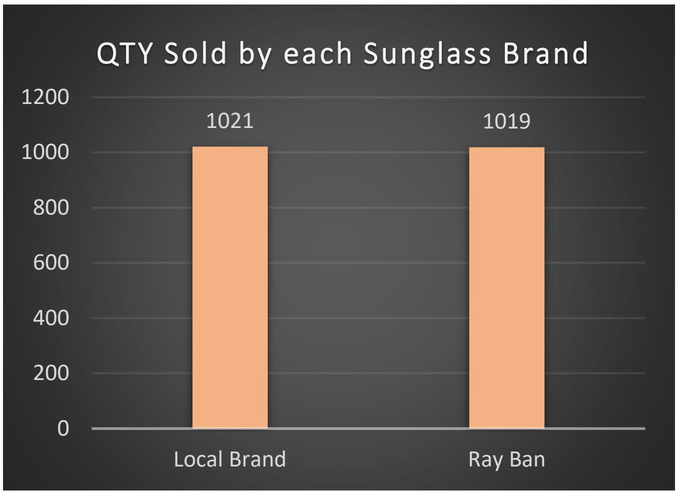
NOVA | **Ziess** | **Oakley** | **Baush & Lomb**

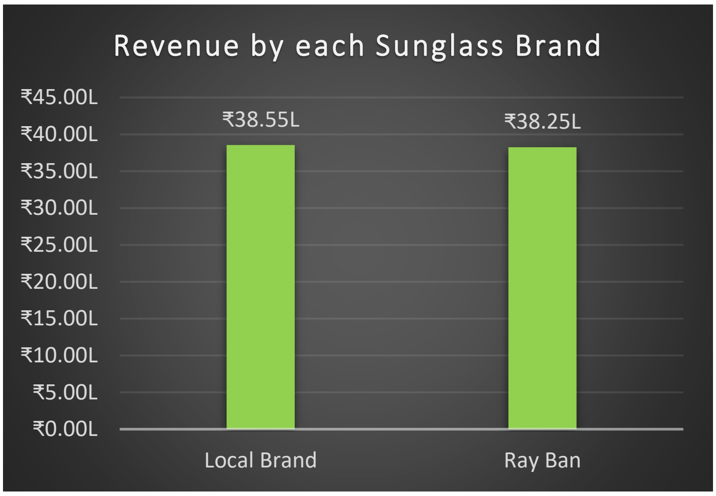


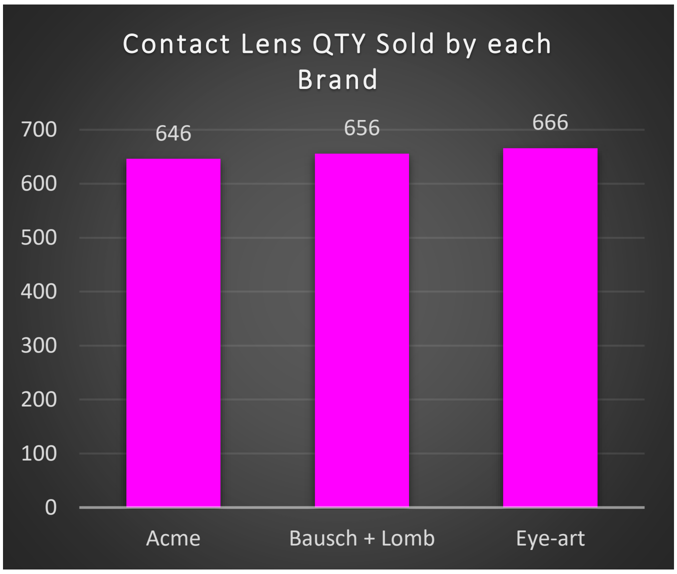


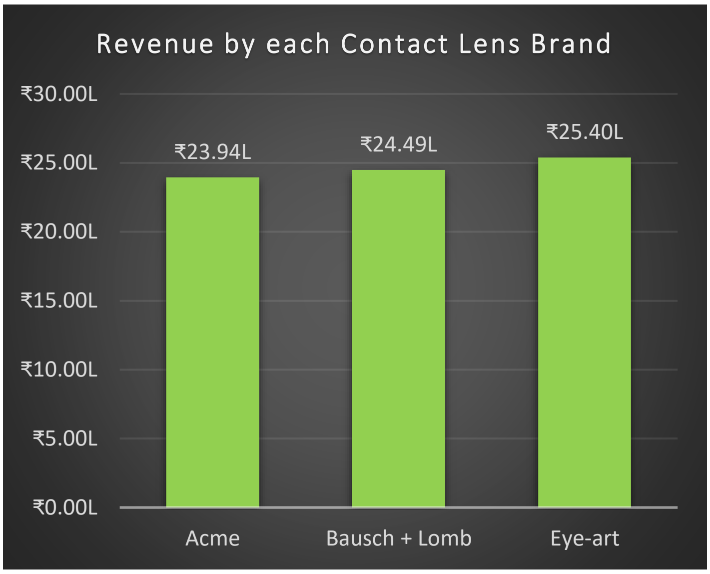


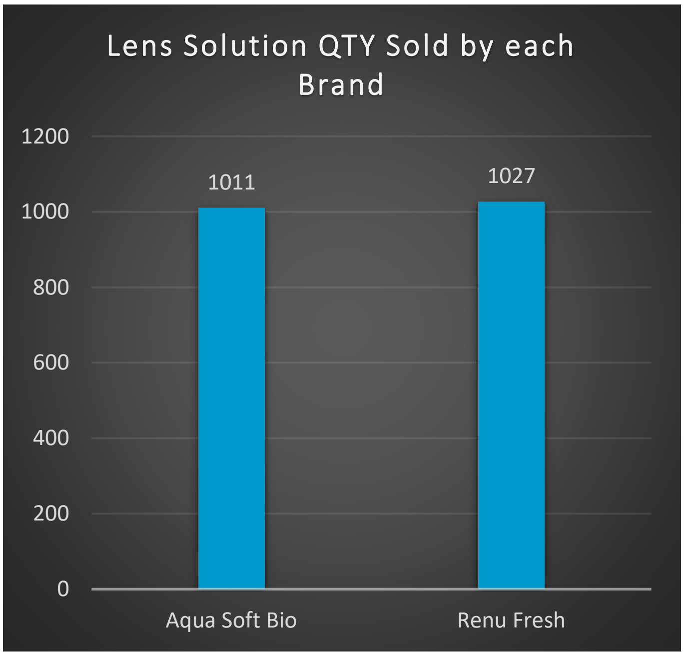


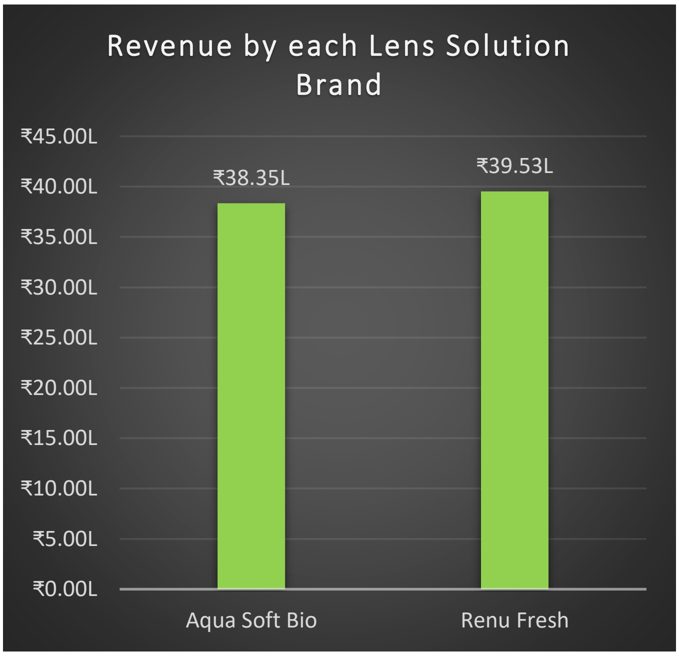

QTY Sold by each Sunglass Brand
1200
1000
800
600
400
200
0
1021
1019
Local Brand
Ray Ban


Revenue by each Sunglass Brand
₹45.00L
₹40.00L
₹35.00L
₹30.00L
₹25.00L
₹20.00L
₹15.00L
₹10.00L
₹5.00L
₹0.00L
₹38.55L
₹38.25L
Local Brand
Ray Ban


Contact Lens QTY Sold by each Brand
700
600
500
400
300
200
100
0
646
656
666
Acme
Bausch + Lomb
Eye-art


Revenue by each Contact Lens Brand
₹30.00L
₹25.00L
₹20.00L
₹15.00L
₹10.00L
₹5.00L
₹0.00L
₹23.94L
₹24.49L
₹25.40L
Acme
Bausch + Lomb
Eye-art


Lens Solution QTY Sold by each Brand
1200
1000
800
600
400
200
0
1011
1027
Aqua Soft Bio
Renu Fresh


Revenue by each Lens Solution Brand
₹45.00L
₹40.00L
₹35.00L
₹30.00L
₹25.00L
₹20.00L
₹15.00L
₹10.00L
₹5.00L
₹0.00L
₹38.35L
₹39.53L
Aqua Soft Bio
Renu Fresh

Product Brand Analysis

Unlike the product category analysis, the brand-level analysis reveals a different and equally positive business story.

Instead of being dependent upon one dominant brand, the business maintains a healthy and diversified brand portfolio across nearly all product categories.

Key Findings & Business Insights

1. Frame Sales are Distributed Across Multiple Brands

The Frame category demonstrates excellent brand diversification.

Brands such as:

- Calvin Klein
- Essilor
- Local Brand
- KD
- Prada

appear consistently among both:

- Highest quantity sold
- Highest revenue generated

This indicates that customer demand is spread across multiple brands rather than concentrated around a single premium label.

Such diversification reduces supplier dependency while giving customers a wider range of purchasing options across different price segments.

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2. Glass Lens Brands Show Balanced Customer Demand

One of the strongest operational findings appears within the Glass category.

Sales quantities across major lens brands remain remarkably similar.

Brands including:

- Essilor
- Crizal
- Prime
- Zeiss

- NOVA
- Bonzer

all contribute comparable sales volumes and revenue.

This indicates that customers trust multiple lens brands rather than demanding only one premium option.

For inventory planning, this is an excellent outcome because inventory investment remains evenly distributed instead of being concentrated into one supplier.

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3. Local Sunglasses Successfully Compete Against Ray-Ban

Perhaps the most interesting competitive insight emerges from the Sunglasses category.

The analysis shows that Local Brand Sunglasses perform almost identically to Ray-Ban in both:

- Quantity Sold
- Revenue Generated

This suggests that customers are highly receptive to affordable alternatives when product quality and styling meet expectations.

For management, this provides an opportunity to continue promoting high-margin local brands while maintaining premium international brands for customers seeking branded products.

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4. Contact Lens Brands Demonstrate Stable Demand

The Contact Lens category also exhibits a balanced distribution.

Brands such as:

- Acme
- Bausch & Lomb
- Eye-Art

all generate comparable sales quantities and revenue.

This indicates consistent customer acceptance across manufacturers and reduces operational dependence on any single supplier.

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5. Lens Solutions Generate Stable Supporting Revenue

Although Lens Cleaning Solutions contribute a smaller share compared to prescription products, they collectively generate approximately **₹80 Lakhs** in revenue.

This is a significant supporting revenue stream.

Because every spectacle and contact lens user requires regular cleaning solutions, this category represents an excellent opportunity for:

- Bundle offers
- Add-on sales
- Checkout recommendations
- Subscription-style replacement reminders

Increasing attachment rates for these products requires minimal selling effort while improving average order value.

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6. Brand Portfolio Risk is Well Diversified

Perhaps the strongest overall conclusion from the brand analysis is that:

The business is not overly dependent on any single brand across any product category.

Revenue generation remains well distributed among multiple suppliers, reducing procurement risk while providing customers with greater choice across different pricing segments.

This diversification strengthens inventory stability and improves the business's resilience against supplier-related disruptions.

Overall Product Analysis Conclusion

The product analysis demonstrates that the business possesses a strong and balanced merchandising strategy.

While prescription **Glasses** remain the primary revenue engine driven largely by loyal senior customers the remaining product categories contribute consistently to overall business performance.

Similarly, brand analysis shows healthy diversification across suppliers, reducing operational risk while ensuring broad customer choice.

The most valuable business opportunities identified include:

- Increasing Progressive Lens adoption
- Cross-selling premium spectacle frames
- Recovering churned Glass buyers through proactive follow-ups
- Promoting high-margin local brands alongside premium brands
- Bundling Lens Solutions with eyewear purchases to increase average bill value

Overall, the product portfolio appears both commercially strong and operationally balanced, providing an excellent foundation for sustainable long-term revenue growth.

8) Business Insights & Key Performance Indicators (KPIs)

Business Insights

The Retail Optical Store Analysis reveals several strategic business findings that explain the company's current performance and identify future growth opportunities.

1. Senior Customers are the Primary Revenue Engine

Customers above 60 years of age consistently outperform every other age group across the entire customer lifecycle. They generate the highest customer volume, revenue, repeat purchases, referrals, and referral revenue, making them the business's most valuable customer segment.

2. Customer Retention Drives Long-Term Profitability

One of the most significant findings is that although only 22% of buyers become repeated customers, they contribute approximately 47% of total business revenue. This demonstrates that long-term profitability depends far more on customer retention than continuously acquiring new customers.

3. Referral Customers are the Highest ROI Acquisition Channel

Referral customers account for approximately 10% of buyers while contributing nearly ₹47.6 lakh (10%) of total revenue without requiring any direct marketing expenditure. Strengthening referral programs can therefore generate highly cost-effective business growth.

4. First-Time Buyer Churn is the Business's Biggest Challenge

The analysis shows that 51.69% of first-time buyers churn, compared to only 22.36% of repeated customers. This indicates that the primary business challenge is not retaining loyal customers but successfully converting first-time buyers into long-term repeat customers.

5. Glasses are the Core Product Category

Prescription Glasses represent the highest-selling product category, contributing approximately ₹1.48 Crore and nearly 33% of total product revenue. This confirms that prescription eyewear remains the primary revenue driver of the business.

6. Progressive Lenses Offer the Highest Revenue Opportunity

Although customer demand exists across all vision types, Progressive lenses generate the highest revenue and profit margins. Investing in premium progressive lens dispensing and consultation services can substantially improve customer lifetime value.

7. Staff Performance is Consistent Across the Organization

Revenue generation, conversion rates, retention rates, and churn rates remain remarkably similar across all staff members. This indicates standardized customer service practices and balanced operational performance throughout the organization.

8. Product Portfolio and Brand Mix are Well Diversified

The business is not overly dependent on any single product brand or supplier. Revenue and sales remain well distributed across multiple brands, reducing inventory risk while providing customers with broader product choices.

9) Solution, Feedback & Recommendations

Based on the analysis performed throughout this project, the following recommendations can help improve customer retention, increase revenue, and strengthen long-term business growth.

Customer Relationship Management

- Prioritize retention of Senior Customers through regular eye-checkup reminders, personalized follow-ups, and loyalty programs.
- Improve the first-time customer experience to encourage repeat purchases through post-purchase engagement and personalized communication.
- Develop structured CRM campaigns to reconnect churned customers before they permanently disengage.

Referral Growth

- Introduce a formal referral rewards program for existing customers.
- Encourage family-based referral campaigns, particularly among senior customers.
- Promote referral initiatives through WhatsApp, SMS, and in-store communication.

Product Strategy

- Continue expanding Progressive Lens adoption through premium consultation services.
- Increase cross-selling opportunities by recommending premium spectacle frames whenever prescription lenses are replaced.
- Bundle Lens Solutions with spectacles and contact lenses to increase average order value.
- Continue maintaining a diversified product portfolio to minimize dependency on individual suppliers.

Customer Retention

- Monitor repeat customers approaching 90–180 days since their last visit and proactively engage them before they churn.
- Create personalized reminder campaigns for annual eye examinations and prescription renewals.
- Pay special attention to high-value repeat customers who contribute a significant share of business revenue.

Operations & Staff

- Maintain standardized staff training and customer service practices.
- Continue monitoring operational KPIs to identify future performance deviations.
- Focus future improvement efforts on customer acquisition, CRM, marketing, and retention rather than restructuring staff performance.

10. Conclusion

This project demonstrates how data analytics can be applied to transform raw transactional data into meaningful business insights that support strategic decision-making.

Using SQL, customer behaviour, sales performance, referrals, churn, staff efficiency, product performance, and brand contribution were analysed to understand the key drivers of business growth.

The findings reveal that the business has established a strong position among senior customers, who consistently contribute the highest revenue, repeat purchases, and referrals. Furthermore, repeat customers and referral customers together generate a disproportionately large share of business revenue, emphasizing the importance of long-term customer relationships over short-term customer acquisition.

From a product perspective, prescription Glasses remain the primary revenue driver, while Progressive Lenses represent the highest-value growth opportunity. The business also benefits from a diversified product portfolio, balanced brand distribution, and consistent staff performance, reducing operational risk and supporting sustainable growth.

Overall, the analysis indicates that the business is operationally stable with strong customer loyalty, balanced workforce performance, and healthy product diversification. Future growth should focus on improving customer retention, strengthening referral programs, enhancing CRM initiatives, expanding premium product adoption, and increasing customer lifetime value.

This project illustrates how structured SQL analysis and business intelligence techniques can provide actionable insights that help management make informed, data-driven decisions while improving overall business performance.